

Project Euler Solutions

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Problem 1: Multiples of 3 or 5

Solution: 233168

Add the integers below 1000 that are divisible by 3 or by 5, looping once and testing each.

Problem 2: Even Fibonacci Numbers

Solution: 4613732

Generate Fibonacci numbers up to four million and sum the even ones.

Problem 3: Largest Prime Factor

Solution: 6857

Factorise 600851475143 by trial division and take the largest prime factor.

Problem 4: Largest Palindrome Product

Solution: 906609

Check every product of two three-digit numbers and keep the largest that reads the same in reverse.

Problem 5: Smallest Multiple

Solution: 232792560

Scan upward for the first number divisible by every integer from 2 to 20.

Problem 6: Sum Square Difference

Solution: 25164150

Subtract the sum of the squares of 1 to 100 from the square of their sum, using 5050 for the latter.

Problem 7: 10001st Prime

Solution: 104743

Step through the odd numbers, testing primality, until the 10001st prime is reached.

Problem 8: Largest Product in a Series

Solution: 23514624000

Slide a window of thirteen digits across the given 1000-digit number and take the largest product.

Problem 9: Special Pythagorean Triplet

Solution: 31875000

Search for $a + b + c = 1000$ with $a^2 + b^2 = c^2$ and return the product abc .

Problem 10: Summation of Primes

Solution: 142913828922

Sum every prime below two million using a sieve.

Problem 11: Largest Product in a Grid

Solution: 70600674

For each cell and each of the eight directions, multiply four adjacent entries and keep the maximum.

Problem 12: Highly Divisible Triangular Number

Solution: 76576500

Generate triangular numbers until one has more than 500 divisors.

Problem 13: Large Sum

Solution: 5537376230

Add the fifty fifty-digit numbers and report the first ten digits.

Problem 14: Longest Collatz Sequence

Solution: 837799

Compute the Collatz chain length for every start below one million and report the longest.

Problem 15: Lattice Paths

Solution: 137846528820

The monotonic routes through a 20×20 grid number $\binom{40}{20}$.

Problem 16: Power Digit Sum

Solution: 1366

Sum the decimal digits of 2^{1000} .

Problem 17: Number Letter Counts

Solution: 21124

Count the letters used to spell every number from 1 to 1000, using small per-range letter tables (with the British "and").

Problem 18: Maximum Path Sum I

Solution: 1074

Work up the triangle from the bottom: each cell takes its value plus the larger of the two cells below, leaving the answer at the apex.

Problem 19: Counting Sundays

Solution: 171

Walk a running day count month by month from 1901 to 2000, counting the months that begin on a Sunday.

Problem 20: Factorial Digit Sum

Solution: 648

Sum the decimal digits of $100!$.

Problem 21: Amicable Numbers

Solution: 31626

Sum the amicable numbers below 10000, where $d(a)$ is the proper-divisor sum and $a \neq d(a) = b$ with $d(b) = a$.

Problem 22: Names Scores

Solution: 871198282

Sort the names alphabetically, then sum each name's position times its alphabetical letter value.

Problem 23: Non-Abundant Sums

Solution: 4179871

Sum the positive integers that cannot be written as the sum of two abundant numbers (numbers whose proper divisors sum to more than themselves). Every integer greater than 28123 can be so written, so that bound is the cap.

Sieve the proper-divisor sums for all $i \leq 28123$ (walk each i across its multiples, adding i to each). The abundant numbers are those with divisor-sum exceeding themselves. Mark every $a + b$ for abundant $a \leq b$, then add up the integers left unmarked.

Problem 24: Lexicographic Permutations

Solution: 2783915460

Generate the permutations of the digits 0 to 9 in lexicographic order and take the millionth.

Problem 25: 1000-digit Fibonacci Number

Solution: 4782

Find the index of the first Fibonacci number with one thousand digits.

Problem 26: Reciprocal Cycles

Solution: 983

For each denominator below 1000, run long division while tracking remainders; the recurring cycle length is found by the first repeated remainder. Report the longest.

Problem 27: Quadratic Primes

Solution: -59231

Search coefficients $a, b \in (-1000, 1000)$ for the quadratic $n^2 + an + b$ that produces the longest run of primes from $n = 0$, returning ab .

Problem 28: Number Spiral Diagonals

Solution: 669171001

Sum the diagonals of a 1001×1001 spiral using the closed form for each ring's four corners.

Problem 29: Distinct Powers

Solution: 9183

Count the distinct values a^b for $2 \leq a, b \leq 100$ with a set.

Problem 30: Digit Fifth Powers

Solution: 443839

Sum the numbers equal to the sum of the fifth powers of their digits (the search bound is $6 \cdot 9^5$).

Problem 31: Coin Sums

Solution: 73682

Count the ways to make two pounds from the coin denominations by recursing over the coin list.

Problem 32: Pandigital Products

Solution: 45228

Sum the distinct products c for which $a \times b = c$ and the three numbers together use each digit 1–9 exactly once. The digit counts can only split as $1 \times 4 = 4$ or $2 \times 3 = 4$, so the smaller factor is below 100. Iterate a and b , concatenate a , b and ab , and keep the product whenever those nine characters are exactly $\{1, \dots, 9\}$.

Problem 33: Digit Cancelling Fractions

Solution: 100

Find the non-trivial two-digit fractions whose value is unchanged by cancelling a shared digit, and reduce the product of the four.

Problem 34: Digit Factorials

Solution: 40730

Sum the numbers equal to the sum of the factorials of their digits.

Problem 35: Circular Primes

Solution: 55

Count the primes below one million for which every digit rotation is also prime.

Problem 36: Double-base Palindromes

Solution: 872187

Sum the numbers below one million that are palindromic in both base ten and base two.

Problem 37: Truncatable Primes

Solution: 748317

Find the eleven primes that remain prime as digits are stripped from the left and from the right.

Problem 38: Pandigital Multiples

Solution: 932718654

Search for the largest 1 to 9 pandigital formed by concatenating $n, 2n, \dots$ for some integer.

Problem 39: Integer Right Triangles

Solution: 840

For perimeters up to 1000, count integer right-triangle solutions and report the perimeter with the most.

Problem 40: Champernowne's Constant

Solution: 210

Concatenating the positive integers gives $0.123456789101112\dots$; the task is the product of the digits at positions $1, 10, 100, \dots, 10^6$. Building a string of a million digits is unnecessary. The c -digit numbers form a block of $9 \cdot 10^{c-1}$ numbers contributing $c \cdot 9 \cdot 10^{c-1}$ digits, so subtract whole blocks from the target position until it lands inside the c -digit block. The number holding it is then $10^{c-1} + \lfloor (n-1)/c \rfloor$ and the wanted digit is at offset $(n-1) \bmod c$. The seven digits are 1, 1, 5, 3, 7, 2, 1, whose product is 210.

Problem 41: Pandigital Prime

Solution: 7652413

Test the permutations using digits 1 to n for primality and keep the largest prime.

Problem 42: Coded Triangle Numbers

Solution: 162

A word's letter value is triangular exactly when $1 + 8v$ is a perfect square (for value v); count the words that qualify.

Problem 43: Sub-string Divisibility

Solution: 16695334890

Sum the 0-to-9 pandigital numbers whose sliding three-digit windows $d_2d_3d_4, d_3d_4d_5, \dots, d_8d_9d_{10}$ are divisible by 2, 3, 5, 7, 11, 13, 17 respectively. Instead of testing all $10!$ permutations, build from the right: start with the three-digit multiples of 17 that have distinct digits, then repeatedly prepend a digit so each new leading window is divisible by the next prime down the list. Finally prepend the one unused digit and sum the results.

Problem 44: Pentagon Numbers

Solution: 5482660

Search pairs of pentagonal numbers whose sum and difference are both pentagonal, minimising the difference, using the inverse formula to test pentagonality.

Problem 45: Triangular, Pentagonal, and Hexagonal

Solution: 1533776805

Find the next number after 40755 that is simultaneously triangular, pentagonal, and hexagonal. The triangular condition comes for free: $H_n = n(2n - 1) = (2n - 1)(2n)/2 = T_{2n-1}$, so every hexagonal number is already triangular. It therefore suffices to walk the hexagonal numbers (starting just past $H_{143} = 40755$) and test each for pentagonality, which holds when $1 + 24H$ is a perfect square whose root, increased by one, is divisible by 6. The first hit is 1533776805.

Problem 46: Goldbach's Other Conjecture

Solution: 5777

Find the smallest odd composite that cannot be written as a prime plus twice a square.

Problem 47: Distinct Primes Factors

Solution: 134043

Find the first run of four consecutive integers each having exactly four distinct prime factors.

Problem 48: Self Powers

Solution: 9110846700

Sum $1^1 + 2^2 + \dots + 1000^{1000}$ and report the last ten digits.

Problem 49: Prime Permutations

Solution: 296962999629

Find the arithmetic progression of three four-digit primes that are digit permutations of one another, other than the given 1487 case.

Problem 50: Consecutive Prime Sum

Solution: 997651

Track running sums of consecutive primes below one million and report the prime that is the longest such sum.

Problem 51: Prime Digit Replacements

Solution: 121313

Find the smallest prime belonging to an eight-member family formed by replacing one fixed set of digit positions with each value 0–9. Sieve the primes for constant-time tests and, for each prime, try the position masks. Only a mask replacing a number of positions divisible by 3 can produce eight primes: otherwise, as the replacement digit runs 0–9, the digit sum modulo 3 cycles and culls too many candidates as multiples of 3. Return the smallest member of the first eight-family found.

Problem 52: Permuted Multiples

Solution: 142857

Find the smallest x whose multiples $2x$ through $6x$ are all digit permutations of x .

Problem 53: Combinatoric Selections

Solution: 4075

Count the binomial coefficients $\binom{n}{r}$ with $n \leq 100$ that exceed one million.

Problem 54: Poker Hands

Solution: 376

Given a thousand pairs of five-card hands, count how often the first player wins. Rather than encode poker's branching rules with nested comparisons and kicker logic, collapse each hand into one comparable key `(category, tiebreak)`. The category is the usual ranking (straight flush down to high card), read off from whether the hand is a flush, a straight, and from the sorted pattern of rank multiplicities `([4, 1], [3, 2], [3, 1, 1], ...)`. The tiebreak is the list of card values ordered by frequency first and value second, so that quads, trips, and pairs float to the front; ordinary tuple comparison then resolves every kicker for free. Comparing the two keys with a single `>` gives the winner, and the first player takes 376 hands.

Problem 55: Lychrel Numbers

Solution: 249

Count the numbers below 10000 that fail to produce a palindrome within fifty reverse-and-add steps.

Problem 56: Powerful Digit Sum

Solution: 972

Maximise the digit sum of a^b over $a, b < 100$.

Problem 57: Square Root Convergents

Solution: 153

Iterate the continued-fraction convergents of $\sqrt{2}$ and count those whose numerator has more digits than the denominator.

Problem 58: Spiral Primes

Solution: 26241

Grow the diagonals of a number spiral until the proportion of prime diagonal entries falls below ten percent.

Problem 59: XOR Decryption

Solution: 129448

The ciphertext was XOR-encrypted with a repeating three-letter key; recover the key and sum the decrypted byte values. Brute-forcing all 26^3 keys and judging which output "looks like English" is avoidable. Since the key has length three, the message is really three independent streams, each XOR-ed by a single byte. The most common character in English text is the space (32), so within each stream the most frequent ciphertext byte is an encrypted space, and the key byte is simply that value XOR 32. Three frequency counts recover the key (it spells `exp`) with no searching; decrypting and summing gives 129448.

Problem 60: Prime Pair Sets

Solution: 26033

Find five primes for which concatenating any two (in either order) is also prime, minimising their sum. This is a minimum-weight 5-clique search in the graph whose edges join compatible primes, which two ideas tame. First, a residue filter: concatenating p and q gives $p \cdot 10^k + q \equiv p + q \pmod{3}$, so unless a prime equals 3, every prime in the set must share one residue mod 3; the search therefore runs separately within each residue class. Second, a sum bound: exploring each class in increasing order, a partial clique is abandoned as soon as adding the next (and hence every later) prime would reach the best sum found so far. The minimal set is $\{13, 5197, 5701, 6733, 8389\}$, summing to 26033.

Problem 61: Cyclical Figurate Numbers

Solution: 28684

Find the six four-digit numbers — one triangular, one square, one pentagonal, one hexagonal, one heptagonal, one octagonal — that form a cycle in which each number's last two digits are the next's first two, with the cycle closing back to the start. Generate the four-digit numbers of each type and index them by leading two digits, so the overlap constraint becomes an instant lookup. A depth-six search then assigns one number per type, at each step only following numbers whose first two digits match the previous number's last two; the constraint is tight enough that the tree barely branches. Anchoring on the octagonal numbers fixes the cycle's rotation. The unique ring is 1281, 8128, 2882, 8256, 5625, 2512, summing to 28684.

Problem 62: Cubic Permutations

Solution: 127035954683

Group cubes by their sorted digits and find the smallest cube belonging to a group of exactly five digit-permutation cubes.

Problem 63: Powerful Digit Counts

Solution: 49

Count the n -digit positive integers that are also n th powers.

Problem 64: Odd Period Square Roots

Solution: 1322

Count the $n \leq 10000$ whose continued fraction for $\sqrt{n}$ has an odd period. No decimal expansion is needed: the continued fraction of a quadratic irrational is generated exactly by an integer recurrence. Starting from $a_0 = \lfloor \sqrt{n} \rfloor$, $m_0 = 0$, $d_0 = 1$, iterate

$$m_{k+1} = d_k a_k - m_k, \quad d_{k+1} = (n - m_{k+1}^2)/d_k, \quad a_{k+1} = \lfloor (a_0 + m_{k+1})/d_{k+1} \rfloor.$$

The period always closes exactly when a term equals $2a_0$, so counting the steps to that point gives the period length (perfect squares having none). Of the non-squares up to 10000, 1322 have an odd period.

Problem 65: Convergents of e

Solution: 272

Build the hundredth convergent of the continued fraction for e and sum the digits of its numerator.

Problem 66: Diophantine Equation

Solution: 661

For each non-square $D \leq 1000$, the equation $x^2 - Dy^2 = 1$ (Pell's equation) has a smallest positive solution; find the D whose minimal x is largest. These minimal solutions can be astronomically large (for $D = 661$ the value of x has 26 digits), so no direct search works. Instead the fundamental solution sits among the convergents h/k of the continued fraction of $\sqrt{D}$: running the same exact integer recurrence as Problem 64 while accumulating convergents via $h_i = a_i h_{i-1} + h_{i-2}$ and $k_i = a_i k_{i-1} + k_{i-2}$, the first convergent satisfying $h^2 - Dk^2 = 1$ gives the minimal x . Scanning D up to 1000, the largest minimal solution occurs at $D = 661$.

Problem 67: Maximum Path Sum II

Solution: 7273

The same bottom-up dynamic programming as Problem 18, applied to the hundred-row triangle read from file.

Problem 68: Magic 5-gon Ring

Solution: 6531031914842725

Arrange 1 to 10 on a five-pointed ring (five outer nodes, five inner) so that every line — one outer node plus two adjacent inner nodes — has the same total, then read the lines off as a string starting from the lowest outer node. Find the maximum 16-digit string. Two observations remove the search. A 16- rather than 17-digit string forces the two-digit 10 onto an outer node (outer nodes appear once in the string, inner nodes twice). To maximise a string that begins at the smallest outer node, the outer ring must be the five largest values $\{6, 7, 8, 9, 10\}$ and the inner pentagon $\{1, 2, 3, 4, 5\}$; the common line total is then forced to $(\text{sum outer}) + 2(\text{sum inner}) = 70 = 5 \cdot 14$. Each outer value is therefore determined by the inner arrangement, $\text{outer}_j = 14 - \text{inner}_j - \text{inner}_{j+1}$, so it suffices to try the 120 inner pentagons, keep those whose outers are exactly $\{6, \dots, 10\}$, and take the largest string: 6531031914842725.

Problem 69: Totient Maximum

Solution: 510510

Maximise $n/\varphi(n)$ for $n \leq 10^6$; the optimum is the primorial 510510.

Problem 70: Totient Permutation

Solution: 8319823

We want the n with $1 < n < N$, where $N = 10^7$, that minimises the ratio $\frac{n}{\varphi(n)}$ subject to $\varphi(n)$ being a digit permutation of n .

Computing every totient at once

Factorising each n on its own is the slow part. Instead we compute φ for every value below N in a single sieve, using the product formula

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

Start with $\varphi[i] = i$ for all i . Sweep $p = 2, 3, 4, \dots$; whenever $\varphi[p]$ is still equal to p its entry has never been modified, so p must be prime. For each such p we fold in the factor $(1 - \frac{1}{p})$ across all of its multiples:

$$\varphi[k] \leftarrow \varphi[k] - \frac{\varphi[k]}{p}, \quad k = p, 2p, 3p, \dots$$

After the sweep $\varphi[i]$ holds $\varphi(i)$ for every $i < N$. The cost is $O(N \log \log N)$, versus factorising N numbers separately.

Searching for the minimum ratio

Two observations keep the search cheap.

First, a digit permutation leaves the digit sum unchanged, and every integer is congruent to its digit sum modulo 9. So a necessary condition for $\varphi(n)$ to be a permutation of n is

$$n \equiv \varphi(n) \pmod{9},$$

which rejects roughly $\frac{8}{9}$ of all candidates before any further work. For the survivors we confirm a genuine permutation by tallying digits: increment a length-10 array once per digit of n , decrement it once per digit of $\varphi(n)$, and accept only if every entry is back to zero.

Second, the ratios are compared exactly, without floating point. The best ratio so far is kept as a fraction $n_{\text{best}}/d_{\text{best}}$, and a new candidate $n/\varphi(n)$ improves on it precisely when

$$n \cdot d_{\text{best}} < n_{\text{best}} \cdot \varphi(n),$$

a comparison of two integer products.

The search itself assumes nothing about the form of the answer, but the winner turns out to be $8319823 = 2339 \times 3557$, a product of two primes straddling $\sqrt{N} \approx 3162$. That is exactly where $\frac{n}{\varphi(n)} = \prod_{p|n} p/(p-1)$ is smallest: few prime factors, each as large as possible, while still leaving room for $\varphi(n)$ to be a permutation of n .

Problem 71: Ordered Fractions

Solution: 428570

Scan denominators up to one million for the reduced fraction immediately below $\frac{3}{7}$.

Problem 72: Counting Fractions

Solution: 303963552391

We count the reduced proper fractions n/d with $1 \leq n < d$, $\gcd(n, d) = 1$, and $d \leq N$ for $N = 10^6$.

Fix a denominator d . The valid numerators are exactly the integers in $[1, d)$ that are coprime to d , and the count of those is by definition Euler's totient $\varphi(d)$. Summing over every denominator gives

$$\sum_{d=2}^N \varphi(d).$$

(The denominator $d = 1$ contributes nothing, since $n < d$ leaves no numerator.)

So the whole problem reduces to one totient sieve: build $\varphi(d)$ for all $d \leq N$ with the same $O(N \log \log N)$ sieve used in Problem 70, then add up the entries from 2 to N .

Problem 73: Counting Fractions in a Range

Solution: 7295372

Count the reduced fractions strictly between $\frac{1}{3}$ and $\frac{1}{2}$ with denominator at most 12000.

Problem 74: Digit Factorial Chains

Solution: 402

Count the starting numbers below one million whose digit-factorial chain has exactly sixty non-repeating terms.

Problem 75: Singular Integer Right Triangles

Solution: 161667

Count the wire lengths $L \leq 1.5 \cdot 10^6$ that can be bent into exactly one integer-sided right triangle. Rather than search each length for triangles, generate the triangles. Euclid's formula gives every primitive Pythagorean triple from coprime $m > n > 0$ of opposite parity, with legs $m^2 - n^2$ and $2mn$ and hypotenuse $m^2 + n^2$, hence perimeter $2m(m + n)$. Every non-primitive triple is an integer multiple of a primitive one, so for each primitive perimeter p , increment a counter at $p, 2p, 3p, \dots$ up to the limit. This is a sieve over triangles rather than a scan over lengths; the answer is the number of perimeters whose counter ends at exactly one, namely 161667.

Problem 76: Counting Summations

Solution: 190569291

Count the ways to write 100 as a sum of two or more positive integers, via memoised recursion over the largest part.

Problem 77: Prime Summations

Solution: 71

Find the first value expressible as a sum of primes in more than five thousand ways.

Problem 78: Coin Partitions

Solution: 55374

Find the least n for which $p(n)$, the number of partitions of n , is divisible by one million. Partition numbers grow super-polynomially, so neither enumerating partitions nor storing the full values is viable. Two ideas handle it. Euler's pentagonal number theorem gives a recurrence with only $O(\sqrt{n})$ terms,

$$p(n) = \sum_{k \geq 1} (-1)^{k+1} (p(n - g_k) + p(n - g_{k'})),$$

where $g_k = k(3k - 1)/2$ and $g_{k'} = k(3k + 1)/2$ are the generalized pentagonal numbers (with $p(0) = 1$ and $p(\text{negative}) = 0$). And since only divisibility by 10^6 matters, every value is kept modulo 10^6 , so the numbers never grow. Marching n upward until a residue of zero appears gives $n = 55374$.

Problem 79: Passcode Derivation

Solution: 73162890

From fifty three-digit login attempts, each revealing three characters of a secret passcode in order, reconstruct the shortest passcode consistent with all of them. Each attempt abc asserts that a comes before b and b before c , so the attempts are precedence edges of a directed acyclic graph on the digits, and the passcode is a topological ordering of it. Building the graph and running a topological sort (repeatedly emitting a digit with no unmet predecessor) recovers the order. The digits turn out to be fully constrained into one sequence with no repeats, giving the passcode 73162890.

Problem 80: Square Root Digital Expansion

Solution: 40886

For each non-square up to 100, compute one hundred digits of its square root with integer arithmetic and sum them.

Problem 81: Path Sum: Two Ways

Solution: 427337

Find the minimum-sum path through an 80×80 matrix from the top-left to the bottom-right, moving only right or down. The number of such paths is $\binom{158}{79}$, far too many to enumerate. But to stand on any cell you must have arrived from the cell above it or the cell to its left, so the cheapest cost to reach a cell is its own value plus the smaller of those two predecessors' costs. Filling this table once, top-left to bottom-right, leaves the answer in the final corner: 427337.

Problem 82: Path Sum: Three Ways

Solution: 260324

Using the same 80×80 matrix, find the minimum-sum path from any cell in the left column to any cell in the right column, now moving up, down, or right. The upward move breaks the single sweep of Problem 81, since a cell's cost can depend on one below it. The fix is to separate movement between columns from movement within a column: process columns left to right, and for each new column first record the cost of stepping right into each row, then relax vertical moves with two passes over the column — one downward, one upward — which resolves arriving at any row from any entry point. The minimum over the final column is 260324.

Problem 83: Path Sum: Four Ways

Solution: 425185

Using the same matrix once more, find the minimum-sum path from top-left to bottom-right, now moving freely in all four directions. With unrestricted movement there is no ordering of cells in which each depends only on earlier ones — the dependency graph has cycles — so the column sweeps of the previous problems no longer apply. This is a shortest-path problem: each cell is a node, and the cost of moving into a cell is its value. Dijkstra's algorithm with a priority queue repeatedly finalizes the cheapest-reachable unvisited cell (valid because all values are non-negative) and relaxes its neighbours, giving the minimum cost to the bottom-right corner as 425185.

Problem 84: Monopoly Odds

Solution: 101524

Playing Monopoly with four-sided dice, find the three squares most likely to be occupied and concatenate their two-digit indices. The board is a 40-square loop with rules that perturb a plain random walk: landing on Go To Jail sends you to jail, three consecutive doubles sends you to jail, and the Community Chest and Chance squares draw from decks that may relocate the player (to Go, jail, specific squares, the next railway or utility, or back three). A Monte Carlo simulation rolls the dice many millions of times, applies every rule (tracking the consecutive-doubles count, and re-resolving the lone case where “go back three” from square 36 lands on a Community Chest), and tallies each resulting square. The three most-visited are Jail (10), R2 (15), and E3 (24) – well separated from the rest – giving 101524.

Problem 85: Counting Rectangles

Solution: 2772

Search grid dimensions for the rectangle count nearest two million; a $w \times h$ grid contains $\sum (w - i)(h - j)$ sub-rectangles.

Problem 86: Cuboid Route

Solution: 1818

A fly walks across the surface of an $a \times b \times c$ room between opposite corners; find the least maximum dimension M for which the number of integer-sided cuboids (up to M) with an integer shortest route first exceeds one million. Unfolding the box turns the surface path into a straight line, and for $a \leq b \leq c$ the shortest unfolding has length $\sqrt{(a+b)^2 + c^2}$, so an integer route means $(a+b)^2 + c^2$ is a perfect square – a Pythagorean condition. For each largest dimension c , range over the split $s = a + b$, test whether $s^2 + c^2$ is square, and add the number of pairs (a, b) with $1 \leq a \leq b \leq c$ summing to s (a short closed count depending on whether $s \leq c + 1$). Accumulating c upward, the total passes one million at $M = 1818$ (and equals 2060 at $M = 100$, as stated).

Problem 87: Prime Power Triples

Solution: 1097343

Count the numbers below 5×10^7 expressible as $p^2 + q^3 + r^4$ for primes p, q, r . The roots are small ($r \leq 84, q \leq 368, p \leq 7071$), so sieve the primes once and mark every reachable sum in a boolean array, breaking each loop as soon as the partial sum passes the limit. The answer is the number of distinct marked sums.

Problem 88: Product-Sum Numbers

Solution: 7587457

A product-sum number for set size k is the smallest N that is both the sum and the product of some k natural numbers; sum the distinct minimal product-sum numbers for $2 \leq k \leq 12000$. Searching sets of up to twelve thousand numbers directly is hopeless, but two observations make it finite. Padding a factorization with 1s leaves the product unchanged while raising the count and the sum, so a product P formed from t factors (each ≥ 2) summing to S realizes $N = P$ for set size $k = t + (P - S)$. And the minimal value for any k is at most $2k$ (take the factors $\{2, k\}$), so products beyond $2 \cdot 12000$ never need checking. A depth-first enumeration of all factorizations up to that bound assigns each its k , keeps the smallest N per k , and the distinct minima sum to 7587457 (and to 30 for k up to 6, as stated).

Problem 89: Roman Numerals

Solution: 743

Given a thousand Roman numerals written in valid but not necessarily minimal form, count the characters saved by rewriting each minimally. Rather than reason about Roman-numeral rules directly, pass through the integer value, which is unambiguous: parse each numeral (subtracting a symbol whenever a larger one follows it, adding it otherwise), then re-emit greedily from a value table that includes the six subtractive pairs (CM, CD, XC, XL, IX, IV), taking the largest symbol that fits at each step. The saving is the length difference, and over all the numerals it totals 743.

Problem 90: Cube Digit Pairs

Solution: 1217

Two cubes each carry six of the ten digits; count the cube pairs that can display all nine two-digit squares (01, 04, 09, 16, 25, 36, 49, 64, 81) by placing the cubes side by side in either order. There are only $\binom{10}{6} = 210$ possible cubes, so the search is small – the real subtlety is that a 6 can be turned upside down to serve as a 9 and vice versa. Modelling that as a single capability (a cube “shows” 6 or 9 if it bears either), a pair is valid when, for every square ab , one cube shows a while the other shows b in some arrangement. Counting valid unordered pairs gives 1217.

Problem 91: Right Triangles with Integer Coordinates

Solution: 14234

Count the right triangles with one vertex at the origin and the other two at lattice points in a 50×50 grid. The right angle may sit at any of the three vertices, and degenerate (collinear) triples must be excluded. No trigonometry or distances are needed: two edge vectors are perpendicular exactly when their dot product is zero, an exact integer test. So for each unordered pair of grid points P, Q (origin excluded), discard the pair if O, P, Q are collinear (cross product zero), then check the three vertices with three integer dot products. Sweeping every pair once counts each triangle exactly once, giving 14234.

Problem 92: Square Digit Chains

Solution: 8581146

Every chain under repeated squared-digit-sum ends at 1 or 89; count the starts below ten million that reach 89.

Problem 93: Arithmetic Expressions

Solution: 1258

Using four distinct digits with $+$, $-$, $\times$, $\div$ and parentheses, each digit once, find the set producing the longest unbroken run of target integers $1, 2, 3, \dots, n$; report the digits in increasing order. Two ideas handle the explosion of expressions. All reachable values are generated by a single recursion: from a multiset of numbers, pick any ordered pair, replace it with each of $a + b$, $a - b$, $a \times b$, $a \div b$, and recurse on the smaller multiset – which automatically covers every ordering and parenthesization. And the arithmetic is done with exact fractions, so division never introduces rounding error and a value counts as a target only when it is exactly a positive integer. Across the $\binom{10}{4} = 210$ digit sets, $\{1, 2, 5, 8\}$ yields the longest consecutive run (reaching 51), so the answer is 1258.

Problem 94: Almost Equilateral Triangles

Solution: 518408346

Sum the perimeters of all almost-equilateral triangles – sides $a, a, a \pm 1$ – that also have integer area, with perimeter at most 10^9 . Searching side lengths up to a third of a billion is far too slow, but the integer-area condition is a Pell equation in disguise. Writing the third side as $b = a + e$ with $e = \pm 1$, an integer area needs $4a^2 - b^2 = k^2$; substituting $u = 3a - e$ turns this into $u^2 - 3k^2 = 4$. The solutions of that Pell equation are produced by the recurrence $u' = 2u + 3k$, $k' = u + 2k$, each roughly 3.7 times the last, so the rare valid triangles are generated directly rather than hunted. About fifteen steps reach the billion limit; their perimeters sum to 518408346.

Problem 95: Amicable Chains

Solution: 14316

Find the smallest member of the longest amicable chain whose every element stays at or below 10^6 (a chain is $n \rightarrow s(n) \rightarrow s(s(n)) \rightarrow \dots \rightarrow n$, where s sums proper divisors).

Sieve $s(i)$ for all $i \leq 10^6$, then from each start follow the sequence. A walk that drops below the start (its true minimum was already handled) or climbs above 10^6 is discarded; one that returns to the start is a chain. A generous length cap stops the walk from spinning forever inside a foreign cycle whose minimum exceeds the start. Track the longest.

Problem 96: Su Doku

Solution: 24702

Solve fifty Sudoku puzzles and sum the three-digit number in the top-left corner of each solution. A blind cell-by-cell guess faces up to 9 choices per blank across dozens of blanks, an astronomical tree. Backtracking with a minimum-remaining-values heuristic tames it: always fill the empty cell with the fewest legal digits first, so forced cells are resolved immediately and branching happens only where unavoidable. Row, column, and box constraints are kept as bitmasks, making each cell's candidate set a couple of bitwise operations. The search then resolves each grid in milliseconds, and the corner numbers sum to 24702.

Problem 97: Large Non-Mersenne Prime

Solution: 8739992577

Report the last ten digits of $28433 \cdot 2^{7830457} + 1$.

Problem 98: Anagramic Squares

Solution: 18769

Among anagram word pairs, a bijective letter-to-digit substitution may turn both words into square numbers; find the largest such square. Searching all letter-to-digit assignments per word is wasteful. Instead, let the squares supply the mappings: group words by their sorted letters to find anagram pairs, and for each pair of a given length, scan the squares of that length. Aligning a square's digits with the first word reads off the implied letter-to-digit map (rejecting it unless it is a clean bijection); applying that same map to the anagram partner and checking whether the result is also a square (with no leading zero) confirms an anagramic pair. The largest square obtained this way is 18769.

Problem 99: Largest Exponential

Solution: 709

Compare the base-exponent pairs from file by exponent times $\ln$ of base and report the line of the largest.

Problem 100: Arranged Probability

Solution: 756872327473

A box of blue and red discs gives probability exactly $1/2$ of drawing two blue discs without replacement; find the number of blue discs in the first arrangement with more than 10^{12} discs in total. The condition $2b(b-1) = n(n-1)$ is, after substituting $u = 2n-1$ and $v = 2b-1$, the negative Pell equation $u^2 - 2v^2 = -1$. Its solutions are generated by $(u, v) \rightarrow (3u + 4v, 2u + 3v)$, each about 5.8 times the last, producing the arrangements $3/4, 15/21, 85/120, \dots$ directly rather than by search. After about twenty steps the total passes a trillion, with 756872327473 blue discs.

Problem 101: Optimum Polynomial

Solution: 37076114526

A degree-ten polynomial generates a sequence; if only the first k terms were known, one would fit the unique degree- $(k - 1)$ polynomial through them and predict the next term. Sum these first incorrect predictions over all k before the fit becomes exact. There is no need to solve for polynomial coefficients: Lagrange interpolation evaluates the fitted polynomial directly at a target point as a weighted combination of the known values, done in exact fractions. Since a fit through the first k points reproduces the sequence exactly there, its first disagreement is always at term $k + 1$; summing $OP(k, k + 1)$ for $k = 1$ to 10 gives 37076114526.

Problem 102: Triangle Containment

Solution: 228

Given a thousand triangles by their vertex coordinates, count how many contain the origin. The cross product gives the signed side of a directed edge, and the origin lies inside a triangle exactly when it is on the same side of all three edges traversed in order. So for each triangle, compute the three cross products of the origin against edges AB , BC , CA ; if their signs are all the same the origin is enclosed, and if mixed it is outside. Three integer multiplications per triangle, with no floating point or special cases, give 228 containing the origin.

Problem 103: Special Subset Sums: Optimum

Solution: 20313839404245

A special sum set has all subset sums distinct **and** larger subsets always summing to more than smaller ones; find the minimum-sum special set of size seven, written as its concatenated elements. Two simplifications make the conditions cheap to test: "disjoint subsets have distinct sums" is equivalent to "all 2^n subset sums are distinct," and the size-monotonic rule reduces to checking that the smallest $k + 1$ elements outweigh the largest k , for a few small k . The optimum is not searched from scratch but built: from an optimum set of size n with median m , the set $\{m\} \cup \{m + a_i\}$ is a near-optimal size- $(n + 1)$ candidate. Applied to the size-six optimum $\{11, 18, 19, 20, 22, 25\}$ this yields $\{20, 31, 38, 39, 40, 42, 45\}$, and a small local search confirms nothing nearby is both smaller and special, giving 20313839404245.

Problem 104: Pandigital Fibonacci Ends

Solution: 329468

Find the Fibonacci index whose first nine and last nine digits are each 1 to 9 pandigital, keeping the tail modulo 10^9 to stay fast.

Problem 105: Special Subset Sums: Testing

Solution: 73702

From a file of around a hundred sets, identify which are special sum sets and sum the totals of those. This reuses the test from Problem 103: a set is special when all 2^n subset sums are distinct (equivalent to the disjoint-subset condition) and the $k + 1$ smallest elements always outweigh the k largest. Applying that check to each set and adding the totals of the special ones gives 73702.

Problem 106: Special Subset Sums: Meta-testing

Solution: 21384

For a set of n elements whose size rule already holds, how many equal-size disjoint subset pairs actually need an equal-sum test? Unequal sizes are settled by the size rule, so only equal-size pairs (B, C) matter, and even most of those are pre-decided: if B sorted is element-wise smaller than C sorted, then B obviously sums less.

The count of such pre-decided (dominating) splits of $2k$ chosen elements is the Catalan number $C_k = \binom{2k}{k} / (k+1)$, because "one sorted sequence stays ahead of the other" is exactly the ballot condition Catalan numbers count. So for subset size k the pairs needing a test number $\binom{2k}{k} / 2 - C_k$, and summing over the ways to pick the $2k$ elements gives

$$\sum_{k=2}^{\lfloor n/2 \rfloor} \binom{n}{2k} \left(\binom{2k}{k} / 2 - C_k \right) = 21384$$

for $n = 12$.

Problem 107: Minimal Network

Solution: 259679

Given a 40-vertex weighted network as a symmetric adjacency matrix, find the greatest weight that can be removed while keeping every vertex connected. The cheapest connected subgraph is the minimum spanning tree, so the saving is the total of all edge weights minus the MST weight. Parsing the upper triangle into an edge list and running Kruskal's algorithm (sort edges ascending, add each edge whose endpoints lie in different components via union-find) gives an MST; subtracting it from the total yields 259679.

Problem 108: Diophantine Reciprocals I

Solution: 180180

We want the least n for which $1/x + 1/y = 1/n$ (positive integers) has more than 1000 distinct solutions.

Counting solutions

Clearing denominators turns $1/x + 1/y = 1/n$ into $xy = nx + ny$, and adding n^2 to both sides factors it (Simon's favourite factoring trick):

$$(x - n)(y - n) = n^2.$$

Every factorisation $n^2 = uv$ with $u \leq v$ gives a solution $x = n + u$, $y = n + v$, so the number of unordered solutions is

$$\frac{d(n^2) + 1}{2},$$

where d counts divisors. The $+1$ then halving accounts for the single symmetric pair $u = v = n$. Requiring more than 1000 solutions means

$$d(n^2) \geq 2001.$$

Finding the smallest such n

Write $n = \prod_i p_i^{a_i}$. Then $n^2 = \prod_i p_i^{2a_i}$ and

$$d(n^2) = \prod_i (2a_i + 1).$$

So we need the smallest integer whose exponents satisfy $\prod (2a_i + 1) \geq 2001$. For a fixed multiset of exponents the product $\prod p_i^{a_i}$ is smallest when the largest exponents sit on the smallest primes, so it suffices to search exponent tuples that are non-increasing across $2, 3, 5, 7, \dots$

A short depth-first search does exactly this: at prime p_i try each exponent a (capped by the previous prime's exponent), divide the remaining divisor target by $(2a + 1)$, and recurse. The search space is tiny and returns instantly. The minimum is

$$n = 180180 = 2^2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13, \quad d(n^2) = 5 \cdot 5 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = 2025,$$

giving $(2025 + 1)/2 = 1013$ solutions.

Problem 109: Darts

Solution: 38182

Count the ways to check out a score below 100, where a checkout uses at most three darts and the final dart must be a double. The board offers 62 distinct scoring segments (singles 1–20 and 25, doubles of those plus the bullseye 50, trebles 3–60), of which 21 are doubles. Fixing each possible finishing double, the zero, one, or two preceding darts form an unordered multiset; iterating the preceding darts with $i \leq j$ to avoid double-counting permutations, and tallying every combination whose total stays under 100, gives 38182.

Problem 110: Diophantine Reciprocals II

Solution: 9350130049860600

Find the least n for which $1/x + 1/y = 1/n$ has more than four million solutions. The number of solutions equals $(d(n^2) + 1)/2$, where d counts divisors, so the requirement is $d(n^2) > 7999999$. Writing $n = \prod p_i^{a_i}$ gives $d(n^2) = \prod (2a_i + 1)$, a product of odd factors. Minimizing n means assigning non-increasing exponents to the smallest primes; a depth-first search over exponent patterns (each prime's exponent capped by the previous one, pruning once n exceeds the best found) locates the minimum, 9350130049860600.

Problem 111: Primes with Runs

Solution: 612407567715

Among 10-digit primes, for each digit d find $M(d)$, the most times d can repeat, then sum every 10-digit prime achieving that maximum; the answer is the total over all d . Rather than scan all primes, fix d and try repetition counts from 10 downward: place d in m of the positions, fill the remaining $10 - m$ with non- d digits (a small space, since m is large), and primality-test each candidate. The first count m that yields any primes is $M(d)$; their sum contributes to the running total of 612407567715.

Problem 112: Bouncy Numbers

Solution: 1587000

Count bouncy numbers (digits neither wholly non-decreasing nor non-increasing) until they make up ninety-nine percent of all numbers seen.

Problem 113: Non-Bouncy Numbers

Solution: 51161058134250

Count the non-bouncy numbers below 10^{100} – those whose digits are entirely non-decreasing or entirely non-increasing. No search is needed; each family is a stars-and-bars count. Non-decreasing numbers of up to d digits (digits 1–9) number $\binom{d+9}{9} - 1$, and non-increasing ones (digits 0–9, discarding the all-zero string at each length) number $\binom{d+10}{10} - 1 - d$. Repdigits are counted in both, $9d$ of them, so by inclusion-exclusion the total is $\binom{d+9}{9} + \binom{d+10}{10} - 10d - 2$, which at $d = 100$ equals 51161058134250.

Problem 114: Counting Block Combinations I

Solution: 16475640049

A special case of Problem 115 with minimum block length three.

Problem 115: Counting Block Combinations II

Solution: 168

Count the arrangements of red blocks of length at least m separated by single gaps via memoised recursion, then find the row length where the count first passes one million.

Problem 116: Red, Green or Blue Tiles

Solution: 20492570929

Count the ways to lay tiles of a single length (2, 3 or 4) on a row of fifty squares, summed over the three colours, excluding the empty arrangement.

Problem 117: Red, Green, and Blue Tiles

Solution: 100808458960497

Count the ways to tile a row of 50 units using grey unit tiles together with red (length 2), green (length 3), and blue (length 4) tiles. Classifying tilings by the final tile gives the tetranacci recurrence $f(n) = f(n-1) + f(n-2) + f(n-3) + f(n-4)$ with $f(0) = 1$ and $f(n) = 0$ for $n < 0$. Rolling it forward to $n = 50$ yields 100808458960497.

Problem 118: Pandigital Prime Sets

Solution: 44680

We count the distinct sets of primes whose digits, taken together, are exactly 1, 2, ..., 9 each used once (for example {2, 5, 47, 89, 631}).

Two stages: count, then combine

Such a set is a partition of the nine digits into blocks, where each block's digits are arranged into a prime. Crucially, a set of primes determines its own partition (group the primes by which digits they use), so we may count

(set partitions of $\{1, \dots, 9\}$) $\times$ (a prime arrangement chosen per block)

without any risk of double counting, as long as each partition is enumerated once.

Stage 1 — primes per digit subset

For every non-empty subset S of the digits, let $f(S)$ be the number of permutations of S that are prime. Two cheap rules cut the work sharply:

- if $|S| > 1$ and the digit sum of S is divisible by 3, then every arrangement is divisible by 3, so $f(S) = 0$;
- a prime longer than one digit must end in 1, 3, 7 or 9, so only those arrangements are tested.

We store $f(S)$ for all 511 subsets, indexing each subset by a 9-bit mask.

Stage 2 — combining blocks

Let $g(M)$ be the number of distinct prime sets using exactly the digits in mask M .

Splitting off the block that contains the lowest set bit makes every partition unique:

$$g(M) = \sum_{B \subseteq M, \ell \in B} f(B) \cdot g(M \setminus B), \quad g(0) = 1,$$

where ℓ is the lowest set bit of M and B ranges over sub-masks containing it.

Enumerating sub-masks costs $O(3^9)$ overall. The answer is g of the full mask, $g(111111111_2) = 44680$.

Problem 119: Digit Power Sum

Solution: 248155780267521

Find the 30th number (at least two digits) that equals a power of its own digit sum, such as $512 = 8^3$ and $2401 = 7^4$. Searching the numbers directly is wasteful; instead generate candidates as b^e for integer base $b \geq 2$ and exponent $e \geq 2$, keeping those where the digit sum of b^e equals b . Since a value below 10^{16} has digit sum at most 144, bases up to about 150 suffice. Collecting the matches, sorting, and taking the 30th gives 248155780267521.

Problem 120: Square Remainders

Solution: 333082500

For each a with $3 \leq a \leq 1000$, maximize the remainder r of $(a-1)^n + (a+1)^n$ modulo a^2 . Expanding both powers binomially, every term with a^2 or higher vanishes modulo a^2 , leaving only the constant and linear terms. Even n gives remainder 2; odd n gives $2an \bmod a^2 = a \cdot ((2n) \bmod a)$. The factor $(2n) \bmod a$ can reach $a-1$ when a is odd (2 is invertible) but only $a-2$ when a is even, so the maximum remainder is $a(a-1)$ or $a(a-2)$ accordingly. Summing over the range gives 333082500.

Problem 121: Disc Game Prize Fund

Solution: 2269

A bag starts with one red and one blue disc; each of 15 turns draws a disc (returned afterward) and then adds another red, so turn i holds one blue among $i + 1$ discs and $P(\text{blue}) = 1/(i + 1)$. The player wins by drawing strictly more blue than red. Scaling turn i by $i + 1$ turns the chances into integer weights (blue 1, red i), so the coefficient of x^k in $\prod_{i=1}^{15} (i + x)$ counts the weighted outcomes with exactly k blue draws. Summing the coefficients for $k \geq 8$ gives the winning weight; the win probability is that over $16!$, and the largest prize keeping the game profitable is $\lfloor 16! / \text{wins} \rfloor = 2269$.

Problem 122: Efficient Exponentiation

Solution: 1582

For each k up to 200, find the fewest multiplications needed to compute x^k , then sum these counts. Each multiplication forms a new power by adding two exponents already obtained, so a method for x^k is exactly an addition chain ending at k , and the minimum is the shortest such chain. The shortest chain is found by iterative deepening: search for a chain of length 1, then 2, and so on, at each step extending by the sum of the current term with an earlier one, pruning whenever repeated doubling of the largest term still cannot reach k . Summing the minimal lengths over $k = 1$ to 200 gives 1582.

Problem 123: Prime Square Remainders

Solution: 21035

Using $(p-1)^n + (p+1)^n \bmod p^2$, find the first prime index whose remainder exceeds 10^{10} .

Problem 124: Ordered Radicals

Solution: 21417

Sort 1 to 100000 by radical (the product of distinct prime factors), breaking ties by value, and read off the 10000th.

Problem 125: Palindromic Sums

Solution: 2906969179

Sum the palindromes below 10^8 that can be written as a sum of two or more consecutive squares. Rather than test every number, generate the sums: for each starting base accumulate consecutive squares until the running total reaches 10^8 , recording each palindromic total in a set so that a number expressible in several ways is still counted once. Sum the set.

Problem 126: Cuboid Layers

Solution: 18522

Cubes are wrapped in layers around an $a \times b \times c$ cuboid; the n -th layer uses $2(ab + bc + ca) + 4(n - 1)(a + b + c) + 4(n - 1)(n - 2)$ cubes. The task is the least layer size produced by exactly 1000 distinct cuboids. Iterating cuboids with $a \geq b \geq c$ (so each is counted once) and all layers n , tallying every layer size up to a bound, then scanning for the smallest size with a count of 1000, yields 18522.

Problem 127: abc-Hits

Solution: 18407904

Summing c over all abc-hits with $c < 120000$, where an abc-hit is a coprime triple $a + b = c$ ($a < b$) with $\text{rad}(abc) < c$ and rad the product of distinct primes. Since $a + b = c$ with $\text{gcd}(a, b) = 1$ forces a, b, c pairwise coprime, $\text{rad}(abc) = \text{rad}(a) \text{rad}(b) \text{rad}(c)$. A radical sieve precomputes every rad , and the values are sorted by radical so that, looping over a and ascending $\text{rad}(b)$, the inner loop breaks as soon as $\text{rad}(a) \text{rad}(b) \geq 120000$ – already too large to undercut any c . The surviving triples passing the coprimality and radical tests sum to 18407904.

Problem 128: Hexagonal Tile Differences

Solution: 14516824220

Hexagonal tiles spiral outward, and $PD(n)$ counts how many of a tile's six neighbour-differences are prime; the 2000th tile with $PD = 3$ is sought. Brute-forcing the spiral on small rings shows $PD = 3$ can only occur at ring corners. The first tile of ring n , numbered $3n^2 - 3n + 2$, has differences $6n - 1, 6n + 1, 12n + 5$; the last tile, $3n^2 + 3n + 1$ (for $n \geq 2$), has $6n - 1, 6n + 5, 12n - 7$. $PD = 3$ holds exactly when those three are all prime. Tile 1 is the first such tile and ring 1's last tile (7) is a non-qualifying special case. Enumerating corners in order until the 2000th gives 14516824220.

Problem 129: Repunit Divisibility

Solution: 1000023

For n coprime to 10, let $A(n)$ be the length of the smallest repunit (a number of the form $111\dots 1$) divisible by n ; find the least n with $A(n) > 10^6$. Since $A(n) \leq n$ always, any such n must itself exceed 10^6 , so the search starts at 1000001. For each candidate coprime to 10, the repunit residue is grown one digit at a time via $x \rightarrow (10x + 1) \bmod n$, stopping early once the length passes 10^6 without hitting zero. The first qualifying n is 1000023.

Problem 130: Composites with Prime Repunit Property

Solution: 149253

With $A(n)$ the least repunit length divisible by n (from Problem 129), every prime p outside $\{2, 3, 5\}$ satisfies $A(p) \mid p - 1$. The task is the sum of the first 25 composite n (coprime to 10) for which $A(n)$ likewise divides $n - 1$. Scanning odd non-multiples of 5, skipping primes, computing $A(n)$ by the residue recurrence $x \rightarrow (10x + 1) \bmod n$, and testing $(n - 1) \bmod A(n) = 0$ collects them; the first 25, beginning with 91, sum to 149253.

Problem 131: Prime Cube Partnership

Solution: 173

Count the primes $p < 10^6$ for which some positive integer n makes $n^3 + n^2p$ a perfect cube. Writing this as $n^2(n + p)$ and trying $n = k^3$ turns it into $k^6(k^3 + p)$, a cube precisely when $k^3 + p$ is the next cube $(k + 1)^3$, i.e. $p = (k + 1)^3 - k^3 = 3k^2 + 3k + 1$. These differences of consecutive cubes are the only p that work, so the answer is the number of primes among them below 10^6 , which is 173.

Problem 132: Large Repunit Factors

Solution: 843296

Find the sum of the first 40 prime factors of the repunit $R(10^9)$, a number with a billion ones. Since $R(k) = (10^k - 1)/9$, a prime p outside $\{2, 3, 5\}$ divides $R(k)$ exactly when $10^k \equiv 1 \pmod{p}$. This is checked instantly by modular exponentiation, never forming the giant number. Testing primes in increasing order and accumulating the first 40 that satisfy the congruence gives 843296.

Problem 133: Repunit Nonfactors

Solution: 453647705

Sum the primes below 10^5 that never divide $R(10^n)$ for any n . A prime p (not 2 or 5) divides some $R(10^n)$ iff its order $\text{ord}_{p(10)}$ divides 10^n for some n , which happens exactly when that order has the form $2^a 5^b$. Equivalently, p is a divisor iff $10^{10^n} \equiv 1 \pmod{p}$ for some small n (the exponents a, b are bounded since the order is below p), testable directly. The primes failing this for all n – the nonfactors, with $p = 3$ a special case since $3 \nmid 9$ – sum to 453647705.

Problem 134: Prime Pair Connection

Solution: 18613426663617118

For each pair of consecutive primes $p_1 < p_2$ with $5 \leq p_1 \leq 10^6$, find the least n whose final digits are p_1 and which is divisible by p_2 , then sum these. Writing $n = p_1 + k \cdot 10^d$ where d is the digit count of p_1 , divisibility by p_2 requires $k \equiv -p_1(10^d)^{-1} \pmod{p_2}$. Since 10^d is coprime to p_2 , its modular inverse exists, giving k and hence n directly for each pair. Summing over all pairs yields 18613426663617118.

Problem 135: Same Differences

Solution: 4989

Count the $n < 10^6$ for which $x^2 - y^2 - z^2 = n$ has exactly ten solutions in positive integers x, y, z that are consecutive terms of an arithmetic progression. Since the AP gives $z = 2y - x$, the form simplifies to $y(4x - 5y)$. Setting $k = 4x - 5y$ makes $n = yk$ with $x = (5y + k)/4$ and $z = (3y - k)/4$, so a valid solution requires $k < 3y$ and $y + k \equiv 0 \pmod{4}$. Sweeping all such (y, k) and tallying the resulting n in a sieve, then counting indices whose tally is exactly ten, gives 4989.

Problem 136: Singletons

Solution: 2544559

Count the $n < 5 \times 10^7$ for which $x^2 - y^2 - z^2 = n$ (with x, y, z consecutive terms of an arithmetic progression) has exactly one solution. This is the Problem 135 reduction reused: solutions correspond to pairs $n = yk$ with $k < 3y$ and $y + k \equiv 0 \pmod{4}$. The same sweep tallies each n , saturating the count at two since only the cases none/one/many matter, and the number of n with a tally of exactly one is 2544559.

Problem 137: Fibonacci Golden Nuggets

Solution: 1120149658760

The Fibonacci power series $A_{F(x)} = \sum_{i \geq 1} F_i x^i$ equals $x/(1 - x - x^2)$. Demanding $A_{F(x)} = n$ for a positive integer n gives $nx^2 + (n+1)x - n = 0$, which has a rational root exactly when its discriminant $5n^2 + 2n + 1$ is a perfect square. The integers n that arise (the golden nuggets) are precisely the products $F_{2k} \cdot F_{2k+1}$: 2, 15, 104, 714, ... The 15th is $F_{30} \cdot F_{31} = 1120149658760$.

Problem 138: Special Isosceles Triangles

Solution: 1118049290473932

Consider isosceles triangles with base b and equal legs L whose height h differs from the base by exactly one ($h = b \pm 1$). From $L^2 = (b/2)^2 + h^2$ the condition rearranges to $(5b \pm 4)^2 - 20L^2 = -4$, a Pell-type equation. Its solutions make the leg lengths satisfy the recurrence $L_n = 18L_{n-1} - L_{n-2}$ with $L_1 = 17$ (the smallest triangle, base 16) and $L_2 = 305$. Summing the first twelve leg lengths gives 1118049290473932.

Problem 139: Pythagorean Tiles

Solution: 10057761

Count Pythagorean triples $a^2 + b^2 = c^2$ with perimeter below 10^8 for which four copies of the triangle tile a $c \times c$ square leaving a central hole of side $|b - a|$ that itself tiles the square, i.e. $(b - a) \mid c$. Generating primitives by Euclid's formula ($a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$), the divisibility test is scale-invariant, so each qualifying primitive of perimeter P contributes $\lfloor (10^8 - 1)/P \rfloor$ scaled copies. Summing over all primitives gives 10057761.

Problem 140: Modified Fibonacci Golden Nuggets

Solution: 5673835352990

For the modified Fibonacci sequence $1, 4, 5, 9, 14, \dots$, the series $A_{G(x)} = \sum G_k x^k$ equals $(x + 3x^2)/(1 - x - x^2)$. Setting $A_{G(x)} = n$ gives $(n + 3)x^2 + (n + 1)x - n = 0$, rational exactly when the discriminant $5n^2 + 14n + 1$ is a perfect square. The resulting golden nuggets $2, 5, 21, 42, 152, 296, 1050, \dots$ obey the recurrence $n_k = 7n_{k-2} - n_{k-4} + 7$; summing the first thirty gives 5673835352990.

Problem 141: Investigating Progressive Perfect Squares

Solution: 878454337159

A number n is progressive if dividing it by d leaves quotient q and remainder r with q, d, r forming a geometric progression. Sum the progressive perfect squares below 10^{12} . Since $r < d$, the quotient q is the geometric mean, so writing $r = ac^2$, $q = ace$, $d = ae^2$ with $e > c$ coprime and core a gives $n = qd + r = a^2ce^3 + ac^2$. Sweeping these parameters within the bound and keeping the perfect squares (deduplicated) sums to 878454337159.

Problem 142: Perfect Square Collections

Solution: 1006193

Find the smallest $x + y + z$ (with $x > y > z > 0$) for which $x + y$, $x - y$, $x + z$, $x - z$, $y + z$, $y - z$ are all perfect squares. The pairs $x \pm y$ and $x \pm z$ being square means both y and z are values v with $x \pm v$ square, i.e. $2x = c^2 + d^2$ and $v = (c^2 - d^2)/2$. Collecting these candidate values for each x , then searching each x for a pair $y > z$ whose sum and difference are also squares (scanning x in increasing order and stopping once x alone exceeds the best sum), yields 1006193.

Problem 143: Investigating the Torricelli Point of a Triangle

Solution: 30758397

The Torricelli point joins to the three vertices by segments p, q, r meeting at 120 degrees, so each triangle side is $\sqrt{p^2 + pq + q^2}$. Integer sides require every pair among p, q, r to satisfy $p^2 + pq + q^2 = \text{square}$ – a "120-pair", generated from Eisenstein triples $A = m^2 - n^2$, $B = 2mn + n^2$ and their multiples. Building the graph whose edges are 120-pairs and summing the distinct $p + q + r \leq 120000$ over all triangles (mutually paired triples) gives 30758397.

Problem 144: Investigating Multiple Reflections of a Laser Beam

Solution: 354

A laser bounces inside the elliptical mirror $4x^2 + y^2 = 100$, entering through a tiny gap at the top and reflecting until it escapes. The simulation tracks the contact point and travel direction: at each impact the inward normal is parallel to $(4x, y)$, so the reflected direction is $d - 2(d \cdot n / n \cdot n)n$. Substituting the reflected ray into the ellipse equation and discarding the current root gives the next contact via $t = -(8xr_x + 2yr_y)/(4r_x^2 + r_y^2)$. Counting contacts until the beam reaches the gap ($|x| \leq 0.01, y > 0$) yields 354.

Problem 145: Reversible Numbers

Solution: 608720

A number n is *reversible* if $n + \text{reverse}(n)$ has only odd decimal digits, with no leading zero in either n or $\text{reverse}(n)$. We count reversible $n < 10^9$.

Instead of testing a billion numbers, count by digit length d . Write the digits of n as $a_1 a_2 \dots a_d$. When n and $\text{reverse}(n)$ are added, column j (counting from the right) always pairs the same two digits, $a_j + a_{d+1-j}$, so the column sums are symmetric. Let $S_j = a_j + a_{d+1-j}$, and let carries propagate right to left.

Even length d

The units column gives $a_1 + a_d = S_1$; for that output digit to be odd, S_1 must be odd. The top column outputs $S_1 + c$ where c is its incoming carry; since S_1 is already odd, c must be even, i.e. 0. Pushing this inward forces **every** column sum to be odd and below 10, so there are no carries anywhere. Counting digit pairs (x, y) with $x + y$ odd and $x + y < 10$:

- the outer pair needs $x, y \geq 1$ (no leading zeros): 20 choices;
- each of the remaining $d/2 - 1$ pairs allows a zero: 30 choices each.

So an even length contributes $20 \cdot 30^{d/2-1}$.

Odd length d

Now there is a middle column that adds a digit to itself, $2a_m$, which is even. Its output can only be odd if a carry of 1 arrives from the right. That single required carry cannot coexist with the carry-free pattern above unless the lengths line up: the consistency of the inward and outward carries closes only when $d \equiv 3 \pmod{4}$. Those lengths contribute $100 \cdot 500^{(d-3)/4}$ (for $d = 3$ this is the $a_1 + a_d \in \{11, 13, 15, 17\}$ outer pair, 20 ways, times the 5 middle digits 0..4, giving 100); lengths with $d \equiv 1 \pmod{4}$ contribute nothing.

Total

Summing over $d = 1, \dots, 9$ (the digit lengths below 10^9):

$$\underbrace{20 + 600 + 18000 + 540000}_{\text{even}} + \underbrace{100 + 50000}_{d \equiv 3} = 608720.$$

A direct brute-force count confirms each per-length value (for instance 120 reversible numbers below 1000).

Problem 146: Investigating a Prime Pattern

Solution: 676333270

Find $n < 1.5 \times 10^8$ for which $n^2 + 1, 3, 7, 9, 13, 27$ are consecutive primes. Parity and a residue check mod 5 force n to be a multiple of 10, and small-prime divisibility of any $n^2 + \text{offset}$ prunes the vast majority of candidates in a fast modular sieve. Survivors are confirmed by full primality tests on the six values, plus checks that the intermediate offsets 5, 11, 15, 17, 19, 21, 23, 25 give composites (so the six are genuinely consecutive). Summing the valid n gives 676333270.

Problem 147: Rectangles in Cross-Hatched Grids

Solution: 846910284

Count all rectangles – axis-aligned and diagonal – fitting in every cross-hatched $m \times n$ grid for $1 \leq m \leq 47$, $1 \leq n \leq 43$, and sum. Axis-aligned rectangles number $\binom{m+1}{2}\binom{n+1}{2}$. Diagonal ones use lines $c = x - y$ (slope +1) and $d = x + y$ (slope -1); a tilted rectangle is an axis box in (c, d) space, valid when its corners stay in the grid. Writing $s = c_1 + d_1$, $t = d_1 - c_1$ (equal parity), the box of spans dc, dd fits when $s \in [0, 2m - dc - dd]$ and $t \in [dc, 2n - dd]$; counting same-parity (s, t) gives the diagonal total. The formula was checked against brute force (a 3 by 2 grid gives 37). Summing over all grids yields 846910284.

Problem 148: Exploring Pascal's Triangle

Solution: 2129970655314432

Count the entries in the first 10^9 rows of Pascal's triangle not divisible by 7. By Kummer's theorem the number of such entries in row n is the product of (base-7 digit +1) over the digits of n . Summing over a complete block of 7^j rows gives 28^j (since $1 + 2 + \dots + 7 = 28$), so a base-7 digit DP over 10^9 accumulates the total 2129970655314432 with no triangle ever built.

Problem 149: Searching for a Maximum-Sum Subsequence

Solution: 52852124

A 2000×2000 grid is filled by a lagged Fibonacci generator. Find the greatest sum of any contiguous run along a row, column, or either diagonal. After generating the grid, Kadane's algorithm runs along every line in each of the four directions, tracking the best running sum, which is 52852124.

Problem 150: Searching a Triangular Array for a Sub-Triangle with Minimum Sum

Solution: -271248680

A 1000-row triangle is filled by a linear congruential generator. Find the minimum sum over all sub-triangles (an apex plus all cells beneath it). With per-row prefix sums, each apex (i, j) is grown one row at a time, adding the segment $j \dots j + d$ of row $i + d$ in constant time, giving an $O(n^3)$ sweep whose minimum is -271248680.

Problem 151: Paper Sheets of Standard Sizes

Solution: 0.464399

Starting from one A1 sheet, a worker repeatedly draws a random sheet; a sheet larger than A5 is cut, adding one of each smaller size, while an A5 is used up. Find the expected number of draws where the envelope holds exactly one sheet, excluding the first and last. A memoised recursion over the count-state $f(s) = \mathbb{I}[s| = 1] + \sum p_k f(s')$ gives the expectation; subtracting the always-present first (the A1) and last (the lone A5) occurrences yields 0.464399.

Problem 152: Writing $1/2$ as a Sum of Inverse Squares

Solution: 301

Count the ways to write $1/2$ as a sum of distinct $1/k^2$ with $2 \leq k \leq 80$. For each prime $p \geq 11$ (so $p^2 > 80$), the chosen multiples of p must combine p -integrally: the numerator of $\sum 1/(k/p)^2$ over them must be divisible by p^2 . Multiples lying in no such group are dropped, cutting 79 candidates to 42. With surviving denominators using only the primes up to 13 the common denominator stays within machine range, so a meet-in-the-middle over the 42 values counts the 301 subsets hitting exactly half.

Problem 153: Investigating Gaussian Integer Divisors

Solution: 17971254122360635

Sum, over $n \leq 10^8$, of the real parts of every Gaussian-integer divisor $a + bi$ with $a > 0$. Writing each divisor as $g(a' + b'i)$ with (a', b') primitive, the geometric sum over g collapses via $\sum_g g \lfloor K/g \rfloor = \sum_{m \leq K} \sigma(m) =: D(K)$, giving $T = \sum_{\text{primitive } (a', b')} a' \cdot D(N/(a'^2 + b'^2))$. Each D is evaluated in $O(\sqrt{K})$ by hyperbola blocking while iterating primitive vectors, yielding 17971254122360635.

Problem 154: Exploring Pascal's Triangle (Trinomials)

Solution: 479742450

Count the trinomial coefficients $200000!/(i!j!k!)$ (with $i + j + k = 200000$) divisible by $10^{12} = 2^{12} \cdot 5^{12}$. By Legendre, v_p of the coefficient is $v_{p(n!)} - v_{p(i!)} - v_{p(j!)} - v_{p(k!)}$, so divisibility needs $v_2 \geq 12$ and $v_5 \geq 12$. Precomputing $v_{p(m!)}$ cumulatively and sweeping $i \leq j \leq k$ (weighting each triple by its 1, 3, or 6 permutations) counts 479742450.

Problem 155: Counting Capacitor Circuits

Solution: 3857447

Count the distinct capacitances obtainable from up to 18 identical capacitors. Let S_k be the set of values built from exactly k units; each value of S_k comes from joining a network of i units with one of $k - i$ units, in parallel ($a + b$) or in series ($ab/(a + b)$). Building $S_1, \dots, S_{18}$ as sets of reduced fractions and taking the union gives 3857447 distinct values.

Problem 156: Counting Digits

Solution: 21295121502550

Let $f(n, d)$ count the occurrences of digit d in writing $1, 2, \dots, n$. For each d we want every n with $f(n, d) = n$, then sum them. The function f is evaluated in $O(\log n)$ by a place-value formula, and the roots of $g(n) = f(n, d) - n$ are found by recursive bisection: since each step changes g by at most $L - 1$ upward and 1 downward, an interval with $g(\text{lo}) > \text{len}$ holds no root and is pruned. Summing all roots over $d = 1, \dots, 9$ gives 21295121502550.

Problem 157: Solving the Diophantine Equation $1/a + 1/b = p/10^n$

Solution: 53490

Count the solutions of $1/a + 1/b = p/10^n$ for $n = 1, \dots, 9$. Writing $a = gx$, $b = gy$ with $\gcd(x, y) = 1$ forces $xy \mid 10^n$, and then $p = (10^n/(xy))(x + y)/g$, so each coprime pair (x, y) contributes $d((10^n/(xy))(x + y))$ solutions (one per valid g). Coprime divisors of 10^n just assign the whole power of 2 and the whole power of 5 to one side each, so a short enumeration sums to 53490.

Problem 158: Exploring Strings with One Left-to-Right Ascent

Solution: 409511334375

Count length- n strings of distinct letters from a 26-letter alphabet having exactly one position where a letter exceeds its left neighbour. Choosing the n letters fixes their relative order, so the arrangements with exactly one ascent number the Eulerian number $\langle n, 1 \rangle = 2^n - n - 1$. Thus $p(n) = \binom{26}{n}(2^n - n - 1)$, and its maximum over n is 409511334375.

Problem 159: Digital Root Sums of Factorisations

Solution: 14489159

For each n , $\text{mdrs}(n)$ is the largest sum of digital roots over all factorisations into integers > 1 . Since every split $n = ab$ has $a, b < n$, processing n in increasing order makes $\text{mdrs}(n) = \max(\text{dr}(n), \max_{a|n} \text{mdrs}(a) + \text{mdrs}(n/a))$ a clean dynamic program, where $\text{dr}(n) = 1 + (n - 1) \bmod 9$. Summing over $2 \leq n < 10^6$ gives 14489159.

Problem 160: Factorial Trailing Digits

Solution: 16576

Find the last five non-zero digits of $(10^{12})!$. The trailing zeros number $z = v_5(n!)$, and we want $(n!/10^z) \bmod 10^5$ via CRT on 2^5 and 5^5 . Peeling multiples of 5 gives the recursion $P(n) = \left(\prod_{5 \nmid k \leq n} k \right) \cdot P(\lfloor n/5 \rfloor)$ for $n!/5^{v_5}$ modulo 5^5 , each full period of residues multiplying to -1 . Dividing out the matched twos (invertible mod 5^5) and combining with $0 \bmod 2^5$ (the surplus of twos exceeds 5) yields 16576.

Problem 161: Triominoes

Solution: 20574308184277971

Count the tilings of a 9×12 grid by triominoes (the straight piece in 2 orientations and the L piece in 4). A broken-profile dynamic program scans the cells in row-major order, keeping a bitmask of the occupancy of the window of cells from the current position forward. At the first empty cell a piece must be placed to cover it; otherwise the cell is already filled and the window slides on. Carrying the masks across all 108 cells counts 20574308184277971 tilings.

Problem 162: Hexadecimal Numbers

Solution: 3D58725572C62302

Count hexadecimal numbers of at most 16 digits (no leading zero) containing at least one 0, one 1 and one A. By inclusion-exclusion over which of the three required digits are forbidden, each term counts strings over a reduced alphabet: if 0 is forbidden the first digit is unconstrained among the remaining symbols, otherwise it must avoid 0. Summing over lengths 1 to 16 and reporting in hexadecimal gives 3D58725572C62302.

Problem 163: Cross-Hatched Triangles

Solution: 343047

Count all triangles in a size- n cross-hatched equilateral triangle. The figure has $9n - 3$ lines in six direction families. Triangle counting is affine-invariant, so shearing to the right triangle $\{x \geq 0, y \geq 0, x + y \leq n\}$ gives lines $x = k, y = k, x + y = k, x - y = c, 2x + y = c, x + 2y = c$ with rational intersections. A triangle is any triple of pairwise non-parallel, non-concurrent lines whose three intersections lie in the region; integer cross-products test this. The model reproduces $T(1) = 16, T(2) = 104$, and gives $T(36) = 343047$.

Problem 164: Numbers for Which No Three Consecutive Digits Have a Sum Greater Than 9

Solution: 378158756814587

Count 20-digit numbers (no leading zero) in which every three consecutive digits sum to at most 9. A digit DP carries the last two digits as state: from (a, b) the next digit c is allowed whenever $a + b + c \leq 9$. Seeding with all valid leading pairs and advancing 18 times sums to 378158756814587.

Problem 165: Intersections

Solution: 2868868

A Blum Blum Shub generator produces 5000 segments; count the distinct true intersection points, where "true" means the unique common point of two segments lying strictly interior to both. Each pair is tested with integer cross products (parallel pairs skipped, endpoints excluded by strict inequalities), and the crossing point is stored as a reduced exact fraction $(X/g, Y/g, \text{den}/g)$ so coincident crossings collapse. The set then holds 2868868 distinct points.

Problem 166: Criss-Cross

Solution: 7130034

Count 4×4 grids of digits where every row, column and both diagonals share one sum s . Fixing rows 0 and 1, the column sums force row 3 from row 2, while the two diagonal conditions and the row-2 sum reduce row 2 to a single free cell: $c_1 - c_0$ and $c_2 - c_3$ are determined, and $c_0 + c_3$ is fixed, so the valid choices of c_0 are counted directly within the digit bounds. Summed over all s and top rows: 7130034.

Problem 167: Investigating Ulam Sequences

Solution: 3916160068885

Sum the 10^{11} -th term of the Ulam sequences $U(2, 2n + 1)$ for $n = 2, \dots, 10$. Each such sequence has exactly two even terms, 2 and $e_2 = 2(v + 1)$; afterwards every term is odd, and an odd x belongs to the sequence exactly when one of $x - 2$, $x - e_2$ is already present, giving $O(1)$ generation. The sequence of gaps is eventually periodic, so a KMP failure function recovers the period P and per-period increase, and the far term is reached arithmetically. The nine terms sum to 3916160068885.

Problem 168: Number Rotations

Solution: 59206

Sum the last five digits of every n with $10 \leq n < 10^{100}$ such that moving the last digit to the front yields a multiple of n . Writing $n = 10a + d$, the rotation $d \cdot 10^{k-1} + a = mn$ gives $a = d(10^{k-1} - m)/(10m - 1)$. Looping over digit length k , multiplier m and last digit d , every integer a in the right range yields a solution; the running sum mod 10^5 is 59206.

Problem 169: Sums of Powers of Two

Solution: 178653872807

Count the ways to write $n = 10^{25}$ as a sum of powers of 2 with each power used at most twice. The count obeys $h(2m) = h(m) + h(m-1)$ and $h(2m+1) = h(m)$, a recursion whose memoised evaluation touches only $O(\log n)$ distinct arguments, giving 178653872807.

Problem 170: Pandigital Concatenated Products

Solution: 9857164023

Find the largest 0-to-9 pandigital concatenated product $P = \text{concat}(ma_1, \dots, ma_k)$ whose inputs $\text{concat}(m, a_1, \dots, a_k)$ are also 0-to-9 pandigital. Iterating pandigital P from the top downward, each is split into product-parts in every way; for a split, any common factor m of the parts yields candidate multipliers $a_i = \text{part}/m$, and the first P for which $\text{concat}(m, a_i)$ is pandigital is the answer, 9857164023.

Problem 171: Squared Digit Sums That Are Squares

Solution: 142989277

Sum every $0 < n < 10^{20}$ whose digits' squares sum to a perfect square, reporting the last nine digits. A digit DP over the 20 positions tracks the square-sum together with how many numbers reach it and their running total mod 10^9 ; summing the totals over square-valued states gives 142989277.

Problem 172: Investigating Numbers with Few Repeated Digits

Solution: 227485267000992000

Count 18-digit numbers (no leading zero) in which no digit appears more than three times. Building strings digit by digit and interleaving each digit's copies, the number of length-18 strings with every digit capped at three is computed, and the strings beginning with zero (digit 0 capped at two over the remaining 17 places) are subtracted, leaving 227485267000992000.

Problem 173: Square Laminae

Solution: 1572729

Count the square laminae (hollow square frames) that can be built with at most 10^6 tiles. A frame of outer width a and hole width b (same parity, $b < a$) uses $a^2 - b^2$ tiles. For each a , start from the widest hole $b = a - 2$ and shrink it; the tile count rises monotonically, so stop the moment it exceeds 10^6 . The outer loop ends once even the thinnest frame, costing $4a - 4$ tiles, no longer fits.

Problem 174: Counting the Number of Hollow Square Laminae

Solution: 209566

A square lamina of t tiles satisfies $t = a^2 - c^2$ with $a > c \geq 1$ and $a \equiv c \pmod{2}$, equivalently $t = 4ds$ with $1 \leq d < s$. The number of laminae using exactly t tiles is therefore the number of such factor pairs, computed by a pair-sieve. Counting how many $t \leq 10^6$ admit between 1 and 10 laminae gives 209566.

Problem 175: Fractions and Sum of Powers of Two

Solution: 1,13717420,8

With $f(n)$ the hyperbinary count, the ratio $f(n)/f(n-1)$ ranges over every positive rational, and the run lengths of n 's binary expansion (its shortened binary expansion) are exactly the continued-fraction quotients of that ratio. Taking the Euclidean quotients of $123456789/987654321$, adjusting for parity, dropping zeros and reversing gives the shortened binary expansion 1,13717420,8.

Problem 176: Rectangular Triangles That Share a Cathetus

Solution: 96818198400000

A right triangle with integer leg n comes from $n^2 = (c - b)(c + b)$, so the number of such triangles is $(\tau(N^2) - 1)/2$ where $N = n$ for odd n and $N = n/2$ for even n . Requiring exactly 47547 triangles forces $\tau(N^2) = 95095 = 5 \cdot 7 \cdot 11 \cdot 13 \cdot 19$. Writing this as a product of odd factors $2e_i + 1$ and assigning the largest exponents to the smallest primes minimises N ; the smallest n is the even case 96818198400000.

Problem 177: Integer Angled Quadrilaterals

Solution: 129325

The diagonals of convex $ABCD$ meet at P , forming four triangles. With $\varphi = \angle APB$, the eight corner angles satisfy $a_1 + a_2 = c_1 + c_2 = 180 - \varphi$ and $b_1 + b_2 = d_1 + d_2 = \varphi$, together with the law-of-sines closure

$(\sin a_2 / \sin a_1)(\sin b_2 / \sin b_1)(\sin c_2 / \sin c_1)(\sin d_2 / \sin d_1) = 1$. Iterating the integer angles a_1, a_2, b_1, c_1 and solving the closure for d_1 yields all labelled quadrilaterals; canonicalising each under the dihedral relabelling group (cyclic shifts by two and reversal) and counting distinct classes gives 129325.

Problem 178: Step Numbers

Solution: 126461847755

Count numbers below 10^{40} that are step numbers (consecutive digits differ by exactly one) and pandigital (use every digit). A digit DP carries the current last digit and the bit-set of digits seen; from each state the next digit is the last ± 1 , and states reaching the full mask of all ten digits are summed across lengths, yielding 126461847755.

Problem 179: Consecutive Positive Divisors

Solution: 986262

We count integers $1 < n < 10^7$ for which n and $n + 1$ have the same number of divisors.

Computing $d(n)$ by factorising each n is wasteful. Instead sieve all divisor counts at once: for every i from 1 to N , walk its multiples $i, 2i, 3i, \dots$ and add one to each, since i divides every one of them. After the sweep $d[k]$ holds the divisor count of k . The work is the harmonic sum

$$\sum_{i=1}^N \frac{N}{i} \approx N \ln N,$$

versus a per-number factorisation. A compact `int32` array keeps the strided increments cache-friendly.

Then a single linear scan counts the positions where $d[n] = d[n + 1]$ for $2 \leq n \leq N - 1$, giving 986262.

Problem 180: Rational Zeros of a Function of Three Variables

Solution: 285196020571078987

After simplification $f_n(x, y, z) = (x + y + z)(x^n + y^n - z^n)$, so a rational zero needs $x^n + y^n = z^n$; by Fermat only $n \in \{-2, -1, 1, 2\}$ qualify, giving $z \in \{x + y, \sqrt{x^2 + y^2}, xy/(x + y), xy/\sqrt{x^2 + y^2}\}$. Ranging x, y over rationals a/b with $0 < a < b \leq 35$ and keeping the golden triples whose z is also such a rational, the distinct values of $x + y + z$ sum to $t = u/v$ with $u + v = 285196020571078987$.

Problem 181: Grouping Objects of Two Colours

Solution: 83735848679360680

Count the ways to split 60 black and 40 white objects into unordered groups. Each possible group type $(b, w) \neq (0, 0)$ is an item usable any number of times, so the answer is an unbounded two-dimensional knapsack: iterating each group type and accumulating $dp[i][j] += dp[i - b][j - w]$ over the 61×41 grid gives 83735848679360680.

Problem 182: RSA Encryption

Solution: 399788195976

For $p = 1009$, $q = 3643$ the number of unconcealed messages under exponent e is $(1 + \gcd(e - 1, p - 1))(1 + \gcd(e - 1, q - 1))$. Scanning all valid e coprime to $\varphi = (p - 1)(q - 1)$, the minimum value is 9; summing every e that attains it gives 399788195976.

Problem 183: Maximum Product of Parts

Solution: 48861552

Splitting N into k equal parts gives product $(N/k)^k$, maximised near $k = N/e$, so only the two integer neighbours need checking. The maximum is a terminating decimal exactly when the reduced denominator $k/\gcd(N, k)$ has only the prime factors 2 and 5; summing $-N$ for terminating cases and $+N$ otherwise over $N = 5, \dots, 10000$ gives 48861552.

Problem 184: Triangles Containing the Origin

Solution: 1725323624056

Count triangles on lattice points inside a disk of radius 105 that strictly contain the origin. A triangle fails this exactly when its three direction-angles fit in a closed half-plane: either the maximum angular gap exceeds π (the points lie in an open half-plane) or equals π (an antipodal pair places the origin on an edge). Counting both via integer cross-products and direction classes and subtracting from $\binom{n}{3}$ leaves 1725323624056.

Problem 185: Number Mind

Solution: 4640261571849533

Twenty-two guesses each report how many of their sixteen digits sit in the correct position. Treating a candidate's total deviation from the required match counts as an energy, a steepest-descent local search with random restarts (changing the single digit that most reduces the deviation, restarting from a random sequence at each local minimum) drives the energy to zero, recovering the unique secret 4640261571849533.

Problem 186: Connectedness of a Network

Solution: 2325629

Calls are produced by the lagged-Fibonacci generator $S_k = (100003 - 200003k + 300007k^3) \bmod 10^6$ for $k \leq 55$ and $S_k = (S_{k-24} + S_{k-55}) \bmod 10^6$ thereafter, paired into caller and called. Misdials (caller equal to called) are skipped and the rest fed to a union-find; the number of successful calls when the prime minister's component first reaches 99% of the million users is 2325629.

Problem 187: Semiprimes

Solution: 17427258

We count the composite numbers $n < N$ (with $N = 10^8$) that are a product of exactly two primes, $n = pq$ with $p \leq q$.

Writing each such n with its smaller factor first makes the count a sum over p : for a fixed prime p , the partners are the primes q with $p \leq q$ and $pq < N$, i.e. $p \leq q \leq \lfloor (N - 1)/p \rfloor$. Counting unordered pairs this way (insisting $q \geq p$) visits each semiprime once. Only primes with $p^2 < N$, that is $p < \sqrt{N} \approx 10^4$, can be the smaller factor, so there are barely more than a thousand outer terms.

The largest partner occurs at $p = 2$, where q can reach $(N - 1)/2$, so we sieve every prime up to $N/2$ once. For each small prime p a binary search in that sorted list returns the number of primes up to $\lfloor (N - 1)/p \rfloor$; subtracting the primes below p leaves the count of valid q . Summing gives 17427258 (and 10 below 30, matching the example).

Problem 188: Hyperexponentiation

Solution: 95962097

Compute the tetration $1777 \uparrow\uparrow 1855$ modulo 10^8 by repeated modular exponentiation from the top down.

Problem 189: Tri-Colouring a Triangular Grid

Solution: 10834893628237824

Colour the 64 triangles of an eight-row grid with three colours so that edge-sharing triangles differ. A transfer-matrix DP carries the colours of a row's upward triangles; building row $k + 1$ as $U_0 D_0 U_1 \dots U_k$, each downward D_j must avoid its two neighbours and the triangle above, leaving $3 - |\{U_j, U_{j+1}, \text{above}_j\}|$ choices. Summing the DP gives 10834893628237824.

Problem 190: Maximising a Weighted Product

Solution: 371048281

To maximise $x_1 x_2^2 \dots x_m^m$ with $\sum x_i = m$, weighted AM-GM gives $x_i = 2i/(m+1)$.
Evaluating the resulting product exactly as a rational and summing its floor over $m = 2, \dots, 15$ yields 371048281.

Problem 191: Prize Strings

Solution: 1918080160

Count length-30 strings over $\{O, A, L\}$ with at most one A and no three consecutive L .
A DP over (position, absences used, trailing run of lates) tallies the valid strings to 1918080160.

Problem 192: Best Approximations

Solution: 57060635927998347

For each non-square $n \leq 100000$, the best rational approximation to $\sqrt{n}$ with denominator at most 10^{12} is either the largest continued-fraction convergent within the bound or the maximal semiconvergent built on it. Generating the convergents of $\sqrt{n}$ and comparing the two candidates in high precision, the sum of the winning denominators is 57060635927998347.

Problem 193: Squarefree Numbers

Solution: 684465067343069

The squarefree count below 2^{50} is $\sum_d \mu(d) \lfloor (2^{50} - 1)/d^2 \rfloor$. Building the Mobius function up to 2^{25} with a sieve that flips signs on prime multiples and zeros prime-square multiples, the weighted sum gives 684465067343069.

Problem 195: 60-Degree Triangle Inscribed Circles

Solution: 75085391

Integer-sided triangles with exactly one 60-degree angle (Eisenstein triples) split into two primitive families; since the inradius of a primitive triangle scales linearly with the triangle, each coprime parameter pair (p, q) contributes $\lfloor \text{bound} / (pq) \rfloor$ scalings whose inradius stays within the limit. Summing both families reproduces $T(100) = 1234$, $T(1000) = 22767$, $T(10000) = 359912$ and gives $T(1053779) = 75085391$.

Problem 196: Prime Triplets

Solution: 322303240771079935

In the triangle of integers each entry has up to eight neighbours. A prime belongs to a prime triplet iff it is a "centre" with at least two prime neighbours, or it neighbours such a centre. Segment-sieving the five rows around each target and applying a 3×3 convolution over the prime mask to find centres and their neighbours, $S(5678027) + S(7208785) = 322303240771079935$.

Problem 197: Investigating the Behaviour of a Recursively Defined Sequence

Solution: 1.710637717

With $f(x) = \lfloor 2^{30.403243784-x^2} \rfloor \cdot 10^{-9}$ and $u_{n+1} = f(u_n)$, the sequence settles into a 2-cycle, so $u_n + u_{n+1}$ stabilises. Iterating to convergence gives $u_n + u_{n+1} = 1.710637717$.

Problem 198: Ambiguous Numbers

Solution: 52374425

An ambiguous number is the midpoint of a Farey-neighbour pair $a/b < c/d$ (with $bc - ad = 1$), so its reduced denominator is $2bd$. Counting those in $(0, 1/100)$ with $2bd \leq 10^8$ is a walk over the Stern-Brocot subdivision of $(0/1, 1/100)$, where node (b, d) has children $(b, b + d)$ and $(b + d, d)$; each left spine is summed in closed form. Adding the 49 straddling pairs $(0/1, 1/d)$ for $51 \leq d \leq 99$ gives 52374425.

Problem 199: Iterative Circle Packing

Solution: 0.00396087

An outer circle (curvature -1) holds three equal mutually tangent circles. Every gap bounded by three mutually tangent circles is filled by a Soddy circle of curvature $a + b + c + 2\sqrt{ab + bc + ca}$ (Descartes' theorem), each spawning three further gaps.

Summing the areas (as $1/\text{curvature}^2$) of all circles produced through ten iterations, the uncovered fraction is 0.00396087.

Problem 200: Prime-Proof Squbes

Solution: 229161792008

A sqube is p^2q^3 for distinct primes p, q . Generating squbes up to 10^{12} , keeping those whose decimal contains "200" and which are prime-proof (no single-digit change, tested by Miller-Rabin, yields a prime), the 200th in increasing order is 229161792008.

Problem 201: Subsets with a Unique Sum

Solution: 115039000

From $\{1^2, \dots, 100^2\}$ take all 50-element subsets and sum those totals achievable by exactly one subset. A DP over (subset size, running sum) counts realisations, capped at 2 since only counts equal to 1 contribute; summing those sums gives 115039000.

Problem 202: Laserbeam

Solution: 1209002624

Unfolding the mirror triangle into a triangular lattice turns reflection into a straight line. A beam reaching the image (x, y) of C is reflected $2(x + y) - 3$ times, leaves early unless $\gcd(x, y) = 1$, and lands on a C -vertex iff $x \equiv y \pmod{3}$. With $s = x + y = (n + 3)/2$ this counts $x \in [1, s)$ coprime to s with $x \equiv -s \pmod{3}$; a Mobius sum over the squarefree divisors of $s = 6008819575$ gives 1209002624 (matching the stated 2 for 11 bounces and 80840 for 1000001).

Problem 203: Squarefree Binomial Coefficients

Solution: 34029210557338

Sum the distinct squarefree entries in the first 51 rows of Pascal's triangle. The key fact is that every prime factor of $\binom{n}{k}$ with $n \leq 50$ is at most n , so squarefreeness only needs testing against the primes up to 47 — the large binomials themselves never have to be factored. Collect the distinct values from the left half of each row (using $\binom{n}{k} = \binom{n}{n-k}$) and add those divisible by no p^2 .

Problem 204: Generalised Hamming Numbers

Solution: 2944730

A type-100 Hamming number has every prime factor at most 100. Recursively multiplying in the 25 primes ≤ 100 and counting products $\leq 10^9$ gives 2944730.

Problem 205: Dice Game

Solution: 0.5731441

Peter rolls nine 4-sided dice, Colin six 6-sided dice. Convolving each die distribution and summing $P(\text{Peter} = p) \cdot P(\text{Colin} < p)$ over all p yields $P(\text{Peter} > \text{Colin}) = 0.5731441$.

Problem 206: Concealed Square

Solution: 1389019170

The square reads 1_2_3_4_5_6_7_8_9_0. Ending in 0 makes the root a multiple of 10, and the hundreds digit 9 forces $n \equiv 30$ or $70 \pmod{100}$; scanning that residue class over the valid root range finds $n = 1389019170$.

Problem 207: Integer Partition Equations

Solution: 44043947822

Writing $x = 2^t$, every integer $x \geq 2$ yields a partition with $k = x(x - 1)$, perfect exactly when x is a power of two. At threshold $k(x)$ the proportion is (powers of two $\leq x$)/($x - 1$); the first x where this drops below $1/12345$ is $x = 209867$, giving $k = x(x - 1) = 44043947822$.

Problem 208: Robot Walks

Solution: 331951449665644800

The robot faces one of 5 directions; each 72° arc has a displacement that is an integer combination of four irrational basis terms. It returns to the start exactly when all four integer coordinates vanish. A DP over (facing direction, four coordinates) for 70 arcs counts 331951449665644800 closed journeys (and reproduces the stated 70932 for 25 arcs).

Problem 209: Circular Logic

Solution: 15964587728784

The map $\tau(a_1, \dots, a_6) = (a_2, \dots, a_6, a_1 \oplus (a_2 \wedge a_3))$ permutes the 64 inputs into cycles. The constraint $f(x) \wedge f(\tau x) = 0$ is an independent-set condition on each cycle, counted by the Lucas number L_n ; multiplying L_n over all cycle lengths gives 15964587728784.

Problem 210: Obtuse Angled Triangles

Solution: 1598174770174689458

A point B makes triangle OBC obtuse (with $C = (r/4, r/4)$) in three disjoint ways: angle at O gives the half-plane $x + y < 0$, angle at C gives $x + y > r/2$, and angle at B gives the open disk on diameter OC (centre $(r/8, r/8)$, $R^2 = r^2/32$). Intersecting each with the diamond $|x| + |y| \leq r$ and removing the degenerate points on $x = y$ gives $N(10^9) = 1598174770174689458$ (verified against $N(4) = 24$, $N(8) = 100$).

Problem 211: Divisor Square Sum

Solution: 1922364685

$\sigma_2(n)$, the sum of the squares of the divisors of n , is built for all $n < 64 \cdot 10^6$ by a divisor sieve. Summing those n for which $\sigma_2(n)$ is a perfect square (e.g. $\sigma_2(42) = 2500 = 50^2$) gives 1922364685.

Problem 212: Combined Volume of Cuboids

Solution: 328968937309

The 50000 lagged-Fibonacci cuboids have integer coordinates in $[0, 10398]$. Sweeping z over the distinct boundaries, each slab contributes (2D union area of the active boxes) $\times$ (slab height); the area comes from an x -sweep with a Klee segment tree over y , which self-clears each slab. The union volume is 328968937309 (matching the stated 723581599 for the first 100 cuboids).

Problem 213: Flea Circus

Solution: 330.721154

Each flea performs an independent random walk on the 30×30 grid, so a flea starting at s is at cell c after 50 turns with probability $(T^{50})_{s,c}$ for the walk matrix T . Raising T to the 50th power by squaring, the expected number of empty squares is $\sum_c \prod_s (1 - (T^{50})_{s,c}) = 330.721154$.

Problem 214: Totient Chains

Solution: 1677366278943

Sieving Euler's totient up to $4 \cdot 10^7$, the chain length satisfies $L(n) = L(\varphi(n)) + 1$ and is filled in ascending order (chains strictly decrease). Summing the primes p (those with $\varphi(p) = p - 1$) whose chain length is 25 gives 1677366278943.

Problem 215: Crack-free Walls

Solution: 806844323190414

Each row is a way to tile a width-32 strip with bricks of length 2 and 3, described by its set of internal crack positions. Two rows may be stacked iff their crack sets are disjoint; a DP over 10 rows counts the crack-free walls as 806844323190414 (the small case $W(9, 3) = 8$ checks out).

Problem 216: The Primality of $2n^2-1$

Solution: 5437849

$t(n) = 2n^2 - 1$ is composite exactly when some prime p divides it, i.e. $n^2 \equiv (p+1)/2 \pmod{p}$, which is solvable only when 2 is a quadratic residue mod p (so $p \equiv \pm 1 \pmod{8}$). Using Tonelli–Shanks to find the roots and sieving $n \equiv \pm r \pmod{p}$, the survivors for $n \leq 5 \cdot 10^7$ number 5437849 (and 2202 for $n \leq 10^4$).

Problem 217: Balanced Numbers

Solution: 6273134

A k -digit number is balanced when its first and last $\lfloor k/2 \rfloor$ digits have equal sums (an odd length leaves a free middle digit). A digit DP groups each half by digit sum s and tracks both the count and the numeric-value sum of half-strings; combining the leading half (first digit 1..9) with the trailing half over each shared s gives $T(L)$.

Summing $L = 1 \dots 47$ and reducing mod 3^{15} yields 6273134 (with $T(1) = 45$, $T(2) = 540$, $T(5) = 334795890$).

Problem 218: Perfect Right-angled Triangles

Solution: 0

A perfect triangle is a primitive triple with hypotenuse $c = d^2$, so $d^2 = m^2 + n^2$ is itself a primitive triple. Enumerating all of these for $d \leq 10^8$ (i.e. $c \leq 10^{16}$) via Euclid parameters gives 15915492 perfect triangles, and every one has area divisible by $\text{lcm}(6, 28) = 84$. Hence the count of perfect-but-not-super-perfect triangles is 0.

Problem 219: Skew-cost Coding

Solution: 64564225042

An optimal prefix-free code is built greedily: repeatedly split the cheapest current leaf of weight w into children of weight $w + 1$ and $w + 4$, which changes the total leaf-weight sum by $w + 5$. Batching equal-weight leaves makes $\text{Cost}(10^9) = 64564225042$ instant (and $\text{Cost}(6) = 35$).

Problem 220: Highway Dragon

Solution: 139776,963904

The dragon is the L-system $a \rightarrow aRbFR$, $b \rightarrow LFaLb$. Precomputing, for each symbol and depth, its F-step count and net (displacement, turn) lets whole blocks be skipped when they fit in the remaining budget. Walking 10^{12} steps into D_{50} from a northward start reaches (139776, 963904) (the 500th step of D_{10} is (18, 16)).

Problem 221: Alexandrian Integers

Solution: 1884161251122450

An Alexandrian integer satisfies $A = pqr$ with $1/A = 1/p + 1/q + 1/r$, which rearranges to $A = p(p+d)(p+(p^2+1)/d)$ where d ranges over divisors of p^2+1 (whose prime factors are only 2 and primes $\equiv 1 \pmod{4}$). The minimum A for a given p exceeds $4p^3$, so generating all $A \leq 3 \cdot 10^{15}$ for $p \leq 95000$ and sorting yields the 150000th value 1884161251122450 (the first six are 6, 42, 120, 156, 420, 630).

Problem 225: Tribonacci Non-divisors

Solution: 2009

The tribonacci sequence taken mod n is purely periodic, returning to 1, 1, 1; n divides some term iff a 0 appears first. Scanning odd n (the non-divisors begin 27, 81, 91, 103, ...), the 124th such number is 2009.

Problem 226: A Scoop of Blancmange

Solution: 0.11316017

The blancmange curve is $T(x) = \sum_{n \geq 0} s(2^n x)/2^n$. The area inside the circle C (centre $(1/4, 1/2)$, radius $1/4$) and under the curve is $\int_0^{1/2} \max(0, \min(T(x), y_{\text{high}(x)}) - y_{\text{low}(x)}) dx$, where $y_{\text{low}, \text{high}} = 1/2 \mp \sqrt{1/16 - (x - 1/4)^2}$. Midpoint integration on a dyadic grid converges quickly (the curve is self-similar) to 0.11316017.

Problem 227: The Chase

Solution: 3780.618622

Track the gap between the two dice on the cycle of 100 players. Each turn both dice move $-1/0/+1$ with probability $1/6, 4/6, 1/6$, so the gap changes by $\Delta \in [-2, 2]$ with the convolved distribution. Gap 0 is absorbing; solving the linear system for the expected absorption time from gap 50 gives 3780.618622.

Problem 228: Minkowski Sums

Solution: 86226

The Minkowski sum of convex polygons has one edge per distinct edge direction. Each S_n contributes directions $\{90^\circ + 360^\circ k/n\}$, so the total number of sides is the count of distinct $k/n \pmod{1}$ over all $n \in [1864, 1909]$. Distinct reduced fractions with denominator d number $\varphi(d)$, so the answer is $\sum_{d \in D} \varphi(d)$ over all divisors d of numbers in the range, giving 86226 (and $S_3 + S_4$ has 6 sides).

Problem 229: Four Representations Using Squares

Solution: 11325263

For each form $a^2 + kb^2$ with $k \in \{1, 2, 3, 7\}$, a sieve marks a distinct bit at every representable value up to $2 \cdot 10^9$. The integers carrying all four bits number 11325263 (and 75373 up to 10^7).

Problem 230: Fibonacci Words

Solution: 850481152593119296

Treating the two 100-digit seeds A (digits of π) and B as letters, digit position p lies in block $q = (p - 1) \div 100$ at offset $(p - 1) \bmod 100$. Whether block q is A or B follows from a recursion on the Fibonacci block-counts ($F_n = F_{n-2}F_{n-1}$). Summing $10^n D((127 + 19n) \cdot 7^n)$ for $n = 0 \dots 17$ gives 850481152593119296.

Problem 301: Nim

Solution: 2178309

Count the $n \leq 2^{30}$ with $n \oplus 2n \oplus 3n = 0$, the losing positions of this three-heap Nim.

Problem 303: Multiples with Small Digits

Solution: 1111981904675169

For a positive integer n , $f(n)$ is the least positive multiple of n whose decimal digits are all at most 2 (drawn from $\{0, 1, 2\}$); we want $\sum_{n=1}^{10000} f(n)/n$.

For a fixed n , find the smallest qualifying multiple by a breadth-first build of the number one digit at a time, working over the remainders modulo n . A state is a remainder r , and appending a digit $d \in \{0, 1, 2\}$ sends $r \mapsto (10r + d) \bmod n$. Starting from the leading digits 1 and 2 (no leading zero) and always trying digits in increasing order, the queue reaches candidates in order of increasing length and, within a length, increasing value; so the first time a remainder is seen it is via the smallest number reaching it, and the first time remainder 0 is reached gives $f(n)$. Marking remainders visited keeps each search to at most n states. Such a multiple always exists: a repunit R_k is divisible by any n coprime to 10, and trailing zeros absorb the factors 2 and 5. Summing over $n \leq 100$ gives 11363107 as a check.

Problem 304: Primonacci

Solution: 283988410192

With $a(1) = \text{next_prime}(10^{14})$ and $a(n) = \text{next_prime}(a(n-1))$, the $a(n)$ are the consecutive primes just above 10^{14} . Writing f for the Fibonacci sequence and $b(n) = f(a(n))$, we need $\sum_{n=1}^{100000} b(n)$ modulo $m = 1234567891011$.

Two ingredients suffice. The primes are collected by stepping through odd numbers above 10^{14} and applying a deterministic Miller–Rabin test; the gaps average $\ln(10^{14}) \approx 32$, so a hundred thousand primes need only a couple of million candidates. Each Fibonacci value is taken modulo m by fast doubling,

$$F(2k) = F(k)(2F(k+1) - F(k)), \quad F(2k+1) = F(k)^2 + F(k+1)^2,$$

processing the bits of the index from the most significant down. Since $m \approx 1.2 \cdot 10^{12}$, a product of two residues can reach $1.5 \cdot 10^{24}$ and overflow 64 bits, so the multiplications use an overflow-safe modular routine (correct for $m < 2^{62}$). Summing the residues gives the answer.

Problem 306: Paper-strip Game

Solution: 852938

Two players alternately paint two contiguous white squares of a length- n strip black; whoever cannot move loses. We count the $n \leq 10^6$ that are wins for the first player.

Painting two adjacent squares of a length- k strip removes them and splits the rest into two independent strips whose lengths sum to $k - 2$, so the Sprague–Grundy value obeys

$$G(k) = \text{mex} \{G(a) \oplus G(b) : a + b = k - 2, a, b \geq 0\},$$

with $G(0) = G(1) = 0$. A single strip of length n is a first-player win exactly when $G(n) \neq 0$. Computing the sequence is quadratic, but it is eventually periodic: the values settle into a cycle of period 34 after a short preperiod (here the first 53 terms). Generating a few thousand terms, detecting and verifying the period, then extending by $G(n) = G(\text{pre} + (n - \text{pre}) \bmod 34)$, lets the count reach 10^6 at once. For $1 \leq n \leq 5$ the wins are $n = 2, 3, 4$, as stated.

Problem 307: Chip Defects

Solution: 0.7311720251

With k independent, uniformly placed defects spread over n chips, $p(k, n)$ is the probability that some chip carries at least three defects; we need $p(20000, 1000000)$ to ten decimal places.

Work with the complement, where every chip holds 0, 1 or 2 defects. If exactly j chips hold a pair (using $2j$ of the labelled defects) and the other $k - 2j$ defects sit alone, the number of placements is $\binom{n}{j} \binom{n-j}{k-2j} k! / 2^j$; dividing by n^k makes it a probability t_j . The ratio of consecutive terms telescopes to

$$\frac{t_j}{t_{j-1}} = \frac{(k - 2j + 2)(k - 2j + 1)}{2j(n + j - k)},$$

starting from $t_0 = \prod_{i=0}^{k-1} (n - i) / n^k$ (all defects in distinct chips). Then $p = 1 - \sum_{j=0}^{\lfloor k/2 \rfloor} t_j$. The terms span dozens of orders of magnitude (with $t_0 \approx e^{-200}$), so the running sum is carried in extended precision; $p(3, 7) \approx 0.0204081633$ confirms the recurrence.

Problem 309: Integer Ladders

Solution: 210139

Two ladders of integer lengths lean across a street of integer width w , crossing at integer height h above the road. For lengths $0 < x < y < 10^6$, how many triples (x, y, h) admit an integer w ?

Let the ladders meet the opposite walls at heights $X = \sqrt{x^2 - w^2}$ and $Y = \sqrt{y^2 - w^2}$. Similar triangles give the optic equation $1/h = 1/X + 1/Y$, that is $h = (XY)/(X + Y)$. Were X or Y irrational, h could not be rational, so both wall heights are integers and (w, X, x) , (w, Y, y) are Pythagorean triples. Writing the two heights as $X_1 < X_2$, the crossing height is an integer exactly when

$$(X_1 + X_2) \mid X_1^2, \quad \text{because} \quad h = X_1 - \frac{X_1^2}{X_1 + X_2}.$$

So enumerate every Pythagorean triple with hypotenuse below 10^6 . Each triple $\{p, q, c\}$ gives two configurations: width p with wall height q , and width q with wall height p . For a configuration of width w and smaller height X_1 , the admissible partners are $X_2 = D - X_1$ for each divisor D of X_1^2 with $D > 2X_1$ (forcing $X_2 > X_1$); such an X_2 counts when $w^2 + X_2^2$ is itself a perfect square below the limit. Each valid pair is reached once. There are 5 triples below 200 and 146 below 1600, matching the known checks.

Problem 310: Nim Square

Solution: 2586528661783

Nim Square is three-heap normal-play Nim in which a move removes a positive square number of stones from one heap. For positions (a, b, c) with $0 \leq a \leq b \leq c \leq 10^5$, how many are losing for the player to move?

A single heap of size s has Grundy value

$$g(s) = \text{mex } \{g(s - i^2) : i \geq 1, i^2 \leq s\},$$

found for all $s \leq 10^5$ in one sweep — each heap needs only the ≈ 316 square subtractions, and the values stay small (at most 74). As in ordinary Nim, a three-heap position is losing precisely when $g(a) \oplus g(b) \oplus g(c) = 0$.

A direct $O(N^3)$ scan is hopeless, so tally how many heap sizes carry each Grundy value and let T be the number of **ordered** triples with vanishing XOR, read off from the pairwise XOR-distribution of those tallies (the value range is tiny). Convert ordered to sorted counts: an all-equal triple needs $g = 0$; a multiset $\{p, p, q\}$ has XOR $g(q)$, so it vanishes iff $g(q) = 0$ (with p free); and the all-distinct multisets are what remains of T after removing those degenerate orderings and dividing by 6. Summing the three classes gives 1160 for the bound 29 and the stated answer at 10^5 .

Problem 315: Digital Root Clocks

Solution: 13625242

Two seven-segment clocks each display a number, then repeatedly its digit sum down to a single digit, beginning and ending blank. Sam's clock clears and relights every segment at each step; Max's only toggles the segments that change. The answer is Sam's total segment transitions minus Max's, summed over the primes in $[10^7, 2 \cdot 10^7)$.

Sieve that prime range rather than testing each number for primality, and compute both transition counts per prime with integer digit arithmetic. For a step $a \rightarrow b$, Sam's cost is the total lit-segment count of a plus that of b , while Max's cost is the sum over aligned digit positions of the population count of the segment-pattern XOR (a blank screen contributing the all-off pattern).

Problem 317: Firecracker

Solution: 1856532.8455

A firecracker bursts at height $h_0 = 100$; its fragments scatter in every direction at speed $v = 20$ and then fall under gravity $g = 9.81$. We want the volume of the region their paths sweep before landing.

Each fragment follows a parabolic arc, and over all launch directions the family of trajectories has as its envelope the *parabola of safety*,

$$Y(r) = h_0 + \frac{v^2}{2g} - \frac{gr^2}{2v^2},$$

the greatest height reachable at horizontal distance r . The swept solid is everything between the ground and this envelope, revolved about the vertical axis. Writing $A = h_0 + \frac{v^2}{2g}$ for the apex height and $B = \frac{g}{2v^2}$, the envelope meets the ground at $r_{\max}^2 = A/B$, and the shell integral collapses:

$$V = \int_0^{r_{\max}} 2\pi r (A - Br^2) \, dr = \frac{\pi A^2}{2B} = \frac{\pi A^2 v^2}{g} \approx 1856532.8455.$$

Problem 321: Swapping Counters

Solution: 2470433131948040

With n red and n blue counters at opposite ends of a row of $2n + 1$ squares (one empty square between them), $M(n)$ is the fewest slides and hops needed to swap the two colours. We want the sum of the first forty values of n for which $M(n)$ is a triangular number.

First, $M(n) = n^2 + 2n$: each of the $2n$ counters must advance $n + 1$ squares, giving $2n(n + 1)$ square-steps; a hop advances two squares and a slide one, and the n^2 red-blue meetings are exactly the hops, leaving $2n$ slides, so $M(n) = 2n(n + 1) - n^2 = n^2 + 2n$ (indeed $M(3) = 15$). Requiring $n^2 + 2n = k(k + 1)/2$ and completing the square turns this into the Pell-like equation

$$m^2 - 8u^2 = -7, \quad u = n + 1, \quad m = 2k + 1.$$

Its positive solutions split into two classes, generated from the fundamentals $(m, u) = (1, 1)$ and $(5, 2)$ by the unit $3 + \sqrt{8}$, that is $(m, u) \mapsto (3m + 8u, m + 3u)$. Merging the resulting u -values in increasing order (dropping $u = 1$) gives $n = u - 1 = 1, 3, 10, 22, 63, \dots$; the first five sum to 99 and the first forty to the answer.

Problem 323: Bitwise-OR Operations on Random Integers

Solution: 6.3551758451

Each step ORs in a random 32-bit integer; the expected number of steps is $\sum i \cdot p(i)$, where $p(i)$ comes from differencing the closed-form chance that all bits are set by step i .

Problem 329: Prime Frog

Solution: 199740353/29386561536000

A frog hops left or right with equal probability among squares 1 to 500 (bouncing back at the ends), croaking just before each hop: on a prime square it croaks P with probability $2/3$ and N with $1/3$, on a non-prime square the reverse. From a uniformly random start, we want the probability that its first fifteen croaks spell PPPPNNPPPNPPNPN, as a reduced fraction.

Keep the arithmetic exact with an integer weight per square. A croak contributes numerator 2 when the square's prime-ness matches the target letter and 1 otherwise, each over a factor of 3; a hop contributes $1/2$ to each neighbour, and a forced hop off an end has probability $1 = 2 \cdot 1/2$, so it carries numerator 2. Every length-15 path then shares the denominator $500 \cdot 3^{15} \cdot 2^{14}$, and a single forward sweep over the croaks accumulates the path-weight numerators across all squares. Reducing numerator against denominator gives the fraction.

Problem 333: Special Partitions

Solution: 3053105

A partition of n into parts $2^i 3^j$ is *valid* when no part divides another; $P(n)$ counts the valid partitions. We want the sum of the primes $q < 10^6$ with $P(q) = 1$.

Since $2^i 3^j \mid 2^k 3^l$ iff $i \leq k$ and $j \leq l$, a valid partition is an *antichain* in the divisibility poset: its lattice points (i, j) have strictly increasing i as j strictly decreases. Sweeping the powers of three from high to low, each column contributes at most one part, whose exponent i must exceed every i chosen so far. The DP state is (largest i used, running sum); a prefix sum along the "largest i " axis injects all earlier states with a smaller i in one pass, and counts are clamped at 2 since only $P(q) = 1$ matters. As a check,

$\sum_{q < 100, P(q)=1} q = 233$ and $P(11) = 2$, $P(17) = 1$.

Problem 340: Crazy Function

Solution: 291504964

With $F(n) = n - c$ for $n > b$ and $F(n) = F(a + F(a + F(a + F(a + n))))$ otherwise, we need the last nine digits of $S(a, b, c) = \sum_{n=0}^b F(n)$ for $a = 21^7$, $b = 7^{21}$, $c = 12^7$ (note $a > c$).

For $n \in (b - a, b]$ we have $a + n > b$, so the innermost call gives $a + n - c$; since $a > c$ each successive argument also exceeds b , and the four levels unwind to $F(n) = n + 4a - 4c$. Inductively, for $n \in (b - (k + 1)a, b - ka]$,

$$F(n) = n + 4(k + 1)a - (3k + 4)c.$$

Writing $m = \lfloor b/a \rfloor$, the sum splits into the partial bottom block $[0, b - ma]$ (using $k = m$) plus the full blocks $k = 0, \dots, m - 1$, each contributing $P_k = ab - a(a - 1)/2 + (3k + 4)a(a - c)$. Everything is evaluated with exact integers and reduced mod 10^9 only at the end, sidestepping modular-division issues. The example $S(50, 2000, 40) = 5204240$ (with $F(0) = 3240$, $F(2000) = 2040$) matches a direct recursive evaluation over the whole range.

Problem 343: Fractional Sequences

Solution: 269533451410884183

Starting from $1/k$, the map $x/y \rightarrow (x+1)/(y-1)$ (reduced) is iterated until the denominator is 1, giving the integer $f(k)$. We want $\sum_{k=1}^{2 \cdot 10^6} f(k^3)$.

The unreduced step keeps $x+y$ fixed; reducing divides $x+y$ by a common factor. Incrementing x from 1, the first reduction happens at x equal to the smallest prime factor p of the current sum S , which divides S by p and resets x to 1. So $S = k+1$ is repeatedly divided by its smallest prime until it becomes prime, and the run ends there with $f(k) = S - 1$; the surviving prime is the **largest** prime factor of $k+1$, hence $f(k) = \text{lpf}(k+1) - 1$.

Because $k^3 + 1 = (k+1)(k^2 - k + 1)$, $f(k^3) = \max(\text{lpf}(k+1), \text{lpf}(k^2 - k + 1)) - 1$. The first factor is read from a largest-prime-factor sieve; the second (up to $4 \cdot 10^{12}$) is reduced by sieving only the primes that can divide it. A prime p divides $k^2 - k + 1$ exactly when $(2k-1)^2 \equiv -3 \pmod{p}$, which has roots only for $p = 3$ and $p \equiv 1 \pmod{3}$; those roots (via a modular square root of -3) give arithmetic progressions of k to strip. After removing all primes up to $2 \cdot 10^6$ any residue above 1 is itself the largest prime factor. The check $\sum_{k=1}^{100} f(k^3) = 118937$ holds.

Problem 344: Silver Dollar Game

Solution: 65579304332

$W(n, c)$ counts the winning configurations of the silver dollar game on n squares with c worthless coins and one silver dollar; we want $W(10^6, 100) \bmod 1000036000099$, where the modulus is the semiprime $1000003 \cdot 1000033$.

With $m = c + 1$ coins (here odd), a placement is described by the $m + 1$ non-negative gaps summing to $N = n - m$, and the silver dollar may be any of the m coins, so there are $m \binom{n}{m}$ configurations. The game reduces to Nim on the alternating "active" gaps: a position is losing exactly when those gaps xor to zero. Splitting by where the dollar sits gives $L_0 = \text{Count}(N)$ (dollar on the second coin) and $L_1 = \text{Count}(N + 1) - \text{Count}'(N + 1)$ for the dollar further right, where Count is the number of gap vectors of a given sum whose active entries xor to zero, and the primed term drops one active gap to remove a vanished leading gap. The answer is $m \binom{n}{m} - (L_0 + (m - 2)L_1)$.

Count is evaluated with a binary carry DP: per column an even number of active gaps hold a 1, and scanning the bits of the target sum the only state is the carry. Everything runs directly modulo the semiprime (inverses exist since both primes exceed m). The exact values $W(10, 2) = 324$ and $W(100, 10) = 1514704946113500$ both check out.

Problem 345: Matrix Sum

Solution: 13938

Choosing one entry from each row of a 15×15 matrix so that no two chosen entries share a column, maximise the sum.

This is the assignment problem, solved by a bitmask dynamic program over the set of columns already used. Let $\text{dp}(\text{mask})$ be the largest total achievable when the chosen columns form mask ; since exactly one entry is taken per row, the number of set bits in mask is the next row to fill. Extending each state by every still-unused column and keeping the maximum fills all 2^{15} states in $O(2^{15} \cdot 15)$ steps, and the all-columns mask holds the Matrix Sum. The 5×5 example evaluates to 3315 as a check.

Problem 346: Strong Repunits

Solution: 336108797689259276

Collect every number that is a repunit of length at least three in some base up to $\sqrt{n}$, together with all values that are length-two repunits, and sum the distinct ones below 10^{12} .

Problem 347: Largest Integer Divisible by Two Primes

Solution: 11109800204052

For distinct primes p, q let $M(p, q, N)$ be the largest integer $\leq N$ whose only prime factors are p and q (so $M = p^a q^b$ with $a, b \geq 1$), or 0 if none exists. $S(N)$ sums all the distinct values; we need $S(10^7)$.

Each value $p^a q^b$ has a unique prime pair, so summing distinct values is the same as summing $M(p, q, N)$ over the pairs themselves. Iterating n and factorising it is wasteful; instead iterate the pairs. A valid pair needs $pq \leq N$, and since $p < q$ this forces $p < \sqrt{N}$, so only a few hundred primes can be the smaller factor. Sieve the primes up to $N/2$ (the largest a factor can be, paired with 2), then loop p over primes below $\sqrt{N}$ and q over larger primes while $pq \leq N$.

For a fixed pair, M is found by trying each power p^a with $p^a q \leq N$ and, for each, taking the largest power $q^b \leq N/p^a$; the maximum product over all a is $M(p, q, N)$. Summing gives $S(100) = 2262$ as a check and $S(10^7) = 11109800204052$.

Problem 348: Sum of a Square and a Cube

Solution: 1004195061

We want the five smallest palindromes expressible as $a^2 + b^3$, with $a, b > 1$, in **exactly** four ways, and their sum.

The naive route, testing every integer for palindromicity, wastes almost all of its time: only about 10^5 palindromes lie below 10^9 . So we generate palindromes directly and in increasing order. A palindrome is fixed by its first half: mirror a root r of h digits to get an even-length palindrome $r \cdot 10^h + \text{reverse}(r)$, or mirror r minus its last digit for odd length. Iterating length upward, and root upward within each length, yields every palindrome in ascending order, so the first five that qualify are the smallest five.

For each palindrome p we count the representations by trying every cube b^3 with $b \geq 2$ and $b^3 + 4 \leq p$ (the $+4$ leaves room for $a^2 \geq 2^2$), and testing whether the remainder $p - b^3$ is a perfect square. Because the remainder is at least 4, its root is automatically ≥ 2 , so both parts exceed 1 as required. Each palindrome costs $O(p^{1/3})$ checks, and we keep only those with a count of exactly four.

The qualifying palindromes are 5229225 (the example), 37088073, 56200265, 108909801 and 796767697, summing to 1004195061.

Problem 351: Hexagonal Orchards

Solution: 11762187201804552

In a hexagonal orchard of order n (a triangular lattice of points inside a regular hexagon of side n), $H(n)$ counts the points hidden from the centre by a nearer point. We want $H(10^8)$.

A point is hidden exactly when it is not the closest lattice point along its ray, i.e. when its coordinates are not coprime. By the six-fold symmetry the lattice splits into six sectors, and at radial index i the visible points in a sector are the $\varphi(i)$ with coprime coordinates, leaving $i - \varphi(i)$ hidden. Hence

$$H(n) = 6 \sum_{i=1}^n (i - \varphi(i)) = 6 \left(\frac{n(n+1)}{2} - \Phi(n) \right), \quad \Phi(n) = \sum_{i=1}^n \varphi(i).$$

The totient summatory function Φ is evaluated sublinearly from the identity $\sum_{d=1}^n \Phi(\lfloor n/d \rfloor) = n(n+1)/2$, giving

$$\Phi(n) = \frac{n(n+1)}{2} - \sum_{d=2}^n \Phi(\lfloor n/d \rfloor),$$

grouped over equal values of $\lfloor n/d \rfloor$ and seeded by a sieve of φ (hence Φ) for all arguments up to about $n^{2/3}$. The checks $H(5) = 30$, $H(10) = 138$, $H(1000) = 1177848$ all hold.

Problem 357: Prime Generating Integers

Solution: 1739023853137

Sum the integers $n \leq 10^8$ for which $d + n/d$ is prime for every divisor d of n . Taking $d = 1$ forces $n + 1$ to be prime, so n is even and the only candidates are $n = p - 1$ for primes p . Sieve the primes up to 10^8 for constant-time primality tests, then for each candidate check the divisor pairs with $d \leq \sqrt{n}$ (the condition is symmetric in d and n/d), bailing out at the first failure — usually already at $d = 2$.

Problem 358: Cyclic Numbers

Solution: 3284144505

A cyclic number is $c = (10^{p-1} - 1)/p$ for a full-reptend prime p (one for which 10 is a primitive root, so $1/p$ has period $p - 1$); c is precisely the repeating block of $1/p$ and so has $p - 1$ digits, with leading zeros kept. We seek the digit sum of the unique such number beginning 00000000137 and ending 56789.

Its digit sum is $9(p - 1)/2$: the digits pair off into nines (for 142857 the sum is $27 = 9 \cdot 6/2$). The two end conditions pin p down. The leading 00000000137 means $\lfloor 10^{11}/p \rfloor = 137$, hence $10^{11}/138 < p \leq 10^{11}/137$. The trailing 56789 means $c \equiv 56789 \pmod{10^5}$; since $cp = 10^{p-1} - 1 \equiv -1 \pmod{10^5}$, this gives $56789p \equiv 99999 \pmod{10^5}$, fixing p modulo 10^5 . Scanning that short arithmetic progression for the prime whose order of 10 is $p - 1$ yields the unique p , and $9(p - 1)/2$ is the answer.

Problem 359: Hilbert's New Hotel

Solution: 40632119

Person n takes the lowest floor that is empty or whose last occupant m has $m + n$ a perfect square; $P(f, r)$ is the person in room r of floor f . We sum $P(f, r)$ over all $fr = 71328803586048 = 2^{27} \cdot 3^{12}$ and report the last eight digits.

Simulation exposes the structure: along any floor the occupants satisfy $a_i + a_{i+1} = (i + b)^2$, where floor 1 has $b = 1$ and floors $2k, 2k + 1$ share $b = 2k$. The first occupants are $P(1, 1) = 1$, $P(2k, 1) = 2k^2$, $P(2k + 1, 1) = 2k(k + 1)$. Solving the alternating recurrence $a_{i+1} = (i + b)^2 - a_i$ with the identity $\sum_{k=1}^n (-1)^{n-k} k^2 = n(n + 1)/2$ gives the closed form

$$P(f, r) = (-1)^{r-1} \left(a_1 - \frac{b(b+1)}{2} \right) + \frac{(b+r-1)(b+r)}{2},$$

which reproduces $P(10, 20) = 440$, $P(25, 75) = 4863$ and $P(99, 100) = 19454$. Summing over the 364 divisor pairs of $2^{27} \cdot 3^{12}$ and reducing modulo 10^8 gives the result.

Problem 365: A Huge Binomial Coefficient

Solution: 162619462356610313

Let $M(n, k, m) = \binom{n}{k} \bmod m$. We want $\sum M(10^{18}, 10^9, pqr)$ over prime triples $1000 < p < q < r < 5000$.

There are 501 primes in that range. For each prime p , $\binom{10^{18}}{10^9} \bmod p$ is computed by Lucas' theorem: writing both numbers in base p (about five or six digits), the result is the product of $\binom{n_i}{k_i} \bmod p$ over digits, each evaluated from factorial and inverse-factorial tables mod p . With those residues in hand, every triple's value mod pqr follows from the Chinese Remainder Theorem on the squarefree modulus. Precomputing a table of $p_j^{-1} \bmod p_i$ turns the inner CRT into a few multiplications, so the roughly $\binom{501}{3} \approx 2.1 \cdot 10^7$ triples are summed quickly.

Problem 381: (prime-k) Factorial

Solution: 139602943319822

For a prime p , $S(p) = \left(\sum_{k=1}^5 (p-k)! \right) \bmod p$; we want $\sum S(p)$ for $5 \leq p < 10^8$.

By Wilson's theorem $(p-1)! \equiv -1 \pmod{p}$, and dividing successively by $p-1, p-2, \dots$ (each $\equiv -1, -2, \dots$) gives $(p-k)! \equiv (-1)^k / (k-1)! \pmod{p}$. Summing the five terms,

$$S(p) \equiv -1 + 1 - \frac{1}{2} + \frac{1}{6} - \frac{1}{24} = -\frac{3}{8} \pmod{p}.$$

Solving $8x \equiv -3 \pmod{p}$ yields a closed form by residue: $S(p) = (3p-3)/8, (p-3)/8, (7p-3)/8, (5p-3)/8$ for $p \equiv 1, 3, 5, 7 \pmod{8}$ respectively. A sieve to 10^8 then sums these (checked against $\sum S(p) = 480$ for $p < 100$).

Problem 401: Sum of Squares of Divisors

Solution: 281632621

Let $\sigma_2(n)$ be the sum of the squares of the divisors of n and $\text{SIGMA2}(n) = \sum_{i=1}^n \sigma_2(i)$; we need $\text{SIGMA2}(10^{15}) \bmod 10^9$.

Rather than factorise every i , count contributions by divisor. A fixed k divides exactly $\lfloor n/k \rfloor$ of the integers $1, \dots, n$, so it contributes k^2 to each of them:

$$\text{SIGMA2}(n) = \sum_{k=1}^n \lfloor n/k \rfloor \cdot k^2.$$

The quotient $\lfloor n/k \rfloor$ takes only $O(\sqrt{n})$ distinct values, and the k giving a fixed quotient q form a contiguous block $[k_0, \lfloor n/q \rfloor]$. Over such a block the contribution is $q \sum_{k=k_0}^{k_1} k^2$, and the prefix sums of squares are $\sum_{k=1}^m k^2 = m(m+1)(2m+1)/6$. Walking the blocks brings the cost from 10^{15} down to about $2\sqrt{10^{15}} \approx 6.3 \cdot 10^7$ steps.

The modulus 10^9 is not prime, so the $/6$ cannot be done with a modular inverse. Instead it is carried out exactly before reducing: among m and $m+1$ one is even, and among m , $m+1$ and $2m+1$ one is a multiple of 3, so dividing the appropriate factors removes the 6 while everything is still an integer. The three reduced factors are then each taken $\bmod 10^9$ and multiplied with reductions in between, every intermediate product staying below 10^{18} . As a check, $\text{SIGMA2}(6) = 113$.

Problem 407: Idempotents

Solution: 39782849136421

For each n let $M(n)$ be the largest $a < n$ with $a^2 \equiv a \pmod{n}$ — an *idempotent* — and we want $\sum_{n=1}^{10^7} M(n)$.

The congruence $a^2 \equiv a$ means $n \mid a(a-1)$. Since a and $a-1$ are coprime, each prime power p^e dividing n must divide a or $a-1$ entirely, i.e. $a \equiv 0$ or $a \equiv 1 \pmod{p^e}$. By the Chinese remainder theorem an idempotent is exactly a choice of 0 or 1 on each of the $\omega(n)$ prime-power factors, giving $2^{\omega(n)}$ of them.

So the work per n is: factor n into prime powers $q_1, \dots, q_m$ with a smallest-prime-factor sieve; build the CRT basis e_i (the value that is 1 $\pmod{q_i}$ and 0 on the others, computed as $(n/q_i) \cdot (n/q_i)^{-1} \pmod{q_i}$); and take the maximum of the 2^m subset sums $\sum_{i \in S} e_i \pmod{n}$. The all-empty choice gives 0 and the all-ones choice gives 1, so prime powers correctly yield $M = 1$; the largest subset sum below n is $M(n)$. As checks, $M(6) = 4$, $\sum_{n \leq 20} M = 75$ and $\sum_{n \leq 100} M = 2549$. The $2^{\omega(n)}$ subsets sum to roughly 10^8 across the whole range, so the sieve-and-enumerate runs in a few seconds.

Problem 409: Nim Extreme

Solution: 253223948

A position is n distinct pile sizes drawn from $\{1, \dots, 2^n - 1\}$ — equivalently the nonzero vectors of $\mathbb{F}_2^n$ — and it is winning unless its nim-sum (XOR) is zero. The worked value $W(2) = 6$ counts the three two-element subsets of $\{1, 2, 3\}$ in both orders each, so positions are **ordered** tuples. Writing $M = 2^n - 1$, the total ordered count is the falling factorial $P(M, n) = M(M-1)\cdots(M-n+1)$, and

$$W(n) = P(M, n) - n! \cdot Z(n),$$

where $Z(n)$ is the number of n -element **subsets** of the nonzero vectors with XOR zero.

To count $Z(n)$, average the zero-sum indicator over the characters $\chi_w(v) = (-1)^{w \cdot v}$ of $\mathbb{F}_2^n$. For a fixed w the generating polynomial $\prod_{v \neq 0} (1 + x \chi_w(v))$ splits by how many nonzero v lie on each side of the hyperplane $w \cdot v = 0$: the trivial character $w = 0$ gives $(1+x)^M$, and each of the M nonzero characters gives $(1+x)^{2^{n-1}-1} (1-x)^{2^{n-1}}$. Hence

$$Z(n) = \frac{1}{2^n} \left[\binom{M}{n} + M \cdot c_n \right], \quad c_n = [x^n] (1+x)^{2^{n-1}-1} (1-x)^{2^{n-1}}.$$

Combining and using $n! \binom{M}{n} = P(M, n)$,

$$W(n) = P(M, n) - 2^{-n} [P(M, n) + M \cdot n! \cdot c_n] \pmod{10^9 + 7}.$$

The exponents 2^{n-1} are astronomically large, but only the single coefficient $c_n = \sum_{j=0}^n \binom{a}{j} \binom{b}{n-j} (-1)^{n-j}$ is needed (with $a \equiv 2^{n-1} - 1$, $b \equiv 2^{n-1}$ reduced mod p , the binomials evaluated as falling factorials). Precomputing inverses $1..n$, building $\binom{b}{\cdot}$ forward and $\binom{a}{\cdot}$ incrementally turns the whole thing into a handful of $O(n)$ passes. It reproduces $W(1..3) = 1, 6, 168$, $W(5) = 19764360$ and $W(100) \equiv 384777056$.

Problem 425: Prime Connection

Solution: 46479497324

Two primes are *connected* if they have the same number of digits and differ in exactly one position, or if one is obtained from the other by prepending a single digit. A prime P is a *2's relative* if there is a chain of connected primes from 2 to P in which no prime exceeds P , and $F(N)$ sums the primes $p \leq N$ that are *not* relatives.

The "no prime exceeds P " rule is the whole trick: a prime's relative-status depends only on smaller primes. So sieve to N , process the primes in increasing order, and maintain a union-find structure. When prime p is reached, join it to every already-seen (hence smaller) prime it connects to: for the same-length case, vary each of its digits and check whether the resulting smaller number is prime; for the prepend case, strip p 's leading digit and, if the remaining tail is a prime with no leading zero, join the two. Looking only "downward" to smaller primes both enforces the chain constraint automatically and avoids handling each connection twice. After the joins, p is a relative exactly when it shares a component with 2; otherwise its value is added to the running total. This reproduces $F(10^3) = 431$ and $F(10^4) = 78728$.

Problem 429: Sum of Squares of Unitary Divisors

Solution: 98792821

A unitary divisor d of n satisfies $\gcd(d, n/d) = 1$, so it takes either all or none of each prime power in n . The sum of squares of unitary divisors S is therefore multiplicative with $S(p^a) = 1 + p^{2a}$ (the 1 from $d = 1$, the p^{2a} from $d = p^a$). We need $S(10^8!) \bmod (10^9 + 9)$.

For $N!$ the exponent of a prime p is given by Legendre's formula $e_p = \sum_{i \geq 1} \lfloor N/p^i \rfloor$, so

$$S(N!) = \prod_{p \leq N} (1 + p^{2e_p}).$$

Sieve the primes up to 10^8 ; for each, accumulate e_p by the floor-sum, compute $p^{2e_p} \bmod (10^9 + 9)$ by fast exponentiation, and fold $(1 + p^{2e_p})$ into the running product. The check $S(4!) = (1 + 2^6)(1 + 3^2) = 650$ confirms the formula.

Problem 432: Totient Sum

Solution: 754862080

Here $S(n, m) = \sum_{i=1}^m \varphi(ni)$ with $n = 510510 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$, and we want the last nine digits of $S(510510, 10^{11})$. Writing $T(n, m) = \sum_{i=1}^m \varphi(ni)$, peel off the smallest prime factor p of the (squarefree) n . Since $\varphi(pk) = (p-1)\varphi(k)$ when $p \nmid k$ and $p\varphi(k)$ when $p \mid k$,

$$T(n, m) = (p-1)T(n/p, m) + T(n, \lfloor m/p \rfloor).$$

The first term handles the i not divisible by p ; the second collects the $i = pj$ (where $\varphi(ni) = \varphi(nj) \cdot p/(p-1) \cdot (p-1)$ telescopes back into $T(n, \lfloor m/p \rfloor)$). Peeling primes one at a time always removes the smallest, so only the eight "suffix" products of the seven primes ever appear, and the recursion bottoms out at $T(1, m) = \Phi(m) = \sum_{i \leq m} \varphi(i)$.

The summatory totient Φ is evaluated over the *quotient lattice* of m : sieve φ and its prefix sums up to $L \approx m^{2/3}$, and for each large value $v = \lfloor m/i \rfloor$ use

$$\Phi(v) = v(v+1)/2 - \sum_{d \geq 2} \Phi(\lfloor v/d \rfloor),$$

with divisor-block grouping so each call is $O(\sqrt{v})$ and every $\lfloor v/d \rfloor$ is again a lattice point (either sieved or a previously computed larger index). Memoizing T on (prime index, m') and reducing mod 10^9 throughout (taking the even factor of $v(v+1)$ before halving, since 10^9 has no inverse of 2) gives the answer in a few seconds. It reproduces the given $S(510510, 10^6) = 45480596821125120$.

Problem 435: Polynomials of Fibonacci Numbers

Solution: 252541322550

The polynomial $F_n(x) = \sum_{i=0}^n f_i x^i$ uses the Fibonacci numbers as coefficients, and we want $\sum_{x=0}^{100} F_n(x)$ for $n = 10^{15}$, modulo $15!$. Because the Fibonacci generating function is $\sum_{i \geq 0} f_i x^i = x/(1 - x - x^2)$, multiplying the partial sum by $1 - x - x^2$ telescopes (every interior coefficient $f_k - f_{k-1} - f_{k-2}$ vanishes), leaving only boundary terms:

$$F_n(x) = \frac{f_{n+1}x^{n+1} + f_n x^{n+2} - x}{x^2 + x - 1}.$$

The denominator $D = x^2 + x - 1$ is nonzero for every integer x , but it need not be invertible modulo the composite $15!$. The fix exploits that $F_n(x)$ is a genuine integer, so the numerator equals $D \cdot F_n(x)$ exactly: computing it modulo mD yields $D \cdot (F_n(x) \pmod{m})$, and an ordinary integer division by D recovers $F_n(x) \pmod{m}$. Each Fibonacci pair f_n, f_{n+1} comes from fast doubling and each power from fast exponentiation, all mod mD . The closed form reproduces $F_7(11) = 268357683$.

Problem 437: Fibonacci Primitive Roots

Solution: 74204709657207

A Fibonacci primitive root g of p satisfies $g^{n+2} \equiv g^{n+1} + g^n$, i.e. $g^2 \equiv g + 1$, while also being a primitive root. So g must be a root of $x^2 - x - 1$ in $\mathbb{F}_p$, which requires 5 to be a quadratic residue: either $p = 5$ or $p \equiv \pm 1 \pmod{5}$. For such p the two roots are $(1 \pm \sqrt{5})/2$, and p qualifies exactly when at least one of them is a primitive root.

There is no simple congruence test for this (the density is of Artin type), so each candidate prime is checked directly: take $\sqrt{5}$ via Tonelli–Shanks (a single power when $p \equiv 3 \pmod{4}$), form the two roots, factor $p - 1$ by trial division against the primes up to 10^4 (the leftover cofactor, if any, is prime since $p - 1 < 10^8$), and test primitivity through $g^{(p-1)/q} \neq 1$ for every prime $q \mid p - 1$. Sieving the primes below 10^8 and running this in compiled code reproduces the stated 323 primes below 10^4 summing to 1480491, and finishes the full range in a few seconds.

Problem 443: GCD Sequence

Solution: 2744233049300770

The sequence is $g(4) = 13$ and $g(n) = g(n-1) + \gcd(n, g(n-1))$, and we want $g(10^{15})$. Stepping one term at a time is hopeless, but the increments are extremely structured. Consider a stretch where the gcd happens to be 1: then $g(j) = g(j-1) + 1$, so the difference $c = g(j-1) - j$ is *constant* throughout the stretch. While c holds steady,

$$\gcd(j, g(j-1)) = \gcd(j, j+c) = \gcd(j, c),$$

so the run of 1's continues until j first shares a factor with c . The earliest such index past the current n is

$$j^* = \min_{p \mid c} p \cdot (\lfloor n/p \rfloor + 1),$$

taken over the distinct prime factors p of c . At j^* the increment jumps to $d = \gcd(j^*, c)$, after which a fresh constant c takes over. So the whole sequence can be fast-forwarded jump to jump: from state $(n, v = g(n))$ set $c = v - n - 1$, find j^* , and update $v \leftarrow v + (j^* - 1 - n) + d$, $n \leftarrow j^*$ (stopping early with $v + (T - n)$ once j^* would overshoot the target T).

The number of jumps grows glacially – about 116 by 10^6 and only ~ 340 by 10^{15} – so the cost is dominated by factoring each c (up to $\sim 10^{15}$). Trial division to $\sqrt{c}$ is too slow there, so Pollard's rho with a Miller–Rabin primality check supplies the distinct prime factors in roughly $c^{1/4}$ steps each. The fast-forward reproduces $g(1000) = 2524$ and $g(10^6) = 2624152$.

Problem 448: Average Least Common Multiple

Solution: 106467648

Since $\text{lcm}(n, i) = ni / \gcd(n, i)$, the average is $A(n) = \frac{1}{n} \sum_{i=1}^n \text{lcm}(n, i) = \sum_{i=1}^n i / \gcd(n, i)$. Grouping the i by $d = \gcd(n, i)$ and writing $i = dj$ with $\gcd(j, n/d) = 1$ turns the inner term into j , so $A(n) = \sum_{e|n} T(e)$ where $T(e)$ sums the integers in $[1, e]$ coprime to e , namely $T(e) = e\varphi(e)/2$ for $e \geq 2$ and $T(1) = 1$. This gives $A(10) = 32$. Summing over $k \leq n$ and counting how often each e divides k ,

$$S(n) = \sum_{e=1}^n T(e) \lfloor n/e \rfloor = \frac{1}{2} \left(n + \sum_{e=1}^n e\varphi(e) \lfloor n/e \rfloor \right),$$

which reproduces $S(100) = 122726$.

Writing $f(e) = e\varphi(e)$ and $F(x) = \sum_{e \leq x} f(e)$, the Dirichlet identity $(\text{Id} \cdot \varphi) * \text{Id} = \text{Id}^2$ (because $\sum_{d|n} \varphi(d) = n$) yields $\sum_{m \leq x} m \cdot F(\lfloor x/m \rfloor) = \sum_{m \leq x} m^2$, hence the recursion

$$F(x) = \frac{x(x+1)(2x+1)}{6} - \sum_{m \geq 2} m \cdot F(\lfloor x/m \rfloor).$$

Sieving f up to $\sim N^{2/3}$ and memoizing F on the $O(\sqrt{N})$ values $\lfloor N/m \rfloor$ makes the whole evaluation sublinear; the final $\sum_e f(e) \lfloor N/e \rfloor$ is also accumulated over divisor blocks, all modulo the prime 999999017.

Problem 449: Chocolate Covered Candy

Solution: 103.37870096

The candy is the spheroid $x^2/a^2 + y^2/a^2 + z^2/b^2 = 1$, coated with a uniform 1 mm layer; the chocolate occupies the parallel body at distance $t = 1$. By the Steiner tube formula, for a convex body the volume of the layer of thickness t is $St + Mt^2 + (4\pi/3)t^3$, where S is the surface area, $M = \int H dA$ is the integral of the mean curvature, and the cubic coefficient $(4\pi)/3 = \frac{1}{3} \int K dA$ comes from Gauss–Bonnet. At $t = 1$ the answer is simply $S + M + 4\pi/3$ (which gives $28\pi/3$ for the unit sphere).

Treating the spheroid as a surface of revolution with profile $(r, z) = (a \sin u, b \cos u)$ and line element $W = \sqrt{a^2 \cos^2 u + b^2 \sin^2 u}$, the two integrals reduce to

$$S = 2\pi a \int_0^\pi \sin u \cdot W du, \quad M = \pi \int_0^\pi \sin u (a^2 b / W^2 + b) du,$$

using the meridional curvature ab/W^3 and the azimuthal curvature $b/(aW)$. Evaluating these by high-resolution numerical integration reproduces the given $a = 2$ value 60.35475635 and yields the $a = 3$ answer.

Problem 451: Modular Inverses

Solution: 153651073760956

A number m is its own inverse modulo n exactly when $m^2 \equiv 1 \pmod{n}$, and $I(n)$ is the largest such m below $n - 1$. The self-inverse residues come in complementary pairs $(m, n - m)$ – if $m^2 \equiv 1$ then $(n - m)^2 \equiv 1$ too – and both 1 and $n - 1$ are always solutions. So the largest solution below $n - 1$ is the complement of the *smallest* solution above 1:

$$I(n) = n - s(n), \quad s(n) = \min\{m > 1 : m^2 \equiv 1 \pmod{n}\}.$$

By CRT, the roots of $x^2 \equiv 1$ modulo $n = \prod q_i$ (each q_i a prime power) are all combinations of the roots modulo each q_i : just $\{1, q_i - 1\}$ for odd primes, but for powers of two there is one root mod 2, two mod 4, and four mod 2^e ($e \geq 3$), namely $\{1, 2^{e-1} - 1, 2^{e-1} + 1, 2^e - 1\}$.

A smallest-prime-factor sieve to $2 \cdot 10^7$ factors every n quickly. For each, build the CRT basis $b_i \equiv 1 \pmod{q_i}$, $b_i \equiv 0$ modulo the other factors (one modular inverse per factor via the extended Euclidean algorithm), enumerate the at-most-256 root combinations $\sum_i r_i b_i \pmod{n}$, and take the smallest exceeding 1. Summing $n - s(n)$ over $3 \leq n \leq 2 \cdot 10^7$ runs in a few seconds. The method gives $I(15) = 11$, $I(100) = 51$ and $I(7) = 1$.

Problem 463: A Weird Recurrence Relation

Solution: 808981553

The four cases of f combine neatly: summing one block of four gives $f(4n) + f(4n + 1) + f(4n + 2) + f(4n + 3) = 6f(2n + 1) - 2f(n)$. Checking small values reveals what f really is – the *binary bit-reversal* of n : writing n with $b = \lfloor \log_2 n \rfloor + 1$ bits and reversing them. Indeed $f(8) = f(1000_2) = 0001_2 = 1$ and $f(5) = f(101_2) = 101_2 = 5$, matching the recurrence.

So $S(N) = \sum_{i=1}^N \text{rev}(i)$. The b -bit numbers $[2^{b-1}, 2^b - 1]$ reverse onto every b -bit pattern whose leading bit is set, and a short computation shows their reversals sum to exactly 4^{b-1} ; the full groups for $b = 1, \dots, B - 1$ (with $B = \text{bit length of } N$) therefore contribute $\sum_{b=1}^{B-1} 4^{b-1}$. The top group $[2^{B-1}, N]$ is partial, so its reversal sum is taken bit by bit: bit k contributes 2^{B-1-k} times the number of in-range integers with that bit set, each count obtained in closed form. This reproduces $S(8) = 22$ and $S(100) = 3604$, and $S(3^{37}) \pmod{10^9}$ follows instantly.

Problem 469: Empty Chairs

Solution: 0.56766764161831

Knights take random empty chairs around a round table of N , always leaving at least one empty chair beside each occupied one, until no admissible chair remains; $E(N)$ is the expected fraction of empty chairs. This is random sequential adsorption with nearest-neighbour exclusion. The first knight (uniform, but by symmetry any seat) occupies one chair and forbids its two neighbours, leaving a *line* of $N - 3$ free chairs, so the expected number occupied is $M(N) = 1 + g(N - 3)$ where $g(m)$ is the same quantity for a row of m chairs. Splitting a row on its first knight gives

$$g(m) = 1 + \frac{2}{m} \sum_{j=0}^{m-2} g(j),$$

which reproduces $E(4) = (4 - M(4))/4 = 1/2$ and $E(6) = 5/9$.

Asymptotically $g(m) = \theta m + (3\theta - 1) + \varepsilon(m)$ with $\theta = (1 - e^{-2})/2$, and substituting back the constant cancels exactly: $M(N) = \theta N + \varepsilon(N - 3)$, so the empty fraction is $(1 + e^{-2})/2$ minus a correction $\varepsilon(N - 3)/N$. The correction decays faster than any power of N (it already agrees to sixteen digits by $N = 100$), so for $N = 10^{18}$ the answer is $(1 + e^{-2})/2$ rounded to fourteen places.

Problem 479: Roots on the Rise

Solution: 191541795

For each k the three numbers a_k, b_k, c_k solve $1/x = (k/x)^2(k + x^2) - kx$. Multiplying through by x^2 and rearranging turns this into the cubic

$$x^3 - kx^2 + \frac{1}{k}x - k^2 = 0,$$

so Vieta's formulas give $a + b + c = k$, $ab + bc + ca = 1/k$ and $abc = k^2$. The summand depends only on the symmetric product

$$(a + b)(b + c)(c + a) = (a + b + c)(ab + bc + ca) - abc = k \cdot \frac{1}{k} - k^2 = 1 - k^2,$$

which is why $S(n)$ comes out an integer. Hence the triple product raised to the p -th power is simply $(1 - k^2)^p$ and

$$S(n) = \sum_{k=1}^n \sum_{p=1}^n (1 - k^2)^p.$$

The inner sum is a geometric series with ratio $r = 1 - k^2$: it equals $r(r^n - 1)/(r - 1)$ (and vanishes for $k = 1$, where $r = 0$). Evaluating each with fast exponentiation and a modular inverse of $r - 1 = -k^2$ modulo $10^9 + 7$ takes $O(n \log n)$ overall. It reproduces $S(4) = 51160$.

Problem 485: Maximum Number of Divisors

Solution: 51281274340

With $d(n)$ the divisor count, $M(n, k)$ is the maximum of d over the length- k window $[n, n + k - 1]$, and $S(u, k) = \sum_{n=1}^{u-k+1} M(n, k)$. We want $S(10^8, 10^5)$, so the task is the sum of sliding-window maxima of the divisor-count sequence.

First tabulate d up to 10^8 with the harmonic sieve: for every i , add one to each multiple of i . Since $d(n) < 800$ in this range the values fit in 16-bit cells, keeping the array near 200 MB. Then sweep a monotonic deque across it: each index is pushed once after evicting smaller-or-equal values from the back, the front is dropped when it leaves the window, and once k values have been seen the front holds the current window maximum. Held in a circular buffer of capacity $k + 1$, this gives every maximum in amortised $O(1)$, so the whole sum is one linear pass. The construction reproduces $S(1000, 10) = 17176$.

Problem 487: Sums of Power Sums

Solution: 106650212746

With $f_{k(n)} = \sum_{i=1}^n i^k$ and $S_{k(n)} = \sum_{i=1}^n f_{k(i)}$, we want $\sum_p (S_{10000}(10^{12}) \bmod p)$ over the primes p in $(2 \cdot 10^9, 2 \cdot 10^9 + 2000)$. Swapping the order of summation, $S_{k(n)} = \sum_{j=1}^n (n - j + 1)j^k = (n + 1)f_{k(n)} - f_{k+1}(n)$, and since f_k is a polynomial of degree $k + 1$ (Faulhaber), S_k is a polynomial in n of degree $k + 2$.

A degree- $(k + 2)$ polynomial is pinned down by its values at the consecutive nodes $0, 1, \dots, k + 2$, which are cheap to tabulate mod p : run two prefix sums, $f \leftarrow f + i^k$ and $S \leftarrow S + f$, with $i^k \bmod p$ from fast exponentiation. Lagrange interpolation at $x = 10^{12} \bmod p$ then gives $S_{k(10^{12})} \bmod p$,

$$S_{k(x)} = \sum_{i=0}^d y_i \frac{\prod_{j \neq i} (x - j)}{\prod_{j \neq i} (i - j)}, \quad d = k + 2,$$

evaluated in $O(d)$ via prefix/suffix products of $(x - j)$ and the factorial identity $\prod_{j \neq i} (i - j) = (-1)^{d-i} i!(d - i)!$. Because every $p < 2^{31}$, the products $a \cdot b$ stay within 64 bits.

Reducing each integer factor $(10^{12} - j) \equiv (x - j) \bmod p$ is valid even though S_k has rational coefficients, since the Lagrange form never expands into monomials. There are 100 primes in the window; the method reproduces the stated $S_4(100) = 35375333830$.

Problem 491: Double Pandigital Number Divisible by 11

Solution: 194505988824000

A double pandigital number has all twenty digits placed so each of 0..9 appears twice, with no leading zero. Index the twenty positions by their power of ten; since $10 \equiv -1 \pmod{11}$, the number is divisible by 11 when the digits in even powers and odd powers satisfy $E - O \equiv 0 \pmod{11}$. Each group holds ten positions, and the grand total is $E + O = 2(0 + 1 + \dots + 9) = 90$, so $2E \equiv 90 \equiv 2$ and hence $E \equiv 1 \pmod{11}$.

Let $k_d \in \{0, 1, 2\}$ be how many copies of digit d land in the even-power group; a valid assignment needs $\sum_d k_d = 10$ and $\sum_d dk_d \equiv 1 \pmod{11}$. There are only 3^{10} assignments to scan. Each contributes $(10! / \prod_d k_d!)(10! / \prod_d (2 - k_d)!)$ arrangements, the multinomials for filling the two groups. The leading digit lies in the odd-power group, so the arrangements with a zero there are removed: fixing one of its $2 - k_0$ zeros in front leaves $9! / ((2 - k_0 - 1)! \prod_{d \geq 1} (2 - k_d)!)$ ways for the rest. Summing over all valid (k_d) gives the count directly.

Problem 492: Exploding Sequence

Solution: 242586962923928

The sequence $a_1 = 1$, $a_{k+1} = 6a_k^2 + 10a_k + 3$ grows doubly exponentially, but the substitution $w = 6a + 5$ linearises the quadratic map: $w_{k+1} = w_k^2 - 2$ with $w_1 = 11$. Writing $w = \lambda + \lambda^{-1}$, squaring-minus-two just doubles the exponent, so

$$w_n = \lambda^{2^{n-1}} + \lambda^{-2^{n-1}}, \quad \lambda + \lambda^{-1} = 11,$$

where λ is a root of $X^2 - 11X + 1 = 0$ (discriminant $117 = 9 \cdot 13$). This is exactly the Lucas sequence $V_m = \lambda^m + \lambda^{-m}$ with parameters $P = 11$, $Q = 1$, and $a_n = (w_n - 5)/6$.

To get $a_n \pmod{p}$ for $n = 10^{15}$ the exponent 2^{n-1} is astronomically large, but λ lives in $\mathbb{F}_p$ when 13 is a quadratic residue mod p (then $\lambda^{p-1} = 1$) and in $\mathbb{F}_{p^2}$ otherwise (then the conjugate is λ^{-1} , so $\lambda^{p+1} = 1$). Either way only $2^{n-1} \bmod (p \mp 1)$ matters; reduce it by fast exponentiation, evaluate V_E with the Lucas doubling rules $V_{2j} = V_j^2 - 2$, $V_{2j+1} = V_j V_{j+1} - 11$, and divide by 6. A segmented sieve supplies the primes in $[10^9, 10^9 + 10^7]$. The method reproduces $B(10^9, 10^3, 10^3) = 23674718882$ and $B(10^9, 10^3, 10^{15}) = 20731563854$.

Problem 493: Under the Rainbow

Solution: 6.818741802

Seventy balls carry seven colours, ten of each; twenty are drawn at random, and we want the expected number of distinct colours among them. Linearity of expectation sidesteps the messy joint distribution: the expected count is the sum over the seven colours of the probability that each appears. A given colour is absent precisely when all twenty drawn balls come from the other sixty, which happens with probability $\binom{60}{20}/\binom{70}{20}$, so each colour is present with probability $1 - \binom{60}{20}/\binom{70}{20}$. The answer is

$$7 \left(1 - \frac{\binom{60}{20}}{\binom{70}{20}} \right) \approx 6.818741802.$$

Problem 497: Drunken Tower of Hanoi

Solution: 684901360

"Optimal" means fewest disk pickups, which forces the unique minimal Tower-of-Hanoi sequence of $2^n - 1$ moves. Bob's walk is a random walk on the squares $\{1, \dots, k\}$ with reflecting ends, so the expected number of squares to get from one rod to another is the walk's first-passage time. For a step rightward from i to $i + 1$ the expected cost is $2i - 1$, giving a first passage from x to a larger square y of $(y - 1)^2 - (x - 1)^2$; by the mirror symmetry a leftward passage costs $(k - y)^2 - (k - x)^2$. (These sum to the integer answer the problem promises.)

The total expected distance is a sum over Bob's visited rods: starting at b , then for each disk move walking to its source and then to its destination. The number of times each of the six directed rod-to-rod transitions occurs follows the Hanoi recursion. Moving m disks $X \rightarrow Y$ (third rod Z) splits into moving $m - 1$ disks $X \rightarrow Z$ (Bob from his current rod), the big disk $X \rightarrow Y$ (preceded by a hop $Z \rightarrow X$), then $m - 1$ disks $Z \rightarrow Y$ (Bob now at Y). Carrying an 18-state vector (six ordered pairs $\times$ three possible start rods) of transition counts and advancing m from 1 to 10000 gives every $E(n, \dots)$ cheaply; multiplying counts by the first-passage costs and summing mod 10^9 over n finishes it. The recursion reproduces $E(2, 5, 1, 3, 5) = 60$ and $E(3, 20, 4, 9, 17) = 2358$.

Problem 498: Remainder of Polynomial Division

Solution: 472294837

$R_{n,m}(x)$ is the remainder of x^n divided by $(x-1)^m$, and $C(n,m,d)$ is the absolute value of its degree- d coefficient. Substituting $t = x-1$ turns the dividend into $(t+1)^n$ and the divisor into t^m , so the remainder is simply the low-degree truncation $\sum_{i=0}^{m-1} \binom{n}{i} t^i$. Re-expanding $t = x-1$ and collecting the x^d term gives

$$C(n,m,d) = \binom{n}{d} \sum_{k=0}^{m-d-1} \binom{n-d}{k} (-1)^k.$$

The alternating partial sum telescopes via $\sum_{k=0}^j \binom{M}{k} (-1)^k = (-1)^j \binom{M-1}{j}$, leaving the clean product

$$C(n,m,d) = \binom{n}{d} \cdot \binom{n-d-1}{m-d-1},$$

which reproduces $C(6,3,1) = 24$ and $C(100,10,4) = 227197811615775$. For $C(10^{13}, 10^{12}, 10^4)$ modulo the prime 999999937, neither factor is divisible by the prime, and each binomial has arguments far larger than p , so Lucas' theorem reduces it to a product of binomials of base- p digits (each digit's binomial evaluated by the multiplicative formula with one modular inverse).

Problem 500: Problem 500!!!

Solution: 35407281

We need the smallest integer with exactly 2^{500500} divisors, modulo 500500507. The divisor count of $n = \prod p_i^{e_i}$ is $\prod (e_i + 1)$, so to reach 2^{500500} every factor $e_i + 1$ must itself be a power of two, and we must accumulate exactly 500500 doublings.

Each doubling corresponds to multiplying the current number by some p^{2^k} : this lifts the exponent of p from $2^k - 1$ to $2^{k+1} - 1$, turning the local factor 2^k into 2^{k+1} . To keep the result minimal, greedily take the 500500 smallest values from the table $\{p^{2^k} : p \text{ prime}, k \geq 0\}$. A min-heap does this: seed it with the primes themselves, and whenever p^{2^k} is popped, push its square $p^{2^{k+1}}$. Multiplying the 500500 popped values modulo 500500507 gives the answer. Squares of primes above $\sqrt{7376507}$ already exceed the 500500-th prime, so seeding with the first 500500 primes is enough – no prime beyond 7376507 is ever needed.

Problem 501: Eight Divisors

Solution: 197912312715

A number has exactly eight divisors precisely when its prime signature gives $\prod(e_i + 1) = 8$. There are three shapes:

- p^7 ,
- p^3q with distinct primes p, q ,
- pqr with $p < q < r$.

Let $\pi(x)$ be the prime-counting function. The three contributions to $f(N)$ for $N = 10^{12}$ are

$$\pi(N^{1/7}) + \sum_{p^3 \leq N} (\pi(N/p^3) - 1) + \sum_{p < q, pq^2 \leq N} (\pi(N/(pq)) - \pi(q)),$$

where in the second sum the -1 removes the case $q = p$, and in the third we count primes r with $q < r \leq N/(pq)$.

The arguments N/m range over the $O(\sqrt{N})$ distinct values of $\lfloor N/m \rfloor$, so we need π at those points. Computing them with the Lucy_Hedgehog recurrence in $O(N^{3/4})$ time makes the whole evaluation fast. Checks $f(100) = 10$, $f(1000) = 180$, $f(10^6) = 224427$ all hold.

Problem 504: Square on the Inside

Solution: 694687

The vertices $A(a, 0)$, $B(0, b)$, $C(-c, 0)$, $D(0, -d)$ form a kite whose diagonals lie on the axes, so its area is $\frac{1}{2}(a + c)(b + d)$. The boundary lattice points on the four edges total $\gcd(a, b) + \gcd(b, c) + \gcd(c, d) + \gcd(d, a)$. By Pick's theorem $A = I + B/2 - 1$, the strictly interior count is

$$I = \frac{(a + c)(b + d) - (\gcd(a, b) + \gcd(b, c) + \gcd(c, d) + \gcd(d, a))}{2} + 1.$$

We just count quadruples $1 \leq a, b, c, d \leq 100$ for which I is a perfect square.

Precomputing a gcd table and a boolean "is a square" array up to the maximum possible I turns the 10^8 -iteration loop into simple lookups. For $m = 4$ this reproduces the stated 42.

Problem 506: Clock Sequence

Solution: 18934502

The digit stream repeats 1, 2, 3, 4, 3, 2 with period 6 (digit-sum 15 per period), chopped into integers v_n whose digit-sum equals n . We want $S(N) = \sum_{n=1}^N v_n$ modulo $M = 123454321 = 11111^2$ for $N = 10^{14}$, far too many terms to build directly.

Let $T(n) = n(n+1)/2$ be the cumulative digit-sum target. Walking the stream, the position consumed after emitting $v_1 \dots v_n$ is

$$e(n) = 6\lfloor T(n)/15 \rfloor + j(T(n) \bmod 15),$$

where j maps the six attainable partial sums $\{0, 1, 3, 6, 10, 13\}$ to $\{0, 1, 2, 3, 4, 5\}$. The value v_n is the digit block strictly between positions $e(n-1)$ and $e(n)$.

Reading the stream's prefixes as numbers modulo M gives a closed form. With $g(n) = \text{Pr}_0[r] - C \cdot 10^r$, where $r = j(T(n) \bmod 15)$, $C = 123432/999999$, and $\text{Pr}_0 = [0, 1, 12, 123, 1234, 12343]$, one verifies

$$v_n \equiv g(n) - g(n-1) \cdot 10^{L_n} \pmod{M}, \quad L_n = e(n) - e(n-1).$$

Now g is periodic in n with period 30, and $L_{n+30} = L_n + 12$. Hence $S(N) = \Sigma_1 - \Sigma_2$ where $\Sigma_1 = \sum g(n)$ collapses to a 30-term block sum, and $\Sigma_2 = \sum g(n-1)10^{L_n}$ splits into 30 geometric series of common ratio 10^{12} (whose denominator $10^{12} - 1$ is invertible mod M). Everything is $O(1)$ after a 30-term setup. Checks $S(11) = 36120$ and $S(1000) \equiv 18232686$ confirm the formula.

Problem 509: Divisor Nim

Solution: 151725678

This is a three-pile impartial game: from a pile of k stones a player removes a proper divisor of k . By the Sprague-Grundy theorem each pile has a Grundy value $g(k)$, and a position is a win for the player to move iff $g(a) \oplus g(b) \oplus g(c) \neq 0$.

Computing small values reveals $g(k) = \nu_2(k)$, the exponent of 2 in k . An odd k has only odd divisors, so every move lands on an even number (Grundy ≥ 1), giving $g(k) = 0$; for $k = 2^a m$ with m odd the reachable Grundy values are exactly $0, 1, \dots, a-1$, so the mex is a .

So we count triples in $[1, n]^3$ whose valuations XOR to a nonzero value. Let $c_v = \lfloor n/2^v \rfloor - \lfloor n/2^{v+1} \rfloor$ be the count of integers in $[1, n]$ with valuation v . Then

$$S(n) = n^3 - \sum_{v_1, v_2} c_{v_1} c_{v_2} c_{v_1 \oplus v_2},$$

the subtracted term counting losing positions. Valuations only reach about 46 for $n \approx 1.23 \cdot 10^{14}$, so this is a tiny double sum. Checks $S(10) = 692$ and $S(100) = 735494$ hold.

Problem 510: Tangent Circles

Solution: 315306518862563689

Two circles of radii r_A, r_B rest on a line, tangent to each other, and a third circle of radius r_C nestles between them tangent to both and to the line. Descartes' configuration along a line gives the curvature relation

$$\frac{1}{\sqrt{r_C}} = \frac{1}{\sqrt{r_A}} + \frac{1}{\sqrt{r_B}}.$$

Writing the radii in lowest terms shows every integer solution is parametrised by coprime $p \leq q$ and a scale $e \geq 1$:

$$r_A = ep^2(p+q)^2, \quad r_B = eq^2(p+q)^2, \quad r_C = ep^2q^2,$$

and this is a bijection onto the integer triples. We need $S(n) = \sum(r_A + r_B + r_C)$ over all solutions with $r_A, r_B, r_C \leq n$, which here means $r_B = eq^2(p+q)^2 \leq n$ is the binding constraint.

For fixed coprime $p \leq q$ let $u = p^2(p+q)^2 + q^2(p+q)^2 + p^2q^2$ be the per-unit sum and $M = \lfloor n/(q^2(p+q)^2) \rfloor$ the largest scale. The scales $e = 1, \dots, M$ contribute $u \cdot M(M+1)/2$. Summing over coprime pairs with $q^2(p+q)^2 \leq n$ finishes it. Checks $S(5) = 9$ and $S(100) = 3072$ hold.

Problem 512: Sums of Totients of Powers

Solution: 50660591862310323

Let $f(n) = \left(\sum_{i=1}^n \varphi(n^i)\right) \bmod (n+1)$ and $g(N) = \sum_{n=1}^N f(n)$; we need $g(5 \cdot 10^8)$.

Because $\varphi(n^i) = n^{i-1}\varphi(n)$, the inner sum is $\varphi(n)(1 + n + \dots + n^{n-1})$. Working modulo $n+1$, where $n \equiv -1$, the powers alternate $n^{i-1} \equiv (-1)^{i-1}$, so the geometric sum collapses to 1 when n is odd and 0 when n is even. Since $\varphi(n) \leq n-1 < n+1$, no further reduction happens, and

$$f(n) = \begin{cases} \varphi(n) & \text{if } n \text{ odd} \\ 0 & \text{if } n \text{ even} \end{cases}.$$

Thus $g(N) = \sum_{n \text{ odd} \leq N} \varphi(n)$; call it $A(N)$.

To evaluate A in sublinear time, restrict the identity $\sum_{d|m} \varphi(d) = m$ to odd m (whose divisors are all odd). Summing over odd $m \leq N$ and writing $m = dk$ with both odd,

$$\sum_{m \text{ odd} \leq N} m = \sum_{k \text{ odd} \leq N} A(\lfloor N/k \rfloor).$$

The left side is the sum of the first $((N+1)/2)$ odd numbers, namely $((N+1)/2)^2$. Isolating the $k=1$ term gives the recurrence

$$A(N) = (\lfloor (N+1)/2 \rfloor)^2 - \sum_{k \text{ odd}, 3 \leq k \leq N} A(\lfloor N/k \rfloor),$$

which we memoise over the $O(\sqrt{N})$ distinct values of $\lfloor N/k \rfloor$, with a linear φ -sieve supplying the base cases. As a check, $g(100) = 2007$.

Problem 516: 5-smooth Totients

Solution: 939087315

A Hamming number is 5-smooth: of the form $2^a 3^b 5^c$. We want $S(L) = \sum_{n \leq L, \varphi(n) \text{ Hamming}} n$ for $L = 10^{12}$, modulo 2^{32} .

Since φ is multiplicative with $\varphi(p^e) = p^{e-1}(p-1)$, the totient is 5-smooth exactly when every prime-power factor contributes a 5-smooth value. For $p \in \{2, 3, 5\}$ this holds for every exponent (as $p-1 \in \{1, 2, 4\}$). For a prime $p > 5$ the factor p^{e-1} would introduce p itself unless $e = 1$, and then we additionally need $p-1$ to be Hamming. So the admissible n are precisely

$$n = (2^a 3^b 5^c) \cdot Q,$$

where Q is a squarefree product of distinct "special" primes $p > 5$ with $p-1$ Hamming. This factorisation is unique (the $2, 3, 5$ part is itself a Hamming number H), so

$$S(L) = \sum_Q Q \cdot H(\lfloor L/Q \rfloor),$$

summed over all squarefree special-prime products $Q \leq L$, where $H(x)$ is the sum of Hamming numbers $\leq x$. We list the few thousand Hamming numbers up to L (with prefix sums for $H(x)$), collect the special primes by testing $h+1$ for primality, and enumerate the products Q by depth-first search. As a check, $S(100) = 3728$.

Problem 518: Prime Triples and Geometric Sequences

Solution: 100315739184392

We want $S(n) = \sum(a + b + c)$ over prime triples $a < b < c < n$ for which $a + 1, b + 1, c + 1$ form a geometric progression, with $n = 10^8$.

Every positive-integer geometric progression can be written (ky^2, kxy, kx^2) with $x > y \geq 1$ and $\gcd(x, y) = 1$ (the middle term squared equals the product of the outer two).

Setting

$$a = ky^2 - 1, \quad b = kxy - 1, \quad c = kx^2 - 1,$$

the ordering $a < b < c$ is automatic from $y < x$, and we need all three prime with $c < n$.

Sieve primes up to n , then loop over x (up to $\sqrt{n}$), over k with $kx^2 \leq n$, prune on $c = kx^2 - 1$ being prime, and finally over coprime $y < x$ checking a and b . As a check, $S(100) = 1035$ from the eleven listed triples.

Problem 520: Simbers

Solution: 238413705

A *simber* is a positive integer in which every odd digit present occurs an odd number of times and every even digit present an even number of times. With $Q(n)$ the count of simbers of at most n digits, we need $\sum_{u=1}^{39} Q(2^u) \bmod 1000000123$. As 2^{39} has hundreds of billions of digits, only a closed form will do.

Count length- L digit strings (leading zeros allowed) with each even digit occurring an even number of times and each odd digit a 0-or-odd number of times, via exponential generating functions: an even digit contributes $\cosh(x)$ (even counts) and an odd digit $1 + \sinh(x)$ (0 or odd). With five digits of each parity,

$$T(L) = L! [x^L] \cosh(x)^5 (1 + \sinh(x))^5 = \frac{1}{2^{10}} \sum_j a_j j^L,$$

where, with $z = e^x$, the integer $a_j = [z^{j+10}] (z^2 + 1)^5 (z^2 + 2z - 1)^5$ for $-10 \leq j \leq 10$. A simber string that begins with 0 pins one zero in front, leaving $L - 1$ digits in which digit 0 must now appear an odd number of times; that count is $Z(L - 1) = (1/2^{10}) \sum_j b_j j^{L-1}$ with $b_j = [z^{j+10}] (z^2 - 1)(z^2 + 1)^4 (z^2 + 2z - 1)^5$. The L -digit simbers (no leading zero) number $T(L) - Z(L - 1)$, so

$$Q(n) = \frac{1}{2^{10}} \left(\sum_j a_j \sum_{L=1}^n j^L - \sum_j b_j \sum_{t=0}^{n-1} j^t \right),$$

each inner sum a geometric series evaluated modulo 1000000123. As checks, $Q(7) = 287975$ and $Q(100) \equiv 123864868$.

Problem 521: Smallest Prime Factor

Solution: 44389811

Let $\text{smpf}(i)$ be the smallest prime factor of i and $S(n) = \sum_{2 \leq i \leq n} \text{smpf}(i)$; we need $S(10^{12}) \bmod 10^9$.

Group the integers by their smallest prime factor. The $i \leq n$ with $\text{smpf}(i) = p$ are exactly $i = pm$ where m has no prime factor below p , so their number is $\Phi(\lfloor n/p \rfloor, p)$, where $\Phi(x, p)$ counts the integers $m \leq x$ that equal 1 or have every prime factor at least p . Hence

$$S(n) = \sum_{p \text{ prime} \leq n} p \cdot \Phi(\lfloor n/p \rfloor, p).$$

Split at $\sqrt{n}$. For $p > \sqrt{n}$ we have $\lfloor n/p \rfloor < p$, so the only valid m is 1 and $\Phi = 1$; these primes contribute $\sum_{\sqrt{n} < p \leq n} p = \text{PrimeSum}(n) - \text{PrimeSum}(\sqrt{n})$. For $p \leq \sqrt{n}$ we use a Lucy_Hedgehog sieve: maintaining the array $C[v]$ of integers in $[2, v]$ that are prime or have smallest prime factor exceeding the primes sieved so far, the value read just before sieving out p gives $\Phi(\lfloor n/p \rfloor, p) = 1 + C[\lfloor n/p \rfloor] - \pi(p - 1)$. Running a second Lucy pass on prime sums (modulo 10^9) supplies the large-prime tail. Both passes cost $O(n^{3/4})$. As checks, $S(100) = 1257$, and the result matches a direct sieve for n up to 10^6 .

Problem 530: GCD of Divisors

Solution: 207366437157977206

With $f(n) = \sum_{d|n} \gcd(d, n/d)$ and $F(k) = \sum_{n=1}^k f(n)$, we need $F(10^{15})$.

Summing f over $n \leq N$ is the same as summing $\gcd(d, e)$ over every ordered pair with $de \leq N$, since each n contributes exactly its divisor pairs $(d, n/d)$. Using $\gcd(d, e) = \sum_{g|d, g|e} \varphi(g)$ and writing $d = ga, e = gb$ (so $de = g^2 ab$),

$$F(N) = \sum_{de \leq N} \gcd(d, e) = \sum_g \varphi(g) \cdot \#\{(a, b) : ab \leq \lfloor N/g^2 \rfloor\} = \sum_{g^2 \leq N} \varphi(g) D(\lfloor N/g^2 \rfloor),$$

where $D(M) = \sum_{a \leq M} \lfloor M/a \rfloor$ is the divisor-summatory function. Each $D(M)$ is evaluated in $O(\sqrt{M})$ by the hyperbola method, and a linear φ -sieve up to $\sqrt{N}$ supplies the totients, for a total cost of about $O(\sqrt{N} \log N)$. As checks, $F(10) = 32$ and $F(1000) = 12776$.

Problem 531: Chinese Leftovers

Solution: 4515432351156203105

Let $g(a, n, b, m)$ be the smallest non-negative x with $x \equiv a \pmod{n}$ and $x \equiv b \pmod{m}$, or 0 if no such x exists, and let $f(n, m) = g(\varphi(n), n, \varphi(m), m)$. We need $\sum f(n, m)$ over $10^6 \leq n < m < 1005000$.

This is a two-modulus Chinese remainder computation that need not assume coprimality. Writing $G = \gcd(n, m)$, a solution exists iff $G \mid (a - b)$, and then it is unique modulo $\text{lcm}(n, m)$:

$$x = a + n \cdot (((b - a)/G) \cdot (n/G)^{-1} \bmod (m/G)),$$

where the inverse is taken modulo m/G (which is coprime to n/G). We sieve φ over the small window $[10^6, 1005000)$ and evaluate the roughly $1.25 \cdot 10^7$ pairs directly, accumulating the smallest solutions. The given values $g(2, 4, 4, 6) = 10$ and $g(3, 4, 4, 6) = 0$ check the congruence solver.

Problem 540: Counting Primitive Pythagorean Triples

Solution: 500000000002845

$P(n)$ counts primitive Pythagorean triples $a < b < c \leq n$. Primitive triples are parametrised bijectively by pairs (m, k) with $m > k \geq 1$, $\gcd(m, k) = 1$ and m, k of opposite parity, through $c = m^2 + k^2$. So $P(N)$ counts such pairs with $m^2 + k^2 \leq N$.

Look at the coprime pairs with $m > k \geq 1$ and $m^2 + k^2 \leq N$. Being coprime rules out both even, so each is either both-odd or mixed-parity, and the primitive ones are exactly the mixed-parity ones. Therefore $P(N) = C - D$ with C the count of all coprime pairs and D the count of coprime both-odd pairs. Removing the coprimality constraint by Möbius inversion,

$$C = \sum_d \mu(d) T(\lfloor N/d^2 \rfloor), \quad D = \sum_{d \text{ odd}} \mu(d) T_{\text{odd}}(\lfloor N/d^2 \rfloor),$$

where $T(X) = \#\{(a, b) : a > b \geq 1, a^2 + b^2 \leq X\}$ and T_{odd} restricts to both a, b odd. Each of T and T_{odd} is read off a quarter-disc lattice count in $O(\sqrt{X})$ (summing $\lfloor \sqrt{X - a^2} \rfloor$ over a), and the Möbius sums need $\mu(d)$ for $d \leq \sqrt{N}$. Checks: $P(20) = 3$ and $P(10^6) = 159139$.

Problem 549: Divisibility of Factorials

Solution: 476001479068717

Let $s(n)$ be the smallest m with $n \mid m!$, and $S(n) = \sum_{i=2}^n s(i)$; we need $S(10^8)$.

Writing $n = \prod p^e$, the factorial $m!$ must absorb each prime power independently, and the binding requirement is the largest, so $s(n) = \max_{p^e \parallel n} s(p^e)$. For a single prime power, $s(p^e)$ is the smallest multiple m of p at which $v_{p(m!)}$ reaches e (the exponent $v_{p(m!)}$ only grows as m passes multiples of p).

This gives a single modified sieve. A slot still holding 0 marks a prime p ; walking $m = p, 2p, 3p, \dots$ and accumulating $c = v_{p(m!)}$, each time c climbs past an integer k the value $s(p^k)$ equals the current m , so we raise $s[j]$ to m for every multiple j of p^k . Composite slots are filled in by their smallest prime before they are reached, exactly as in the sieve of Eratosthenes, and summing the array yields $S(N)$. The total update work is $\sum_p \sum_k N/p^k = O(N \log \log N)$. As a check, $S(100) = 2012$.

Problem 587: Concave Triangle

Solution: 2240

Let the bottom-left of the rectangle be at $(0, 0)$ and the circles to have radius 1.

The area of the L-section is $1 - \frac{\pi}{4}$.

The diagonal line has slope $\frac{1}{n}$, and the intersection of the line with the first circle has coordinate:

$$x = \frac{-\sqrt{2} \cdot n^{\frac{3}{2}} + n^2 + n}{n^2 + 1}$$
$$y = \frac{x}{n}$$

The concave triangle can be split into a triangle and a curved section.

The area of the triangle is $\text{TriangleArea} = \frac{1}{2}xy$.

Let:

$$\text{CircleInt}(a, b) = \int_a^b (\sqrt{1 - x^2}) \, dx$$
$$= \frac{1}{2} \left(x\sqrt{1 - x^2} + \sin^{-1}(x) \right) \Big|_a^b$$

Then the area of the curved section is $\text{SectionArea} = (1 - x) - \text{CircleInt}(0, 1 - x)$.

And so the ratio is:

$$\frac{\text{TriangleArea} + \text{SectionArea}}{1 - \frac{\pi}{4}}$$

Problem 600: Integer Sided Equiangular Hexagons

Solution: 2668608479740672

An equiangular hexagon has every interior angle 120° . Extending three alternate sides until they meet circumscribes the hexagon in an equilateral triangle of side x , slicing off three corner equilateral triangles of sides a, b, c . The six hexagon sides are then a, b, c together with $x - a - b, x - b - c, x - a - c$, and the perimeter is

$$3x - (a + b + c).$$

A hexagon sits inside two such triangles (one per orientation); always taking the smaller forces $x \geq a + b + c$, so each hexagon is counted exactly once. Counting up to congruence, the three corner cuts are unordered, so impose $1 \leq a \leq b \leq c$.

Group the count by $s = a + b + c$. The number of corner triples with that sum is the number of partitions of s into exactly three positive parts, which is the nearest integer to $s^2/12$. For each such triple the triangle side runs over $s \leq x \leq \lfloor (n + s)/3 \rfloor$ to keep the perimeter $\leq n$, giving $\lfloor (n + s)/3 \rfloor - s + 1$ hexagons. Hence

$$H(n) = \sum_{s=3}^{\lfloor n/2 \rfloor} \text{round}(s^2/12) \cdot (\lfloor (n + s)/3 \rfloor - s + 1),$$

which reproduces $H(6) = 1$, $H(12) = 10$, $H(100) = 31248$ and yields $H(55106)$.

Problem 601: Divisibility Streaks

Solution: 1617243

Define $\text{streak}(n)$ as the smallest $k \geq 1$ for which $k + 1$ does not divide $n + k$. Since $(k + 1) \mid (n + k)$ is the same as $n \equiv 1 \pmod{k + 1}$,

$$\text{streak}(n) \geq s \iff n \equiv 1 \pmod{j} \text{ for } j = 2, \dots, s \iff n \equiv 1 \pmod{L_s},$$

where $L_s = \text{lcm}(1, 2, \dots, s)$. Counting the n with $1 < n < N$ that hit a given residue, the number with $\text{streak}(n) = s$ is

$$P(s, N) = \lfloor (N - 2)/L_s \rfloor - \lfloor (N - 2)/L_{s+1} \rfloor,$$

and the answer is $\sum_{i=1}^{31} P(i, 4^i)$. As checks, $P(3, 14) = 1$ and $P(6, 10^6) = 14286$.

Problem 602: Product of Head Counts

Solution: 269496760

If Alice's first Head lands on her r -th toss (probability $p^{r-1}(1-p)$, with p the Tail probability), every friend has tossed $r-1$ times, so each friend's Head count is Binomial($r-1, 1-p$) with mean $(r-1)(1-p)$. The n friends are independent, so writing $q = 1-p$ and $m = r-1$,

$$e(n, p) = \sum_{m=0}^{\infty} p^m q \cdot (mq)^n = q^{n+1} \sum_{m=0}^{\infty} m^n p^m.$$

The standard generating function $\sum_{m \geq 0} m^n x^m = A_n(x)/(1-x)^{n+1}$, where A_n is the Eulerian polynomial, cancels the q^{n+1} factor exactly, leaving $e(n, p) = A_n(p)$. Therefore $c(n, k)$ is the Eulerian number $\langle\langle n; k-1 \rangle\rangle$, the number of permutations of n elements with $k-1$ ascents. (Check: $A_3(p) = p + 4p^2 + p^3$ matches the given $e(3, p)$.)

Evaluate it with the closed form

$$\langle\langle n; m \rangle\rangle = \sum_{j=0}^m (-1)^j \binom{n+1}{j} (m+1-j)^n,$$

all modulo the prime $10^9 + 7$. The binomial is built incrementally, the inverses $1, \dots, m$ come from the linear recurrence $\text{inv}[i] = -(p/i) \text{inv}[p \bmod i]$, and each $(m+1-j)^n$ is a fast modular power; products stay inside int64 because $(10^9 + 7)^2 < 2^{63}$. This confirms $c(100, 40) \equiv 986699437$ and produces $c(10^7, 4 \cdot 10^6)$.

Problem 603: Substring Sums of Prime Concatenations

Solution: 879476477

For a number with digits $d_0 d_1 \dots d_{L-1}$, the digit d_i appears in every substring whose start lies in $\{0, \dots, i\}$ and whose end $e \geq i$, contributing place value 10^{e-i} . Summing the $(i+1)$ start choices and the geometric series over e gives

$$S = \sum_{i=0}^{L-1} d_i (i+1) \frac{10^{L-i} - 1}{9},$$

which reproduces $S(2024) = 2304$.

Now $C = C(n, k)$ is the period- D repetition of $P = P(n)$, where D is the digit length of P . Write the global index as $i = bD + r$ with block $b \in \{0, \dots, k-1\}$ and offset $r \in \{0, \dots, D-1\}$, so $d_i = p_r$ and $L = kD$. With $g = 10^D$, the place-value term becomes $10^{L-i} = g^{k-b} 10^{-r}$, and substituting $c = k - b$ turns the block sum into the geometric series

$$A = \sum_{c=1}^k g^c, \quad B = \sum_{c=1}^k c g^c,$$

both available in closed form modulo $10^9 + 7$. Separating the parts that depend on r leaves four digit sums over a single copy of P ,

$$s_1 = \sum_r p_r 10^{-r}, \quad s_2 = \sum_r p_r r 10^{-r}, \quad s_3 = \sum_r p_r, \quad s_4 = \sum_r p_r r,$$

and the final value is $(\alpha s_1 + A s_2 - \beta s_3 - k s_4)/9$ with $\alpha = (kD + 1)A - DB$ and $\beta = D \binom{k}{2} + k$. The four sums are accumulated in one pass over the ≈ 6.8 million digits of the first 10^6 primes; everything else is $O(1)$. The block formula was checked against brute force on small repetitions, including $S(C(7, 3))$.

Problem 604: Convex Path in Square

Solution: 1398582231101

A strictly increasing convex function passing through lattice points has step vectors (between consecutive points) with strictly increasing slopes, all with $\Delta x \geq 1$ and $\Delta y \geq 1$. Distinct slopes force distinct primitive vectors (a, b) with $\gcd(a, b) = 1$, and using the primitive representative of each slope minimises displacement. So the task is to pick as many distinct primitive vectors (a, b) , $a, b \geq 1$, as possible subject to $\sum a \leq N$ and $\sum b \leq N$; the answer is that maximum plus one (points = edges + 1).

To maximise the count we add vectors in increasing order of $a + b$. The primitive vectors with $a + b = s$ are the pairs $(a, s - a)$ for a coprime to s , so there are $\varphi(s)$ of them, and by the symmetry $a \leftrightarrow s - a$ they cost $s\varphi(s)/2$ to each of the two budgets. Taking all full levels $s = 2, \dots, L$ that fit leaves an equal leftover $R = N - \sum_{s \leq L} s\varphi(s)/2$ in each budget; at level $s = L + 1$ each additional vector spends a from one axis and $s - a$ from the other, so a balanced selection fits $\lfloor 2R/(L + 1) \rfloor$ more. Hence

$$F(N) = 1 + \sum_{s=2}^L \varphi(s) + \lfloor 2R/(L + 1) \rfloor.$$

A totient sieve to about $2.1 \cdot 10^6$ covers $N = 10^{18}$. The formula was checked against an exact two-dimensional knapsack for small N and reproduces $F(100) = 30$ and $F(50000) = 1898$.

Problem 605: Pairwise Coin-Tossing Game

Solution: 59992576

In each round only the player shared with the next round can possibly win two in a row, so relabel round r by the bit $s_r = 1$ if that shared player wins, $s_r = 0$ otherwise. The s_r are independent fair bits, and the game ends the first time the pattern 1,0 appears (the shared player wins round r then round $r + 1$). A binary string avoids 10 exactly when it is $0^a 1^b$, so the first 10 ends at position $r + 1$ with probability

$$\Pr(T = r) = r \cdot 2^{-(r+1)},$$

(r choices for the $0^a 1^b$ prefix of length r , each of probability 2^{-r} , times $1/2$ for the closing 0). The winner is the player shared between rounds r and $r + 1$, namely player $(r \bmod n) + 1$, so player k wins iff $r \equiv k - 1 \pmod{n}$. Summing the arithmetic-geometric series over $r \equiv m \pmod{n}$ with $m = k - 1$, and writing $D = 2^n - 1$,

$$P_n(k) = \frac{2^{n-m-1}(mD + n)}{D^2}.$$

This reproduces $P_3(1) = 12/49$ and $P_6(2) = 368/1323$. For the reduced fraction, D is odd so the 2^{n-m-1} is coprime to it, and any common prime of numerator and denominator must divide $\gcd(n, D)$. Since $n = 10^8 + 7$ is prime and $2^n \equiv 2 \pmod{n}$ gives $\gcd(n, D) = 1$, the fraction is already in lowest terms. Hence $M_n(k) = 2^{n-m-1}(mD + n)D^2$, evaluated modulo 10^8 .

Problem 606: Gozinta Chains II

Solution: 158452775

A gozinta chain $1 = d_0 \mid d_1 \mid \dots \mid d_t = n$ corresponds to an ordered factorisation of n into factors > 1 , so $g(n)$ depends only on the prime signature (the multiset of exponents).

For a signature with exponents e_i and $k = \sum e_i$,

$$g = \sum_{t=1}^k \sum_{j=0}^t (-1)^j \binom{t}{j} \prod_i \binom{e_i + t - j - 1}{e_i}.$$

For a single exponent k this is 2^{k-1} , which is the minimum over all signatures of that total, so any signature with $k \geq 9$ has $g \geq 256 > 252$. Searching signatures with $k \leq 8$ shows the **unique** one with $g = 252$ is $(3, 3)$: numbers of the form $k = (pq)^3$ for distinct primes p, q .

Thus $k \leq N$ iff $pq \leq \lfloor N^{1/3} \rfloor$, and with $B = \lfloor N^{1/3} \rfloor$,

$$S(N) = \sum_{p < q, pq \leq B} (pq)^3 = \sum_{p \leq \sqrt{B}} p^3 (\Pi_3(\lfloor B/p \rfloor) - \Pi_3(p)),$$

where $\Pi_3(x) = \sum_{q \leq x \text{ prime}} q^3$. For $N = 10^{36}$ we have $B = 10^{12}$, so the inner cube-sums run up to $5 \cdot 10^{11}$; these are obtained for every needed argument $\lfloor B/p \rfloor$ at once by a Lucy_Hedgehog sieve carried out modulo 10^9 in $O(B^{3/4})$ time. The reduction was checked against brute force on $S(10^6)$ and $S(10^{12})$.

Problem 607: Marsh Crossing

Solution: 13.1265108586

Minimise the crossing time over a diagonal marsh of banded speeds by binary-searching the entry angle and applying Snell's law at each band boundary.

Problem 608: Divisor Sums

Solution: 439689828

Write $m = \prod_p p^{a_p}$. A divisor $d \mid m$ has p -exponent $b_p \in [0, a_p]$, and $\sigma_0(kd) = \prod_p (v_p(k) + b_p + 1)$, so summing over $d \mid m$ factors prime-by-prime:

$$\sum_{d \mid m} \sigma_0(kd) = \prod_p h_p(v_p(k)), \quad h_p(e) = \sum_{b=0}^{a_p} (e + b + 1) = \binom{a_p + 2}{2} + (a_p + 1)e.$$

Dividing each h_p by $h_p(0) = \binom{a_p + 2}{2}$ gives a multiplicative $\hat{g}$ with $g(k) = C_0 \hat{g}(k)$, $C_0 = \prod_p \binom{a_p + 2}{2}$. Writing $\hat{g} = 1 * c$ shows $c(p^e) = 2/(a_p + 2)$ (constant in $e \geq 1$), and writing $c = b * 1$ in turn shows b is supported on squarefree products of primes dividing m . Hence $\hat{g} = b * \sigma_0$ and

$$D(m, n) = \sum_{k=1}^n g(k) = \sum_e W(e) \mathcal{D}(\lfloor n/e \rfloor), \quad \mathcal{D}(M) = \sum_{t \leq M} \sigma_0(t),$$

where e runs over squarefree products of distinct primes dividing m with $e \leq n$, and the integer weight is

$$W(e) = \prod_{p \mid e} \left(-\binom{a_p + 1}{2} \right) \prod_{p \mid m, p \nmid e} \binom{a_p + 2}{2}.$$

For $m = 200!$ the a_p come from Legendre's formula over the 46 primes $p \leq 200$. The divisor-summatory $\mathcal{D}(M) = 2 \sum_{i \leq \sqrt{M}} \lfloor M/i \rfloor - \lfloor \sqrt{M} \rfloor^2$ costs $O(\sqrt{M})$. A depth-first search over the 46 primes (pruning once $ep > n$) visits about $4.5 \cdot 10^7$ products with total cost $\sum_e \sqrt{n/e} \approx 5.3 \cdot 10^8$, all modulo $10^9 + 7$. The reduction reproduces $D(3!, 10^2) = 3398$ and $D(4!, 10^6) = 268882292$.

Problem 609: π Sequences

Solution: 172023848

A π sequence starting from u_0 is a prefix of the chain $u_0, \pi(u_0), \pi(\pi(u_0)), \dots$ truncated at any term that is still ≥ 1 ; for $u_0 = 10$ the chain $10 \rightarrow 4 \rightarrow 2 \rightarrow 1$ yields the three sequences $(10, 4)$, $(10, 4, 2)$, $(10, 4, 2, 1)$. With $c(u)$ the number of non-prime entries and $p(n, k) = \#\{u : u_0 \leq n, c(u) = k\}$, the answer is $P(n) = \prod_{k:p(n,k)>0} p(n, k)$.

Group the starting values by $v = \pi(u_0)$. The u_0 with $\pi(u_0) = v$ form the block $[p_v, p_{v+1})$, which contains exactly one prime (p_v) and the remaining $N_v = \min(p_{v+1}, n+1) - p_v - 1$ composites. Every u_0 in the block shares the same downstream chain $v \rightarrow \pi(v) \rightarrow \dots \rightarrow 1$; walking it gives cumulative non-prime counts S_t . The prime $u_0 = p_v$ then contributes one sequence with $c = S_t$ for each prefix length t , while each composite u_0 contributes $c = S_t + 1$. Hence

$$p(n, k) = \sum_{v=1}^{\pi(n)} (\#\{t : S_t(v) = k\} + N_v \cdot \#\{t : S_t(v) + 1 = k\}).$$

Sieving to $n = 10^8$ gives the primality and (via a prefix sum up to $\pi(n) = 5761455$) the values $\pi(w)$ needed to walk each chain, which collapses to 1 in about a dozen steps. Accumulating the histogram and multiplying its positive entries modulo $10^9 + 7$ matches the given $P(10) = 648$ and $P(100) = 31038676032$.

Problem 610: Roman Numerals II

Solution: 319.30207833

Each draw gives a letter (probability 0.14) or # (0.02); invalid appends are skipped and # stops, outputting the current numeral's value. While only leading Ms have been written, all seven letters extend the numeral validly, so the draw never skips: with probability 0.02 we stop (value $1000a$), with 0.14 we add another M, and with 0.14 each of I, V, X, L, C, D we enter the sub-1000 part. Because the thousands add a flat $1000a$, the expected value is linear in a and

$$H(0) = (0.14G_6 + 140)/0.86, \quad G_6 = \sum_{s \in \{1, 5, 10, 50, 100, 500\}} g(s),$$

where $g(s)$ is the expected final sub-1000 value from state s . Appending a letter to a minimal numeral strictly increases its value, so the sub-1000 states 0..999 form a DAG; solving from large s downward,

$$g(s) = \frac{0.02s + 0.14 \sum_t g(t)}{0.02 + 0.14 \deg(s)},$$

with the successors t being the values whose minimal string is $\text{minimal}(s)$ plus one letter. This yields 319.30207833.

Problem 612: Friend Numbers

Solution: 819963842

Two numbers are friends if their base-10 digit sets share a digit. We want $f(10^{18})$, the number of friend pairs $1 \leq p < q < 10^{18}$, modulo 1000267129, and count the complement (pairs sharing no digit).

For a digit set $U \subseteq \{0, \dots, 9\}$, let $D[U]$ be the number of integers in $[1, n)$ whose digits all lie in U . Because $n = 10^{18}$ is a power of ten, $[1, n)$ is exactly the positive integers with at most $L = 18$ digits, so with $a = |U \setminus \{0\}|$ allowed nonzero digits and $b = |U|$ allowed digits,

$$D[U] = \sum_{\ell=1}^L a b^{\ell-1}.$$

D is the subset-sum (zeta) transform of $\text{cnt}[S] = \#\{x \in [1, n) : \text{digit set of } x = S\}$, so cnt is recovered by the Möbius inverse. Two numbers are non-friends exactly when their digit sets are disjoint, and

$$\sum_S \text{cnt}[S] D[\overline{S}]$$

counts ordered disjoint pairs (never pairing x with itself, each unordered pair twice).

Hence $f(n) = \binom{n-1}{2} - \frac{1}{2} \sum_S \text{cnt}[S] D[\overline{S}]$, evaluated exactly and then reduced. Check: $f(100) = 1539$.

Problem 613: Pythagorean Ant

Solution: 0.3916721504

An ant at interior point P moving in a uniformly random direction leaves through a given side with probability equal to the angle that side subtends at P , divided by 2π (the three subtended angles sum to 2π). Averaging over a uniform P , the probability of leaving by the hypotenuse is

$$\frac{1}{2\pi \cdot \text{Area}} \iint_T \angle APB \, dA,$$

with the 3-4-5 triangle placed as $A = (4, 0)$, $B = (0, 3)$, right angle at the origin, $\text{Area} = 6$. Using $\angle APB = \pi/2 + \arctan(x/(3-y)) + \arctan(y/(4-x))$, the inner integral is elementary ($\int \arctan(t/u) \, dt = t \arctan(t/u) - (u/2) \ln(u^2 + t^2)$, whose singular parts cancel), leaving a smooth one-dimensional integral evaluated by Gauss-Legendre quadrature. The result rounds to 0.3916721504.

Problem 614: Special Partitions 2

Solution: 130694090

A special partition uses distinct parts that are odd or divisible by 4, i.e. parts $m \not\equiv 2 \pmod{4}$. Therefore $\sum_{i=0}^n P(i)$ is the number of subsets of such parts with sum $\leq n$, the coefficient sum of $G(x) = \prod_{m \not\equiv 2(4)} (1 + x^m)$. Using $1 + x^m = (1 - x^{2m})/(1 - x^m)$ and grouping the residue classes, this telescopes to

$$G(x) = \frac{E_2^2 E_8}{E_1 E_4^2}, \quad E_k = \prod_{j \geq 1} (1 - x^{kj}) = \sum_i (-1)^i x^{ki(3i-1)/2},$$

each E_k being generalized-pentagonal (only $O(\sqrt{n/k})$ nonzero terms). The series of G to degree $n = 10^7$ is built by sparse multiplications (one sequential sweep per pentagonal term) and divisions (a blocked recurrence: far terms applied by streaming sweeps over already-finalised values, near terms by the in-window recurrence), all modulo $10^9 + 7$ with `int32` storage to limit memory traffic. Summing the coefficients (and subtracting the empty partition) reproduces $P(1) = 1$, $P(10) = 3$, $P(100) = 37076$, $P(1000) = 3699177285485660336$.

Problem 615: The Millionth Number with at Least One Million Prime Factors

Solution: 108424772

Write a number with $\Omega(n) \geq K$ ($K = 10^6$) as $n = 2^a u$ with u odd. Then $\Omega(n) = a + \Omega(u) \geq K$ means $a \geq K - \Omega(u)$, so

$$n/2^K = u \cdot 2^{a-K} = \text{base}(u) \cdot 2^t, \quad \text{base}(u) = u/2^{\Omega(u)}, \quad t \geq 0.$$

Thus the numbers with $\Omega \geq K$ correspond order-preservingly to the values $\text{base}(u) \cdot 2^t$ over odd u and $t \geq 0$. Seeding every odd u with $\text{base}(u) \leq L$ (found by a depth-first product of odd primes, pruning once the base exceeds L , which needs primes up to $2L$) and merging the geometric chains $\text{base}(u), 2 \text{ base}(u), 4 \text{ base}(u), \dots$ in a min-heap enumerates the ratios in increasing order. Comparisons are exact via the integer key $u \cdot 2^{E+\text{off}}$ with $E = t - \Omega(u)$. The 10^6 -th popped ratio gives $n = 2^{K+E}u$, reduced modulo 123454321. With $L = 2 \cdot 10^5$ the millionth ratio is $86597.5 \leq L$, confirming no seed was missed; the verification on small K reproduces the example (the 5-th number with $\Omega \geq 5$ is 80).

Problem 616: Creative Numbers

Solution: 310884668312456458

Alice can only act on $L = \{n\}$ if $n = a^b$ is a perfect power, so non-powers are immediately stuck. Among perfect powers, two families fail to be creative. If $n = p^q$ with p, q both prime, the single split gives $\{p, q\}$, two unsplittable primes that can only recombine, so only finitely many lists are reachable. The single value $16 = 2^4$ also fails: its only splits are $\{2, 4\}$, and since $4 = 2^2 \rightarrow \{2, 2\}$ one can amass at most three 2's, never enough to build 256 and expose an odd factor — everything stays a power of two. Every other perfect power is creative: it can reach a list with an odd prime and enough material to grow without bound, hence reach any $m > 1$. For instance $256 = 2^8 \rightarrow \{2, 8\}$ and $8 = 2^3 \rightarrow \{2, 3\}$ already exposes 3, and from $36 = 6^2 \rightarrow \{6, 2\}$, $2^6 = 64 = 4^3 \rightarrow \{4, 3\}$, $4 \rightarrow \{2, 2\}$ grows the list. Therefore

$$\sum_{\text{creative } \leq 10^{12}} = \sum_{\text{perfect powers}} - \sum_{p, q \text{ prime}} p^q - 16,$$

computed by collecting the distinct perfect powers in a set and subtracting the prime-power-of-a-prime values and the lone exception 16.

Problem 622: Riffle Shuffles

Solution: 3010983666182123972

A perfect riffle (faro out-) shuffle of an even deck of size n moves the card at position i to position $2i \bmod (n-1)$, fixing the two ends. The number of shuffles needed to restore the deck is therefore the multiplicative order of 2 modulo $n-1$:

$$s(n) = \text{ord}_{n-1}(2).$$

We want the sum of all n with $s(n) = 60$. Setting $m = n - 1$ (odd), the condition is $\text{ord}_m(2) = 60$: equivalently $m \mid 2^{60} - 1$ while m divides none of $2^t - 1$ for the maximal proper divisors $t \in \{30, 20, 12\}$ of 60 (any proper divisor of 60 divides one of these). Every divisor of $2^{60} - 1$ is odd, so $n = m + 1$ is automatically even. We factor $2^{60} - 1 = 3^2 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 31 \cdot 41 \cdot 61 \cdot 151 \cdot 331 \cdot 1321$, enumerate its divisors, keep those of exact order 60, and sum $n = m + 1$. The analogous computation reproduces the given $s(n) = 8$ total of 412.

Problem 625: Gcd Sum

Solution: 551614306

We need $G(N) = \sum_{j=1}^N \sum_{i=1}^j \gcd(i, j)$ for $N = 10^{11}$, modulo 998244353.

Replacing $\gcd(i, j) = \sum_{d \mid i, d \mid j} \varphi(d)$ and grouping the pairs by the common divisor d (so $i = da, j = db$ with $1 \leq a \leq b \leq \lfloor N/d \rfloor$, of which there are $T(M) = M(M+1)/2$ for $M = \lfloor N/d \rfloor$),

$$G(N) = \sum_{d \geq 1} \varphi(d) T(\lfloor N/d \rfloor).$$

Grouping d by the value $M = \lfloor N/d \rfloor$, each block contributes $T(M)(\Phi(\text{hi}) - \Phi(\text{lo} - 1))$ where Φ is the summatory totient. The values $\Phi(\lfloor N/d \rfloor)$ at the $O(\sqrt{N})$ distinct arguments are found by the standard recurrence

$$\Phi(v) = T(v) - \sum_{d \geq 2} \Phi(\lfloor v/d \rfloor),$$

evaluated bottom-up with a φ -sieve for small arguments, all modulo the prime. The whole computation is $O(N^{3/4})$. Checks: $G(10) = 122$, $G(1000) = 2475190$, $G(10^4) = 317257140$.

Problem 657: Incomplete Words

Solution: 219493139

A word over an alphabet of α letters is incomplete if it omits at least one letter; $I(\alpha, n)$ counts incomplete words of length at most n . We want $I(10^7, 10^{12})$ modulo $10^9 + 7$.

Classify words by the number i of distinct letters they actually use. The number of words of length $\leq M$ confined to a fixed i -letter sub-alphabet is $\sum_{j=0}^M i^j$, and there are $\binom{N}{i}$ such sub-alphabets; inclusion-exclusion over i then gives

$$I(N, M) = \sum_{i=0}^{N-1} (-1)^{N-1-i} \binom{N}{i} \sum_{j=0}^M i^j.$$

The geometric inner sum is $(i^{M+1} - 1)/(i - 1)$ for $i \geq 2$, equals $M + 1$ for $i = 1$, and equals 1 for $i = 0$. The powers i^{M+1} are taken modulo p with the exponent reduced modulo $p - 1$ by Fermat, and the binomial coefficients are streamed multiplicatively, so the whole sum is one $O(N \log M)$ pass. Checks: $I(3, 0) = 1$, $I(3, 2) = 13$, $I(3, 4) = 79$.

Problem 668: Square Root Smooth Numbers

Solution: 2811077773

A number is square-root smooth when all of its prime factors are strictly below its square root; we count those up to $N = 10^{10}$.

A number n fails to be square-root smooth exactly when it has a prime factor $p \geq \sqrt{n}$. Such a p is unique — two primes that large would multiply past n — and it is the largest prime factor, with $n = pm$ where $m = n/p \leq \sqrt{n} \leq p$. Conversely each pair (prime p , integer $1 \leq m \leq p$ with $pm \leq N$) yields a distinct non-smooth number, so

$$\#\{\text{not smooth} \leq N\} = \sum_{p \leq N} \min(p, \lfloor N/p \rfloor), \quad S(N) = N - \sum_{p \leq N} \min(p, \lfloor N/p \rfloor).$$

Split at $R = \lfloor \sqrt{N} \rfloor$: primes $p \leq R$ contribute p (a direct sum of primes), while primes $p > R$ contribute $\lfloor N/p \rfloor$, grouped as $\sum_{q=1}^{R-1} q(\pi(N//q) - \pi(N//(q+1)))$ with the prime-counting values supplied by a Lucy_Hedgehog sieve in $O(N^{3/4})$. As a check, $S(100) = 29$.

Problem 686: Powers of Two with Leading Digits

Solution: 193060223

Using the fractional part of $j \cdot \log_{10} 2$ to detect a leading block of 123, count the powers of two that start with 123 until the 678910th.

Problem 692: Siegbert and Jo

Solution: 842043391019219959

The game is Fibonacci Nim. Writing the Fibonacci numbers as $F_1 = 1, F_2 = 2, F_3 = 3, F_4 = 5, \dots$, the smallest opening move that still guarantees a win, $H(N)$, is the smallest Fibonacci number appearing in the Zeckendorf representation of N (every integer is uniquely a sum of non-consecutive Fibonacci numbers). The examples confirm this: $18 = 13 + 5$ gives $H(18) = 5$, while $H(8) = 8$ since 8 is itself Fibonacci.

We need $G(n) = \sum_{k=1}^n H(k)$. Let $S(n) = G(n)$ and let F_m be the largest Fibonacci number with $F_m \leq n$. Each $k \in [F_m, n]$ equals $F_m + r$ with $0 \leq r \leq n - F_m < F_{m-1}$; for $r = 0$ the smallest part is F_m , and for $r \geq 1$ the smallest part of k is the smallest part of r . Therefore

$$S(n) = A(m) + F_m + S(n - F_m), \quad A(m) := S(F_m - 1),$$

and $A(m)$ satisfies its own recurrence $A(m) = A(m-1) + F_{m-1} + A(m-2)$ with $A(1) = 0, A(2) = 1$. Both run in $O(\log n)$. Check: $G(13) = 43$.

Problem 694: Cube-full Divisors

Solution: 1339784153569958487

A number is cube-full if every prime dividing it does so to power at least 3 (with 1 cube-full). With $s(n)$ the number of cube-full divisors of n and $S(N) = \sum_{i=1}^N s(i)$, swapping the order of summation gives

$$S(N) = \sum_{d \text{ cube-full}, d \leq N} \lfloor N/d \rfloor,$$

because a fixed cube-full d divides exactly $\lfloor N/d \rfloor$ integers up to N .

Every cube-full number factors uniquely as $d = a^3 b^4 c^5$ with b, c squarefree and $\gcd(b, c) = 1$: split the primes of d by exponent modulo 3, sending $e \equiv 0$ into a , $e \equiv 1$ into the b^4 factor, and $e \equiv 2$ into the c^5 factor. Hence

$$S(N) = \sum_{b, c \text{ "sqfree, coprime"} a \geq 1, a^3 b^4 c^5 \leq N} \lfloor N/(a^3 b^4 c^5) \rfloor,$$

where the inner term is $\lfloor M/a^3 \rfloor$ with $M = \lfloor N/(b^4 c^5) \rfloor$. Since cube-full numbers number about $N^{1/3}$, the total work is roughly $N^{1/3} \zeta(4/3) \zeta(5/3) \approx 7 \cdot 10^6$ for $N = 10^{18}$. Checks: $S(16) = 19$, $S(100) = 126$, $S(10000) = 13344$.

Problem 700: Eulercoin

Solution: 1517926517777556

With $N = 1504170715041707$ and the prime $M = 4503599627370517$, the sequence $Nn \bmod M$ produces an *Eulercoin* whenever a term is strictly smaller than every earlier term. Brute-forcing n upward finds the early (large-valued) coins quickly but stalls long before the small ones appear, so the two ends are searched separately.

Walking $n = 1, 2, \dots$ and tracking the running minimum of $Nn \bmod M$ (updated as $v \leftarrow (v + N) \bmod M$) yields the coins of large value, which occur at small n . For the small-valued coins, observe that a value c first appears at step $n_c = cN^{-1} \bmod M$; the inverse exists because M is prime. Such a c is a coin exactly when it reaches the sequence before every smaller value, i.e. when $n_c < n_{c'}$ for all $0 < c' < c$. Scanning c upward while tracking the running minimum of n_c (updated as $n_c \leftarrow (n_c + N^{-1}) \bmod M$) therefore lists every small-valued coin. Both passes are plain increment-and-compare loops with no multiplication, so nothing overflows; the union of the two coin sets, summed, is the answer. A crossover threshold of $2 \cdot 10^7$ comfortably covers both regimes.

Problem 701: Random Connected Area

Solution: 13.51099836

Each of the $W \times H = 49$ cells is black with probability $1/2$, and we want the expected size of the largest edge-connected black region, $E(7, 7)$. Enumerating all 2^{49} colourings is hopeless, so the grid is swept one row at a time with a connected-component DP.

State

After some rows are processed, only the bottom (frontier) row can still grow into the next row. A state records: the black cells of the frontier row grouped into the connected components that touch it, the accumulated size of each such *active* component, and the largest size among components that have already *finished* (no longer touch the frontier). Each state carries an integer count of the colourings producing it.

Transition

Given a frontier pattern and the black mask of the next row, a union-find merges horizontally adjacent new cells and joins each new black cell to the active component directly above it. New component sizes are the merged old sizes plus the fresh black cells; any old component that no new cell touches is finished, updating the running maximum. The purely structural part of this step – which components merge, which finish – depends only on the frontier pattern and the next mask, not on the sizes, so it is precomputed once for every (pattern, mask) pair. Each state is then packed into one integer (a pattern id, four six-bit component sizes, and a six-bit running maximum) so the hot loop runs under Numba.

After the last row every active component finishes; summing $\text{count} \times (\text{largest size})$ over the final states and dividing by 2^{49} gives $E(7, 7)$. The method reproduces the given $E(2, 2) = 1.875$ and $E(4, 4) = 5.76487732$.

Problem 703: Circular Logic II

Solution: 843437991

The map f sends a length- n bit sequence to its left shift with a new last bit $b_1 \wedge (b_2 \oplus b_3)$, and $S(n)$ counts the functions $T : B^n \rightarrow B$ with $T(x) \wedge T(f(x)) = \text{false}$ for every x . Reading T as the set $\{x : T(x) = \text{true}\}$, the condition says no x and its image $f(x)$ are both chosen: $S(n)$ is the number of *independent sets* in the functional graph of f on 2^n vertices (edges $x - f(x)$). We need $S(20)$ modulo 1001001011.

A functional graph is a disjoint union of components, each a single cycle with in-trees feeding into it, and independent sets multiply across components. Process the trees leaf-first: give every vertex $\text{dp}_0 = \text{dp}_1 = 1$ (ways with it unselected / selected) and, when a vertex v is finished, fold it into its image $p = f(v)$ by

$$\text{dp}_0[p] * = \text{dp}_0[v] + \text{dp}_1[v], \quad \text{dp}_1[p] * = \text{dp}_0[v],$$

the second rule forbidding a selected vertex next to a selected parent. Peeling by in-degree leaves exactly the cycle vertices, each already carrying its trees' contributions $A_v = \text{dp}_0[v]$ and $B_v = \text{dp}_1[v]$. A cycle is then counted by the trace of the product of transfer matrices $M_v = \begin{pmatrix} A_v & B_v \\ A_v & 0 \end{pmatrix}$, which correctly handles self-loops (forcing that vertex unselected) and 2-cycles, and the result is the product over all cycles. Only additions and multiplications occur, so the modulus need not be prime. The method reproduces $S(3) = 35$ and $S(4) = 2118$, and 2^{20} vertices are handled in about a second.

Problem 704: Factors of Two in Binomial Coefficients

Solution: 501985601490518144

Here $g(n, m)$ is the exponent of 2 in $\binom{n}{m}$, $F(n) = \max_m g(n, m)$, and we need $S(N) = \sum_{n=1}^N F(n)$ for $N = 10^{16}$.

By Kummer's theorem $g(n, m)$ equals the number of carries when m and $n - m$ are added in base 2, equivalently $s(m) + s(n - m) - s(n)$, where s is the binary digit sum. Maximising over m asks for the split of n whose binary addition carries the most. A short recursion on the parity of n shows that the maximum of $s(a) + s(b)$ over $a + b = n$, minus $s(n)$, equals the gap between the most and least significant set bits of $n + 1$:

$$F(n) = \lfloor \log_2(n + 1) \rfloor - v_2(n + 1),$$

which matches the given $F(10) = 3$, $F(100) = 6$ and agrees with brute force.

Re-indexing $k = n + 1$ turns the sum into $S(N) = A(N + 1) - B(N + 1)$, where $A(M) = \sum_{k=1}^M \lfloor \log_2 k \rfloor$ and $B(M) = \sum_{k=1}^M v_2(k)$. The first is a sum over geometric blocks of constant $\lfloor \log_2 k \rfloor$, and by Legendre's formula $B(M) = M - \text{popcount}(M)$. Both use exact integer bit operations, reproducing the given $S(100) = 389$ and $S(10^7) = 203222840$.

Problem 705: Total Inversion Count of Divided Sequences

Solution: 480440153

The master sequence $G(N)$ concatenates the decimal digits of all primes below N with every zero digit removed. Replacing each digit by one of its divisors gives a *divided sequence*; $F(N)$ sums the inversion count (pairs $i < j$ whose value at i exceeds the value at j) over all such sequences, and we need $F(10^8) \bmod (10^9 + 7)$.

Treat each position's divisor choice as independent. For a digit d let $\alpha_d(a) = [a \mid d] / \tau(d)$ be the chance of choosing value a , where $\tau(d)$ counts the divisors of d . By linearity over the $T = \prod_i \tau(d_i)$ divided sequences, each potential inversion contributes its probability times T :

$$F = T \sum_{i < j} \sum_{a > b} \alpha_{d_i}(a) \alpha_{d_j}(b).$$

Because only nine digit values occur, this is one left-to-right pass: keep $\text{prefix}[a] = \sum_{i \text{ seen}} \alpha_{d_i}(a)$, and for each new digit d add $\sum_b \alpha_d(b) \sum_{a > b} \text{prefix}[a]$ to the running expectation, then fold d into the prefix and multiply $\tau(d)$ into T . Working modulo the prime $10^9 + 7$ (so the divisions by $\tau(d)$ become modular inverses) and sieving the primes below 10^8 , the Numba digit pass gives the result. The exact-arithmetic version reproduces $F(20) = 3312$ and $F(50) = 338079744$.

Problem 706: 3-Like Numbers

Solution: 884837055

Here $f(n)$ counts the non-empty substrings of the decimal string of n that are divisible by 3, and n is *3-like* when $3 \mid f(n)$. We need $F(10^5)$, the number of 10^5 -digit 3-like numbers, modulo $10^9 + 7$.

A substring is divisible by 3 exactly when its digit sum is, so with prefix sums $S_0 = 0$ and $S_k = (d_1 + \dots + d_k) \bmod 3$ the divisible substrings are the index pairs $a < b$ with $S_a \equiv S_b$. Hence

$$f(n) = \sum_{r=0}^2 \binom{c_r}{2}, \quad c_r = |\{k : S_k = r\}|.$$

Modulo 3 this collapses dramatically: $\binom{c}{2} \equiv 1$ precisely when $c \equiv 2 \pmod{3}$ and is 0 otherwise, so

$$f(n) \equiv |\{r : c_r \equiv 2 \pmod{3}\}| \pmod{3},$$

and n is 3-like iff that count is a multiple of 3 (i.e. 0 or 3). Only the residues $c_r \bmod 3$ matter.

This gives a digit-DP whose state is the current prefix residue together with $(c_0, c_1, c_2) \bmod 3$ — just $3 \cdot 27 = 81$ states. Appending a digit congruent to δ moves the residue and increments the new residue's count; a digit class has weight equal to how many decimal digits lie in it, namely $(4, 3, 3)$ for residues $(0, 1, 2)$, except the leading digit forbids 0 and so uses $(3, 3, 3)$. Iterating 10^5 positions and summing the states whose count of " $\equiv 2$ " residues is a multiple of 3 gives the answer; the recurrence reproduces $F(2) = 30$ and $F(6) = 290898$.

Problem 707: Lights Out

Solution: 652907799

Selecting a cell toggles it and its edge-neighbours, and $F(w, h)$ counts the states of a $w \times h$ grid reducible to all-off. Over $\mathbb{F}_2$ the toggles form a symmetric matrix A , the reachable (solvable) states are exactly its column space, so $F(w, h) = 2^{\text{rank } A} = 2^{wh-d(w,h)}$ with d the nullity. We want $S(199, 199) = \sum_{k=1}^{199} F(199, f_k)$ modulo $10^9 + 7$, where f_k is Fibonacci.

A null vector of A is a "quiet" press pattern; collecting the presses of row i into $r_i \in \mathbb{F}_2^w$ the quiet condition reads $r_{i-1} + Mr_i + r_{i+1} = 0$ with M the $w \times w$ tridiagonal all-ones matrix and boundaries $r_0 = r_{h+1} = 0$. Hence $r_{i+1} = Mr_i + r_{i-1}$ and $r_{h+1} = P_{h+1}(M)r_1$, where the Fibonacci-type polynomials satisfy $P_0 = 0$, $P_1 = 1$, $P_{m+1} = xP_m + P_{m-1}$. So $d(w, h) = \dim \ker P_{h+1}(M)$. An irreducible tridiagonal matrix is non-derogatory, so M is similar to the companion matrix of its characteristic polynomial χ_w and $\mathbb{F}_2^w \cong \mathbb{F}_2[x]/(\chi_w)$ with $P_{h+1}(M)$ acting as multiplication; therefore

$$d(w, h) = \deg \gcd(P_{h+1}(x), \chi_{w(x)}).$$

Here χ_w comes from $D_k = (x+1)D_{k-1} + D_{k-2}$, and $P_{h+1} \bmod \chi_w$ is the top-left entry of $\begin{pmatrix} x & 1 \\ 1 & 0 \end{pmatrix}^h$ in $\mathbb{F}_2[x]/(\chi_w)$, computed by fast exponentiation even though $h = f_k$ is astronomically large. Finally $F = 2^{wh-d}$ is reduced through Fermat's little theorem. The method reproduces every given value, including $F(7, 11) \equiv 270016253$ and $S(5, 7) \equiv 346547294$.

Problem 708: Twos Are All You Need

Solution: 28874142998632109

Replacing every prime factor of n by 2 gives $f(n) = 2^{\Omega(n)}$, where $\Omega(n)$ counts prime factors with multiplicity; this is completely multiplicative with $f(p) = 2$. We need $S(N) = \sum_{n=1}^N f(n)$ for $N = 10^{14}$, far too many terms to sieve directly.

Peel the function with Dirichlet convolutions against the constant 1. Writing $f = 1 * g$ forces $g(p^k) = f(p^k) - f(p^{k-1}) = 2^{k-1}$, and writing $g = 1 * h$ forces $h(p) = 0$ with $h(p^k) = 2^{k-2}$ for $k \geq 2$. Because $h(p) = 0$, the multiplier h is supported only on *squarefull* numbers (every prime exponent at least 2), of which there are merely $O(\sqrt{N})$ below N . Unwinding $f = 1 * 1 * h$ gives

$$S(N) = \sum_{e \text{ squarefull}} h(e) \cdot D(\lfloor N/e \rfloor), \quad D(M) = \sum_{m=1}^M \lfloor M/m \rfloor,$$

where the inner double sum over the two copies of 1 collapses to the divisor-summatory function D , evaluated in $O(\sqrt{M})$ by the hyperbola method.

The squarefull e are enumerated by a depth-first walk over primes $p \leq \sqrt{N} = 10^7$, each taken to an exponent of at least two; since $e \leq N$ has at most about eight distinct prime factors, an explicit fixed-depth frame stack replaces recursion and keeps everything inside Numba. Overflow is avoided by comparing through division ($p^2 > \lfloor N/e \rfloor$) rather than forming ep^2 directly. The method reproduces the given $S(10^8) = 9613563919$ and runs in a few seconds.

Problem 709: Even Stevens

Solution: 773479144

Each day the newest bag is either dropped into the cupboard or used to swallow an even number of the bags currently sitting at top level. A finished packing is therefore a labelled forest in which every bag directly contains an even-sized set of older bags.

We need $f(24680) \bmod 1020202009$, the number of such forests on $n = 24680$ bags.

Relabelling, these are exactly *increasing trees* whose every node has an even number of unordered children. In the increasing-tree framework a single tree has exponential generating function $T(x)$ satisfying $T'(x) = \varphi(T(x))$ with the degree series $\varphi(u) = \sum_{j \geq 0} u^{2j} / (2j)! = \cosh u$, so

$$T' = \cosh T, \quad T(0) = 0.$$

A forest is a set of such trees, so its EGF is e^T and $f(n) = n! [x^n] e^T$.

To avoid composing power series, introduce $C = \cosh T$ and $S = \sinh T$ and differentiate:

$$T' = C, \quad S' = C^2, \quad C' = SC, \quad (e^T)' = C e^T.$$

Reading off coefficients turns each relation into an $O(m)$ convolution for the next term, so all series are built to degree n in $O(n^2)$ time. Every step works modulo the prime 1020202009, with the divisions by $m + 1$ replaced by modular inverses (computed in linear time). Finally $f(n) = n! \cdot [x^n] e^T$. The recurrence reproduces the given $f(4) = 5$ and $f(8) = 1385$, and the full computation runs in a few seconds. The sequence begins 1, 1, 2, 5, 16, 61, 272, 1385,

Problem 710: One Million Members

Solution: 1275000

A *twopal* is a palindromic composition of n that uses at least one part equal to 2, and $t(n)$ counts them; we want the least $n > 42$ with $10^6 \mid t(n)$.

Count by complement: $t(n) = P(n) - Q(n)$, where P counts all palindromic compositions and Q counts those that never use a 2. A palindromic composition is a half-composition followed by its mirror, optionally around a central part, which gives clean generating functions. Allowing every positive part,

$$P(x) = \frac{x + 2x^2}{1 - 2x^2},$$

while forbidding the part 2 (parts drawn from $\{1, 3, 4, 5, \dots\}$) gives, after simplifying the half and centre contributions,

$$Q(x) = \frac{x + x^2 + x^6}{1 - 2x^2 + x^4 - x^6}.$$

Both are rational, so their denominators yield constant-coefficient recurrences

$$P_n = 2P_{n-2}, \quad Q_n = 2Q_{n-2} - Q_{n-4} + Q_{n-6},$$

which are iterated modulo 10^6 from their seed values until $P_n - Q_n \equiv 0$. The series reproduces $t(6) = 4$, $t(20) = 824$ and $t(42) = 1999923$, and the first qualifying index past 42 is 1275000.

Problem 711: Binary Blackboard

Solution: 541510990

Starting from n , the players write binary numbers keeping the running total at most $2n$, so they are jointly splitting a budget of n into a composition; Eric wins when the total count of 1-bits on the board (those of n included) is even. We need $S(N)$, the sum of all $n \leq 2^N$ for which Eric can force a win, modulo $10^9 + 7$, for $N = 12345678$.

Each position is captured by the remaining budget r , whose turn it is, and the parity of move-popcounts so far, so a minimax over r decides every n . Tabulating the winners exposes a purely *syntactic* rule on the binary string of n . If n has even bit-length, Eric wins only for the all-ones number $2^L - 1$. If n has odd bit-length $L = 2j + 1$, drop the leading 1 and read the remaining $2j$ bits as j pairs (most-significant first) with values in $\{0, 1, 2, 3\}$; running the automaton

$$F \xrightarrow{0} F, \quad F \xrightarrow{1} N_3, \quad F \xrightarrow{3} A_3, \quad N_3 \xrightarrow{0} F, \quad N_3 \xrightarrow{1} N_3, \quad A_3 \xrightarrow{3} A_3$$

(every other edge, in particular pair value 2, is fatal), Eric wins exactly when the run finishes in F or A_3 . Intuitively a 1 demands a further valid, non-all-ones tail while a 3 forces all remaining pairs to be 3; this was checked against the exhaustive game for every $n < 2^{16}$.

Summing the winners up to 2^N then has three parts. The odd-length winners are counted by a digit-DP over the pair automaton that carries, per state, both the number of runs and the sum of their values, so each length contributes $(\text{count}) \cdot 2^{L-1} + (\text{value sum})$; advancing one pair is a constant-size update, giving $O(N)$ work overall. The even-length winners contribute $\sum_{\text{even } L \leq N} (2^L - 1)$ in closed form. Finally 2^N itself has bit-length $N + 1$ and is a winner precisely when that length is odd, i.e. when N is even. The method reproduces $S(4) = 46$, $S(12) = 54532$ and $S(1234) \equiv 690421393$.

Problem 712: Exponent Difference

Solution: 413876461

With $\nu_{p(n)}$ the exponent of p in n , $D(n, m) = \sum_p |\nu_{p(n)} - \nu_{p(m)}|$ and $S(N) = \sum_{1 \leq n, m \leq N} D(n, m)$; we want $S(10^{12})$ modulo $10^9 + 7$.

Because D is a sum over primes, $S(N)$ splits prime by prime. Fix p and let $a_e = |\{k \leq N : \nu_{p(k)} = e\}| = \lfloor N/p^e \rfloor - \lfloor N/p^{e+1} \rfloor$. Then the prime's contribution is $\sum_{e, f} a_e a_f |e - f|$. Writing $|e - f|$ as the number of thresholds t separating e and f turns this into $2 \sum_{t \geq 0} L_t (T - L_t)$ with $L_t = \sum_{e \leq t} a_e = N - \lfloor N/p^{t+1} \rfloor$ and total $T = N$. Substituting $s = t + 1$ collapses everything to

$$S(N) = 2 \sum_p \sum_{s \geq 1} \lfloor N/p^s \rfloor (N - \lfloor N/p^s \rfloor),$$

which already reproduces $S(10) = 210$ and $S(10^2) = 37018$.

For $s \geq 2$ only primes $p \leq \sqrt{N} = 10^6$ contribute, so that part is a direct sieve sum. The $s = 1$ part, $\sum_{p \leq N} q(N - q)$ with $q = \lfloor N/p \rfloor$, is handled by grouping primes into the $O(\sqrt{N})$ blocks on which $\lfloor N/p \rfloor$ is constant; each block $(N/(q+1), N/q]$ contributes $q(N - q)$ times the number of primes it contains. Those prime counts at all the values $\lfloor N/i \rfloor$ come from one Lucy_Hedgehog sieve, after which the block sum is immediate. The whole computation runs in a few seconds.

Problem 713: Turán's Water Heating System

Solution: 788626351539895

The heater needs two working fuses, m of the N are good but unknown, and $T(N, m)$ is the fewest pair-tests guaranteeing a try in which both chosen fuses work. We want $L(N) = \sum_{m=2}^N T(N, m)$ for $N = 10^7$.

Model a test of fuses $\{i, j\}$ as an edge. An adversary may keep answering "fails" as long as some m -set of good fuses avoids containing both endpoints of any tested edge, i.e. as long as the tested edges leave an independent set of size m . So the tester must lay down edges until no m vertices are mutually untested-adjacent — the minimum number of edges meeting every m -subset of the N vertices. This is the Turán-type covering number: making the complement K_m -free and as dense as possible (a Turán graph) shows the optimum is to split the N fuses into $k = m - 1$ almost-equal parts and take all within-part pairs,

$$T(N, m) = \sum_{i=1}^{m-1} \binom{s_i}{2}.$$

With $q = \lfloor N/k \rfloor$ the balanced split gives $N \bmod k$ parts of size $q + 1$, and the sum telescopes to the compact form

$$T(N, m) = qN - k \cdot q(q + 1)/2.$$

This reproduces $T(3, 2) = 3$ and $T(8, 4) = 7$. Finally $L(N) = \sum_{k=1}^{N-1} (qN - kq(q + 1)/2)$ is summed over the $O(\sqrt{N})$ blocks on which $q = \lfloor N/k \rfloor$ stays constant, matching $L(10^3) = 3281346$ and running instantly.

Problem 714: Duodigits

Solution: 2.452767775565e20

A *duodigit* uses at most two distinct decimal digits, $d(n)$ is the smallest positive duodigit multiple of n , and we want $D(50000) = \sum_{n=1}^{50000} d(n)$ to 13 significant figures.

The numbers split by whether $10 \mid n$. If $10 \nmid n$, then $d(n)$ turns out to have at most 15 digits, so generating every duodigit of up to 15 digits once (a few million, all fitting in a 64-bit integer), sorting them, and scanning for the first multiple of n yields $d(n)$ directly; a check confirms every such $n \leq 50000$ is covered.

If $10 \mid n$ every multiple ends in 0, so the duodigit must contain 0 and its digit set is $\{0, x\}$ for a single $x \in \{1, \dots, 9\}$; here $d(n)$ can reach 21 digits. For each x a breadth-first search over residues modulo n finds the shortest (then lexicographically least) string of 0s and x s, leading with x , that is divisible by n ; taking the minimum over x gives $d(n)$. Summing both cases as exact big integers and rounding reproduces $D(110) = 11047$, $D(150) = 53312$, $D(500) = 29570988$, and gives $D(50000) \approx 2.452767775565 \times 10^{20}$.

Problem 717: Summation of a Modular Formula

Solution: 1603036763131

For an odd prime p , $f(p) = \lfloor 2^{2^p}/p \rfloor \bmod 2^p$ and $g(p) = f(p) \bmod p$; we want $G(10^7) = \sum_{p < 10^7} g(p)$ over odd primes.

Write $M = 2^p$ and $r = 2^M \bmod p$. The exact quotient satisfies $p \cdot \lfloor 2^M/p \rfloor = 2^M - r$, and because $M \geq p$ we have $2^M \equiv 0 \pmod{2^p}$, so reducing modulo 2^p gives $p \cdot f(p) \equiv -r$. Thus $p \cdot f(p) + r$ is a multiple of 2^p that is smaller than $p \cdot 2^p$, say $p \cdot f(p) + r = 2^p s$ with $0 \leq s < p$. Reading this modulo p and using $2^p \equiv 2$ gives $2s \equiv r$, hence $s = r/2 \pmod{p}$. Finally reducing $p \cdot f(p) = 2^p s - r$ modulo p^2 isolates the wanted residue,

$$g(p) = f(p) \bmod p = ((2^p s - r) \bmod p^2)/p,$$

where $r = 2^{2^p} \bmod p$ is found by reducing the exponent modulo $p - 1$. Each prime needs only a handful of modular exponentiations with operands at most p^2 , so the whole sum to 10^7 runs in seconds. The formula reproduces $g(3) = 2$, $g(31) = 17$, $G(100) = 474$ and $G(10^4) = 2819236$.

Problem 718: Unreachable Numbers

Solution: 228579116

For positive integers a, b, c the equation $17^p a + 19^p b + 23^p c = n$ is solvable for some n ; the rest are *unreachable*, and $G(p)$ sums them. We want $G(6)$ modulo $10^9 + 7$.

Write $A = 17^p$, $B = 19^p$, $C = 23^p$ and $s = A + B + C$ (the value at $a = b = c = 1$).

Substituting $a \rightarrow a + 1$ etc. shows n is reachable iff $n - s$ lies in the numerical semigroup $S = \langle A, B, C \rangle$ of non-negative integer combinations. Since the generators are coprime, S has finitely many gaps, and the unreachable $n > 0$ are exactly $\{1, \dots, s - 1\}$ together with $\{s + g : g \text{ a gap of } S\}$. Therefore

$$G(p) = s(s - 1)/2 + s \cdot (\text{number of gaps}) + (\text{sum of gaps}).$$

The gap statistics come from the Apéry set modulo the smallest generator A : for each residue i , n_i is the least element of S with $n_i \equiv i \pmod{A}$, and writing $q_i = n_i/A$,

$$\text{number of gaps} = \sum_i q_i, \quad \text{sum of gaps} = \sum_i \left(q_i i + A \binom{q_i}{2} \right).$$

The n_i are computed by the Böcker–Lipták round-robin algorithm, which relaxes each residue cycle once per generator in $O(A)$ time and avoids a priority queue, so the $A = 17^6 \approx 2.4 \times 10^7$ residues are handled in a couple of seconds. This matches $G(1) = 8253$ and $G(2) = 60258000$.

Problem 719: Number Splitting

Solution: 128088830547982

An *S-number* is a perfect square n whose decimal digits can be cut into two or more contiguous blocks that add up to $\sqrt{n}$ (for example 6724, since $6 + 72 + 4 = 82 = \sqrt{6724}$).

We want $T(N) = \sum_{n \leq N, n \text{ is an S-number}} n$ for $N = 10^{12}$.

Rather than scan every n , iterate over the root r from 2 to $\sqrt{N} = 10^6$ and test whether r^2 splits to r . The test is a pruned recursion. Splitting a number can only lower its value-sum (the maximum, one block, is the number itself), so for a target t and remaining number m :

- if $m < t$ it is impossible;
- if $m = t$ the remaining number is itself the final block, success;
- otherwise peel off a trailing block $r = m \bmod 10^k$, and if $r < t$ recurse on the prefix $\lfloor m/10^k \rfloor$ with target $t - r$.

The trailing-block extraction handles blocks with leading zeros automatically (so $9801 = 98 + 0 + 1$ is found). Because $r^2 \neq r$ for $r \geq 2$, any successful split uses at least two blocks, as required. As a check, $T(10^4) = 41333$.

Problem 788: Dominating Numbers

Solution: 471745499

A number is *dominating* when one digit value occupies more than half of its places (for example 2022, where three of four digits are 2). Let $D(N)$ count dominating numbers below 10^N ; we need $D(2022) \bmod (10^9 + 7)$.

Counting by length

Fix a digit length d . If a value v fills more than half the d places then no other value can also exceed half, so each dominating number has a **unique** dominant value. We may therefore sum over the count k of that value with $k > d/2$ and never double count.

For a fixed value v appearing exactly k times the placements split by whether v is the leading digit, and there is a separate correction for the leading-zero rule when $v = 0$. Writing all of that out and summing over the ten possible values, the corrections cancel and leave a compact form:

$$A(d) = \sum_{k > d/2} \binom{d}{k} \cdot 9^{d-k+1},$$

the number of dominating numbers with exactly d digits. Then $D(N) = \sum_{d=1}^N A(d)$. As a check, $D(4) = 603$ and $D(10) = 21893256$, matching the problem.

Evaluating modulo $10^9 + 7$

The earlier approach evaluated $\binom{d}{k}$ and 9^{d-k+1} as exact integers; at $d \approx 2000$ those are numbers with hundreds to thousands of digits, which is slow. Working modulo the prime $p = 10^9 + 7$ keeps every value small: precompute factorials and their modular inverses (via Fermat's little theorem) up to N , and the powers of 9 up to $N + 1$. The double sum is then $O(N^2)$ cheap modular multiplications, giving $D(2022) \equiv 471745499 \pmod{p}$.

Problem 800: Hybrid Integers

Solution: 1412403576

A *hybrid-integer* is $p^q q^p$ for distinct primes p, q (for example $800 = 2^5 5^2$). Since the value's prime factorisation is exactly $p^q q^p$, distinct unordered pairs $\{p, q\}$ give distinct hybrid-integers, so $C(n)$ counts pairs $p < q$ with $p^q q^p \leq n$, i.e.

$$q \ln p + p \ln q \leq \ln n, \quad n = 800800^{800800}.$$

For a fixed p the left side increases with q , so for each prime p we binary-search the largest prime q that fits and add the number of primes in $(p, q]$. The whole thing needs primes only up to $q_{\max(2)} \approx (\ln n) / \ln 2 \approx 1.57 \cdot 10^7$ (the bound is loosest at $p = 2$); larger p run out of room once $2p \ln p > \ln n$, around $p \approx 4 \cdot 10^5$.

Floating-point logs place the cutoff to within a prime or two, but the boundary can be hit *exactly* — already in the example $C(800) = 2$, where $2^5 5^2 = 800$ sits on the line. So after the float binary search the boundary prime is re-checked with a 60-digit `Decimal` comparison of $q \ln p + p \ln q$ against $800800 \ln 800800$, nudging the count up or down as needed. This reproduces $C(800) = 2$ and $C(800^{800}) = 10790$.

Problem 801: $x^y \equiv y^x$

Solution: 638129754

For a prime p , count pairs $0 < x, y \leq p^2 - p$ with $x^y \equiv y^x \pmod{p}$, then sum $f(p)$ over the $\approx 2.7 \cdot 10^4$ primes in $[10^{16}, 10^{16} + 10^6]$, modulo $M = 993353399$.

Reducing $f(p)$ to a counting problem

The range $p^2 - p = p(p - 1)$ is the key. Because p and $p - 1$ are coprime, $x \mapsto (x \bmod p, x \bmod (p - 1))$ is a bijection from $\{1, \dots, p(p - 1)\}$ onto all pairs, and the two coordinates vary independently. By Fermat, $x^y \bmod p$ depends only on $(x \bmod p)$ and the exponent $(y \bmod (p - 1))$; likewise $y^x \bmod p$ depends only on $(y \bmod p)$ and $(x \bmod (p - 1))$. Writing $V(b, e)$ for the value of base b raised to exponent e (so $V = 0$ when $b \equiv 0$, else $b^e \bmod p$ with $b^0 = 1$), the condition becomes $V(a, \beta) = V(c, \delta)$ where a, c range over $\mathbb{Z}/p$ and β, δ over $\mathbb{Z}/(p - 1)$, all four free. Hence, grouping by the common value,

$$f(p) = \sum_v N(v)^2, \quad N(v) = \#\{(b, e) : V(b, e) = v\}.$$

Splitting off $v = 0$ (which needs $b \equiv 0$, giving $N(0) = p - 1$) and writing every nonzero base as a power of a primitive root turns the rest into counting solutions of $i\beta \equiv k \pmod{m}$ with $m = p - 1$. If $T(k)$ counts pairs $(i, j) \in (\mathbb{Z}/m)^2$ with $ij \equiv k$, then

$$f(p) = (p - 1)^2 + A(p - 1), \quad A(m) = \sum_{k=0}^{m-1} T(k)^2.$$

Brute force confirms $f(5) = 104$, $f(7) = 366$, $f(11) = 1550$.

A closed form for A

A is multiplicative (CRT splits T as a product), so it suffices to evaluate it on prime powers. Classifying pairs by the p -adic valuation of each factor gives, for $v_{p(k)} = t < e$, the value $T(k) = (t + 1)p^t \varphi(p^{e-t})$, and for $k = 0$ the count Z . Summing $T(k)^2$ over the $\varphi(p^{e-t})$ residues of each valuation class yields

$$A(p^e) = Z^2 + \sum_{t=0}^{e-1} (t + 1)^2 p^{2t} \varphi(p^{e-t})^3, \quad Z = p^{2e} - \sum_{t=0}^{e-1} (t + 1) p^t \varphi(p^{e-t})^2,$$

with $\varphi(p^j) = p^{j-1}(p - 1)$. Everything is additions and multiplications, so the whole computation runs modulo M even though M is not prime. This matches the brute-force $A(p^e)$ and reproduces $f(97) = 1614336$, $S(1, 10^2) = 7381000$ and $S(1, 10^5) \equiv 701331986 \pmod{M}$.

Evaluating over the window

Primes $p \in [10^{16}, 10^{16} + 10^6]$ come from a segmented sieve using base primes up to $\sqrt{10^{16}} = 10^8$. The same base primes drive a parallel factorisation sieve over the shifted window $g = p - 1$: each base prime is divided out of every multiple it hits (recording its exponent and folding $A(q^e)$ into a running product), and any residue left above 1 is necessarily a single prime $> 10^8$ (two such factors would exceed 10^{16}). Summing $(p - 1)^2 + A(p - 1)$ over the sieved primes gives the answer mod M .

Problem 802: Iterated Composition

Solution: 973873727

We are given $f(x, y) = (x^2 - x - y^2, 2xy - y + \pi)$ on $\mathbb{R}^2$, and $P(n)$ sums the x -coordinates of all points of period n under f , where a point has period n when $f^{(n)}$ fixes it. The answer is $P(10^7) \pmod{1020340567}$.

The map is a complex polynomial

Identify (x, y) with $z = x + iy$. Since $(x + iy)^2 = (x^2 - y^2) + 2ixy$, the pair $f(x, y)$ is exactly the real and imaginary parts of

$$g(z) = z^2 - z + i\pi,$$

because $g(z)$ has real part $x^2 - y^2 - x$ and imaginary part $2xy - y + \pi$. So iterating f is iterating the single complex polynomial g .

Sums of x -coordinates by Vieta

A point has period dividing k exactly when it is a root of $g^{(k)}(z) - z$. This is a **monic** polynomial of degree 2^k , so by Vieta the sum of its roots is $-[z^{2^k-1}]g^{(k)}$, and the sum of the x -coordinates is the real part of that. Write

$$S(k) = \sum_{g^{(k)}(z)=z} \Re(z).$$

Composing $g^{(k)} = g \circ g^{(k-1)}$, the subleading coefficient comes only from squaring the leading two terms of $g^{(k-1)}$; the $-z$ term affects only low-degree coefficients for $k \geq 2$. This gives the doubling recurrence $a_k = 2a_{k-1}$ on the subleading coefficient, and tracing the constant in $g(z) - z = z^2 - 2z + i\pi$ yields

$$S(1) = 2, \quad S(k) = 2^{k-1} \quad (k \geq 2).$$

A symbolic check of $g^{(k)}(z) - z$ for $k \leq 5$ confirms $S = 2, 2, 4, 8, 16$.

Möbius inversion over the minimal period

Let $E(m)$ be the sum of x -coordinates over points of *minimal* period m . Then $S(k) = \sum_{m \mid k} E(m)$, and $P(n) = \sum_{m=1}^n E(m)$. Reindexing the double sum over the divisor relation gives the clean form

$$P(n) = \sum_{d=1}^n S(d) \cdot M(\lfloor n/d \rfloor),$$

where M is the Mertens function $M(x) = \sum_{i \leq x} \mu(i)$. This reproduces the given $P(1) = 2$, $P(2) = 2$, $P(3) = 4$.

Computation

A linear sieve gives $\mu(k)$ for $k \leq 10^7$; a prefix sum gives M . A single pass over $d = 1, \dots, 10^7$ accumulates $S(d) \cdot M(\lfloor n/d \rfloor)$ modulo 1020340567, maintaining 2^{d-1} as a running power of two and reducing the (possibly negative) Mertens values into range.

Problem 803: Pseudorandom Sequence

Solution: 9300900470636

The Rand48 generator iterates $a_n = (25214903917a_{n-1} + 11) \bmod 2^{48}$, emits $b_n = \lfloor a_n / 2^{16} \rfloor \bmod 52$, and encodes each b_n as a letter ($0 \rightarrow a, \dots, 25 \rightarrow z, 26 \rightarrow A, \dots, 51 \rightarrow Z$). Given that the stream begins `Puzzle0ne`, we want the first index at which `LuckyText` appears.

Recovering the state behind a 9-letter window

Nine letters carry about $9 \log_2 52 \approx 51$ bits, slightly more than the 48-bit state, so a given 9-letter window is produced by essentially one state a . Write $a = x_{10} + 2^{18}y$ with $x_{10} < 2^{18}$. The contribution of each output modulo 4 is just bits 16, 17 of a_k , which depend only on x_{10} (the low 18 bits propagate among themselves under the recurrence modulo 2^{18}). The nine mod-4 conditions therefore form 18 bits of constraint on the 18-bit x_{10} , leaving about one candidate. For each candidate the high 30 bits y are scanned; the first letter's condition modulo 13 is affine in y (namely $4y + \text{bits} \equiv b_0 \pmod{13}$), fixing y modulo 13 and shrinking the scan by a factor of 13. All arithmetic is done modulo 2^{48} using 64-bit products masked to 48 bits, which is exact because multiplication modulo 2^{64} preserves the low 48 bits.

This recovers a_0 (the state generating `Puzzle0ne`) and a^* (the state generating `LuckyText`). The method reproduces the given facts: `EULERcats` yields $a_0 = 78580612777175$, and the `RxqLBfWzv` state sits 100 steps after $a_0 = 123456$.

Counting the steps between two states

The affine map $T(x) = Ax + C \bmod 2^{48}$ satisfies the Hull–Dobell conditions (C odd, $A \equiv 1 \pmod{4}$), so it has full period 2^{48} and every state is reached exactly once per period. Hence the step count n from a_0 to a^* is unique, and once $T^{n(a_0)}$ and a^* agree modulo 2^k , the count is determined modulo 2^k . Lifting bit by bit: given $n \bmod 2^k$, exactly one of n or $n + 2^k$ makes $T^{n(a_0)} \equiv a^* \pmod{2^{k+1}}$. Each iterate $T^{n(a_0)}$ is evaluated by fast exponentiation of the affine map. The resulting gap is the first occurrence index.

Problem 804: Counting Binary Quadratic Representations

Solution: 4921370551019052

Here $g(n)$ counts integer pairs (x, y) with $x^2 + xy + 41y^2 = n$, and $T(N) = \sum_{n=1}^N g(n)$. Summing g over $1, \dots, N$ simply counts every lattice point $(x, y) \in \mathbb{Z}^2$ whose form value lies in $[1, N]$, so $T(N) = \#\{(x, y) : 1 \leq x^2 + xy + 41y^2 \leq N\}$.

An exact lattice condition

The form is positive definite with discriminant $1 - 4 \cdot 41 = -163$. Multiplying the inequality by 4 and completing the square,

$$4(x^2 + xy + 41y^2) = (2x + y)^2 + 163y^2,$$

so $x^2 + xy + 41y^2 \leq N$ is exactly $(2x + y)^2 \leq 4N - 163y^2$. For a fixed y with $D = 4N - 163y^2 \geq 0$, set $s = \lfloor \sqrt{D} \rfloor$; since $2x + y$ is an integer, the condition becomes $-s \leq 2x + y \leq s$, a contiguous run of x whose length is counted with one floor and one ceiling. Real solutions require $|y| \leq 2\sqrt{N/163}$, about $1.57 \cdot 10^7$ rows for $N = 10^{16}$.

Computation

The row count is even in y (sending $x \rightarrow -x$ maps the y row to the $-y$ row), so it suffices to sum over $y \geq 1$, double, add the $y = 0$ row, and subtract 1 for the origin (where the form vanishes and is excluded). For $N = 10^{16}$ the quantity $4N - 163y^2$ stays below $4 \cdot 10^{16} < 2^{63}$, so an exact 64-bit integer square root keeps every floor correct. This reproduces $T(10^3) = 474$ and $T(10^6) = 492128$.

Problem 805: Shifted Multiples

Solution: 119719335

Let $s(n)$ move the leading digit of n to the back, and let $N(r)$ be the smallest positive n with $s(n) = rn$ (or 0 if none exists). We want $T(200) = \sum N(u^3/v^3)$ over all ordered coprime pairs (u, v) with $1 \leq u, v \leq 200$, reported $(\bmod 10^9 + 7)$.

Reducing $N(r)$ to a modular order

Write a d -digit number as $n = a \cdot 10^{d-1} + m$, where $a \in \{1, \dots, 9\}$ is the leading digit and $0 \leq m < 10^{d-1}$. Moving a to the back gives $s(n) = 10m + a$, so with $r = u^3/v^3$ the equation $s(n) = rn$ rearranges to

$$m = a \cdot (u^3 \cdot 10^{d-1} - v^3) / D, \quad D = 10v^3 - u^3.$$

Since $s(n) < 10n$ always, $D > 0$ is required: if $u^3 \geq 10v^3$ then $N(r) = 0$. From $D = 10v^3 - u^3$ we get $u^3 \equiv 10v^3 \pmod{D}$, hence $u^3 \cdot 10^{d-1} - v^3 \equiv v^3(10^d - 1) \pmod{D}$. Writing $D' = D / \gcd(D, av^3)$, the integrality condition $D \mid a(u^3 10^{d-1} - v^3)$ collapses to

$$D' \mid 10^d - 1, \quad \text{i.e.} \quad 10^d \equiv 1 \pmod{D'}.$$

So a valid digit count d exists only when $\gcd(10, D') = 1$ (or $D' = 1$), and then d must be a positive multiple of $\omega = \text{ord}_{D'}(10)$.

Selecting the smallest n

Two range constraints pin down d for each leading digit a . The bound $m \geq 0$ forces $10^{d-1}u^3 \geq v^3$ (a lower bound on d), and $m < 10^{d-1}$ rearranges to $10^{d-1}(u^3(a+1) - 10v^3) < av^3$; when the coefficient $u^3(a+1) - 10v^3$ is positive this caps d at a tiny value, otherwise it holds for all d . Because n lies in the disjoint, increasing window $[a \cdot 10^{d-1}, (a+1)10^{d-1})$, the candidate with the fewest digits wins, ties broken by the smaller a . For each a the best d is the least multiple of ω meeting the lower bound (then checked against the upper bound).

Computation

For each coprime pair the modulus $D' < 10v^3 \leq 8 \cdot 10^7$, so its multiplicative order is found by trial-factoring with primes up to 9000, taking $\text{ord}_{p^e}(10) \mid p^{e-1}(p-1)$ on each prime power and combining by lcm. The winning $N(r)$ is assembled $(\bmod 10^9 + 7)$ via $m \equiv a(u^3 \cdot 10^{d-1} - v^3)D^{-1}$ and $n \equiv a \cdot 10^{d-1} + m$, where D is invertible because $D < 10^9 + 7$. This reproduces $N(3) = 142857$, $N(1/10) = 10$, $N(2) = 0$, and $T(3) = 262429173$.

Problem 808: Reversible Prime Squares

Solution: 3807504276997394

A reversible prime square is the square of a prime that is not a palindrome and whose digit reversal is also the square of a prime. Because squares grow with their root, iterating primes in increasing order yields these in increasing order too. Sieve primes up to about $3.2 \cdot 10^7$ (where the fiftieth occurs); for each prime p , reverse p^2 , and if the reversal is a perfect square whose root is prime, count p^2 . Stop at the fiftieth and sum.

Problem 810: XOR-Primes

Solution: 124136381

The XOR-product is carry-less binary multiplication: long multiplication in base 2 with the partial results combined by XOR rather than addition. This is exactly multiplication of polynomials over $\text{GF}(2)$, identifying an integer with the polynomial whose coefficients are its binary digits. An XOR-prime, an integer > 1 that is not an XOR-product of two integers > 1 , is therefore an irreducible polynomial over $\text{GF}(2)$ of degree ≥ 1 , read back as an integer. The first few are 2, 3, 7, 11, 13, ... (that is X , $X + 1$, $X^2 + X + 1$, $X^3 + X + 1$, $X^3 + X^2 + 1$), with 41 the tenth.

A polynomial of degree d has integer value in $[2^d, 2^{d+1})$, so ordering the XOR-primes by value is the same as ordering by degree and then by value within a degree. The number of monic irreducibles of degree d over $\text{GF}(2)$ is $L(d) = \frac{1}{d} \sum_{e|d} \mu(e) 2^{d/e}$.

Accumulating $L(d)$, the running total first passes $5 \cdot 10^6$ at degree 26 (the 5{,}000{,}000th lies 2{,}192{,}804 entries into degree 26, value in $[2^{26}, 2^{27})$), so sieving up to $N = 2^{27}$ captures it.

Computation

A carry-less sieve marks every XOR-composite below N . Any composite of degree ≤ 26 has a factor of degree ≤ 13 , so it suffices to run p over the still-unmarked values (the XOR-primes) with $\deg p \leq 13$, and for each mark $p \otimes q$ for every $q \geq 2$ with $\deg p + \deg q \leq 26$. The carry-less product uses the shift-and-XOR loop $r \leftarrow r \oplus a$ while shifting a left and b right. The unmarked values ≥ 2 , taken in increasing order, are the XOR-primes; the 5{,}000{,}000th is 124136381. The sieve runs in a few seconds.

Problem 836: A Bold Proposition

Solution: aprilfoolsjoke

An April Fools' problem; the answer is the literal string hidden in the prompt.

Problem 901: Well Drilling

Solution: 2.364497769

We choose an increasing sequence of depths $0 = d_0 < d_1 < d_2 < \dots$ and drill to each in turn until we strike water. The groundwater depth is $X \sim \text{Exp}(1)$, so $P(X > d) = e^{-d}$, and a drill to depth d_i is only attempted when every earlier drill failed, i.e. with probability $e^{-d_{i-1}}$. Each attempt costs its full depth in hours (we start over each time), so the expected total time is

$$E = \sum_{i \geq 1} d_i e^{-d_{i-1}}.$$

Setting $\partial E / \partial d_j = 0$ gives $e^{-d_{j-1}} = d_{j+1} e^{-d_j}$, that is the optimal depths satisfy the recurrence

$$d_{j+1} = e^{d_j - d_{j-1}}.$$

With $d_0 = 0$ this is a two-point boundary value problem ($d_0 = 0$ and $d_N \rightarrow \infty$). Forward shooting from a guessed d_1 is numerically unstable, because the recurrence amplifies any error exponentially. Instead we solve the whole stationary system at once with Newton's method on the residuals $F_j = d_{j+1} - e^{d_j - d_{j-1}}$, whose Jacobian is tridiagonal, pinning the far end d_N at a large value L (the tail beyond depth L contributes less than e^{-L} and is negligible). The depths converge stably, and the resulting expected time is 2.364497769 hours, robust across choices of N and L .

Problem 905: Now I Know

Solution: 70228218

Three logicians wear hats with positive integers, exactly one of which is the sum of the other two. Each sees the other two numbers but not their own, and in cyclic order A, B, C says either "I don't know" or "Now I know"; $F(A, B, C)$ is the turn on which someone first knows.

The clean way to reason is as public elimination of possible worlds, a world being any valid triple. At turn t the player $p = (t - 1) \bmod 3$ sees the two coordinates other than p ; the value they could hold is either the sum or the absolute difference of what they see, so there are at most two candidate worlds: the real one and its "flip" at position p . A world is eliminated at the first turn on which its scheduled player would know. Calling that turn F of the world,

$$\text{knows}(w, t) \iff (\text{the two seen values are equal}) \vee (F(\text{flip}) \leq t - 1).$$

Taking the minimum over the three players shows the deciding player is always the holder of the sum, and flipping the sum coordinate to $|s - u|$ is exactly one subtractive step of the Euclidean algorithm on the two "part" values, with the sum position migrating to whichever part is larger. So F is assembled bottom-up along the Euclidean descent of the parts: at the base, where the two parts are equal, $F = p_{\text{sum}} + 1$, and climbing one step

$$F = \text{smallest } t > F_{\text{below}} \text{ with } t \equiv p_{\text{sum}} + 1 \pmod{3},$$

where $p_{\text{sum}} \in \{0, 1, 2\}$ is the position of the sum at that step.

A literal subtractive descent is far too long, since a single quotient can take on the order of 10^7 steps. But within one quotient the sum position merely alternates between two slots, and along an alternating run the climb's increment is constant after the first step (each pair of steps raises F by exactly 3), so an entire run is collapsed to $O(1)$ arithmetic. The descent then has only $O(\log)$ runs per triple. Summing $F(a^b, b^a, a^b + b^a)$ over $a \in \{1, \dots, 7\}$ and $b \in \{1, \dots, 19\}$ gives 70228218.

Problem 932: 2025

Solution: 72673459417881349

A 2025-number N splits into a leading part a and a trailing part b (with b having a fixed digit length and no leading zero) such that $N = (a + b)^2$. Writing $s = a + b$, every such N is simply s^2 , so iterate over s rather than over the parts: for each split position k , set $a = \lfloor N/10^k \rfloor$ and $b = N \bmod 10^k$, and accept when $a + b = s$ and b has exactly k digits.

A cheap filter cuts the search by more than four-fifths. Since $10^k \equiv 1 \pmod{9}$, we have $N = a \cdot 10^k + b \equiv a + b = s \pmod{9}$; combined with $N = s^2$ this forces $s^2 \equiv s$, i.e. $s \equiv 0$ or $1 \pmod{9}$. Only those s need be examined.

Problem 997: Dice Box

Solution: 5765993594880

We pack xyz ordinary dice into an $x \times y \times z$ box so that every pair of touching faces shows the same value, the dice being indistinguishable up to rotation, and count the arrangements $f(x, y, z)$. We are told $f(1, 1, 1) = 24$ and $f(2, 3, 4) = 18432$ and want $f(9, 10, 11)$.

A cube has the three opposite-face pairs $\{1, 6\}$, $\{2, 5\}$, $\{3, 4\}$ and 24 rotations, giving $f(1, 1, 1) = 24$. An orientation chooses which pair sits on each axis (a permutation, 6 ways) and which member of each pair faces the positive direction; chirality forces an even number of sign flips, leaving 4 patterns, and $6 \cdot 4 = 24$.

Two dice touching along the x axis must carry the **same** pair on that axis with the orientation flipped. Propagating, every die on a line parallel to an axis shares that axis's pair and the signs alternate. Letting $X(j, k)$, $Y(i, k)$, $Z(i, j)$ be the pairs on the three axes, the three pairs at each cell are distinct while the alternating signs interact through the chirality rule. Counting both the pair assignment and the sign/chirality layer makes f satisfy, in each variable, the recurrence with characteristic roots 2 and 4; fitting the resulting symmetric closed form to small boxes yields

$$f(x, y, z) = 24 \cdot 2^{x+y+z-4} (2^x + 2^y + 2^z - 4),$$

which reproduces the full brute-force table and the given values. Hence $f(9, 10, 11) = 24 \cdot 2^{26} \cdot 3580 = 5765993594880$.

Problem 998: Squaring the Triangle

Solution: 4439835458570

The minimum bounding square of a triangle is the smallest square, in any orientation, that covers it. We need $T(n)$, the sum of perimeters of all non-congruent integer-sided triangles whose minimum bounding square has integer side $s \leq n$, for $n = 10^6$.

For a fixed orientation θ the axis-aligned bounding box has width $W(\theta)$ and height $H(\theta)$, and the bounding square has side $\max(W, H)$. Minimising over θ , the optimum occurs in one of two ways.

Height-dominated (flush). One edge lies along a side of the square and the opposite vertex touches the far side, so s equals the altitude to that edge while the perpendicular extent is no larger. Taking the shortest side a (its altitude h is the largest), the triangle is two integer right triangles glued along the common leg h : legs p, q with $p^2 + h^2$ and $q^2 + h^2$ perfect squares, base $a = p + q$, and $s = h$. Comparing the flush value h with the tilted value of the same edge pair gives the exact regime

$$a \leq h \quad \text{and} \quad h^2 \leq h(p + q) + pq,$$

the latter being $h^2(h - a)^2 \leq (pq)^2$.

Balanced (tilted). Here $W = H = s$; the square is tight on all four sides but the triangle has only three vertices, so one vertex sits at a corner and the other two on the opposite sides. With the corner at $(0, s)$ and the others at $(u, 0)$, (s, w) , the squared side lengths are $u^2 + s^2$, $s^2 + (s - w)^2$ and $(s - u)^2 + w^2$. Writing $l_1 = u$, $l_2 = s - w$, this requires l_1, l_2 to be legs of s (i.e. $l_i^2 + s^2$ a perfect square) together with $(s - l_1)^2 + (s - l_2)^2$ a perfect square.

Both families are generated from the Pythagorean legs of s — the $p < s$ with $p^2 + s^2$ a square — read off from the divisor pairs $d \cdot (s^2/d) = s^2$ of equal parity via $p = (s^2/d - d)/2$. A triangle can have integer s only if its area is rational, so the (rare) tilted hits are confirmed by recomputing the true minimum exactly with rational arithmetic, taking the smallest feasible candidate of the edge-direction system. Summing over $s \leq 10^6$ gives $T(10^6) = 4439835458570$, matching the given $T(40) = 346$, $T(400) = 76402$ and $T(2000) = 3237036$.

Problem 999: Alternating Recurrence

Solution: 801096743

The sequence is given by $a_1 = a_2 = a_3 = 1$, $a_4 = 2$ and the alternating quadratic relation

$$a_n^2 = a_{n+2}a_{n-2} + u \cdot a_{n+1}a_{n-1}, \quad u = \begin{cases} 1 & n \text{ even} \\ 2 & n \text{ odd} \end{cases},$$

equivalently $a_{n+2} = (a_n^2 - ua_{n+1}a_{n-1})/a_{n-2}$. The terms grow like $\exp(cn^2)$: this is an **elliptic divisibility (Somos-4 type) sequence**, with no linear recurrence, so reaching a_N for $N = 10^{18} + 3$ modulo the prime $P = 1234567891$ needs a sub-linear evaluation.

Normalisation

Write $a_n = g(n)W_n$ to absorb the alternating coefficient into a gauge factor, leaving a **constant-coefficient** normalised EDS W with $W_0 = 0$, $W_1 = 1$ and

$$W_{n+2} = (-\sqrt{2}W_{n+1}W_{n-1} + W_n^2)/W_{n-2}.$$

Seeking $a_n = \kappa\rho^n\sigma^{n^2}\omega^{(-1)^n}W_n$ and matching the two coefficient values forces $\rho = i$ (so $\rho^2 = -1$) and $\omega = 2^{-1/8}$, giving the gauge $g(n) = i^{n-1}$ for odd n and $g(n) = i^{n-1}2^{-1/4}$ for even n . The seeds become $W_2 = -i2^{1/4}$, $W_3 = -1$, $W_4 = 2i2^{1/4}$, and $W_5 = -3$ from the recurrence. For **odd** n the eighth-root cancels, so

$$a_n = i^{n-1}W_n,$$

with W_n a rational integer (mod P); this was verified against the direct definition for $n = 1, \dots, 20$, and reproduces the data points $a_{13} = 23321$ and $a_{1003} \equiv 231906014 \pmod{P}$.

Working field

Let $\theta = i2^{1/4}$. Then $\theta^4 = i^4 \cdot 2 = 2$, while $W_2 = -\theta$ and $\sqrt{2} = -\theta^2$. Every quantity that occurs is of the form $a + b\theta + c\theta^2 + d\theta^3$ over $\mathbb{F}_P$, so arithmetic is polynomial multiplication reduced modulo $(\theta^4 - 2, P)$. Odd-index W_n are pure constants; even-index ones are multiples of θ . Crucially the doubling step below divides **only** by $W_2 = -\theta$, and θ is a unit ($\theta^{-1} = \theta^3/2$), so no irreducibility of $\theta^4 - 2$ is required — the ring operations suffice.

Block doubling

Maintain the length-8 window $V = [W_{c-3}, \dots, W_{c+4}]$ with W_c at index 3. The EDS duplication identities send a window centred at c to the nine values $T_j = W_{2c+j}$ for $j = -3, \dots, 5$; for example

$$W_{2c} = (W_{c+2}W_cW_{c-1}^2 - W_{c-2}W_cW_{c+1}^2)/W_2, \quad W_{2c+1} = W_{c+2}W_c^3 - W_{c-1}W_{c+1}^3,$$

and the rest by index shifts. Reading the bits of N below the leading one, bit 0 keeps the sub-window $T[0 : 8]$ (taking $c \rightarrow 2c$) and bit 1 keeps $T[1 : 9]$ (taking $c \rightarrow 2c + 1$). Starting from $c = 1$, this reaches $c = N$ in $O(\log N)$ products and agrees with the direct recurrence on every tested N .

For $N = 10^{18} + 3$ (odd), $N - 1 = 10^{18} + 2 \equiv 2 \pmod{4}$, so $i^{N-1} = -1$ and $a_N = -W_N$. The computation gives $a_N \equiv 801096743 \pmod{P}$ in well under a second.

Problem 1000: Three Sub-Problems and a Meta-Problem

Solution: 891213201

The landmark problem bundles three independent sub-problems whose answers $I(1000)$, $X(1000)$, $C(1000)$ seed a product-recurrence sequence; the final answer is $M(1000) \bmod p$ with $p = 10^9 + 7$.

Sub-problem $I(n)$: Max And

We split $\{1, \dots, n\}$ into A, B and maximise $\sum_{a \in A} \sum_{b \in B} a \wedge b$. Since $a \wedge b$ is symmetric and counted once per crossing ordered pair, this is weighted **max-cut** on the complete graph with edge weight $i \wedge b$. Decompose by bit k (place value 2^k): writing $s_i = \pm 1$ for the side of i , and c_k for how many of $1, \dots, n$ have bit k set, the cut value is

$$\frac{1}{4} \sum_k 2^k (c_k^2 - d_k^2), \quad d_k = \sum_{i: \text{bit } k \text{ of } i} s_i.$$

The $\sum_k 2^k c_k^2$ term is fixed, so maximising the cut means **minimising** $\sum_k 2^k d_k^2$. Each d_k is a sum of c_k terms ± 1 , so it has the parity of c_k and its least possible square is $c_k \bmod 2$. That per-bit minimum is jointly attainable (there is ample freedom in the s_i ; confirmed by brute force for every $n \leq 12$), giving the closed form

$$I(n) = \frac{1}{4} \left(\sum_k 2^k c_k^2 - \sum_k 2^k (c_k \bmod 2) \right).$$

This reproduces $I(10) = 50$ and yields $I(1000) = 61217340$.

Sub-problem $X(N)$: Max Xor Sum

With edge weight $[x, y] = x^2 \oplus y^2$, a valid sequence $a_0, \dots, a_r$ is a walk in K_N whose consecutive edge weights **strictly increase**, and we maximise their sum. Sort the $\binom{N}{2}$ edges by weight and sweep upward, keeping $f[v]$, the best walk-sum that ends at vertex v using only strictly lighter edges. An edge (u, v, w) then offers $f[u] + w$ to v and $f[v] + w$ to u . Edges of equal weight are applied as one batch from the pre-batch f , so two equal-weight edges never chain (the inequality is strict). The answer is $\max_v f[v]$. This gives $X(4) = 71$, $X(10) = 702$, and $X(1000) = 728513240$.

Sub-problem $C(N)$: Unreachable Nim

A status is **unreachable** if it can only ever be an initial position. Every P -position (nim-value 0) is the target of a winning move from one of its (always N -position) predecessors, so all P -positions are reachable. An N -position is reached only from a P -position predecessor, because moving into an N -position is never a winning move. Increasing pile a to $b \oplus c$ is the unique way to turn (a, b, c) 's predecessor into a P -position, and likewise for b, c ; such a predecessor exists iff $b \oplus c > a$ (resp. $a \oplus c > b$, $a \oplus b > c$). Hence (a, b, c) is unreachable exactly when $a \oplus b \oplus c \neq 0$ together with $b \oplus c \leq a$, $a \oplus c \leq b$, $a \oplus b \leq c$. Equality in any of these forces nim-value 0, so the system is equivalently the strict, fully symmetric

$$a \oplus b < c, \quad b \oplus c < a, \quad c \oplus a < b,$$

counted over $[0, N)^3$ by a bit-wise digit DP that tracks, per inequality, whether the prefix is already less, equal, or greater. This matches the given $C(10) = 123$ and gives $C(1000) = 232607184$.

Meta-problem

With $M(0) = I$, $M(1) = X$, $M(2) = C$ and $M(k) = M(k-1)M(k-2)M(k-3)$, every $M(k) = I^{e_0}X^{e_1}C^{e_2}$ where the exponent triples follow the tribonacci recurrence from $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$. Reducing the exponents $\text{mod}(p-1)$ (Fermat) and combining with modular powers gives the result. The supplied check $M(4) = I \cdot X^2 \cdot C^2 \equiv 457587170 \pmod{p}$ holds, confirming all three values, and $M(1000) \equiv 891213201 \pmod{p}$.